

1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

J17 MLB

LAST_MODIFIED=Fri May 31 13:43:58 2013

REV	ECN	DESCRIPTION OF REVISION	CK APPD DATE
A	0002056619	PRODUCTION RELEASED	2013-05-30

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DRAWING
TITLE=J17
ABBREV=DRAWING
LAST_MODIFIED=Fri May 31 13:43:58 2013

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	Apple Inc.		
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BOM VARIANTS

BOM NUMBER	BOM NAME	BOM OPTIONS
639-4206	PCBA, MLB, GOOD, HYNIX, J17	J17_COMMON, J17, CPU:GOOD, GPU:GK187EGX, FB:2G_HYNIX, FBA, FBB, SSD:Y, EEEE:F8YL
639-4418	PCBA, MLB, GOOD, ELPIDA, J17	J17_COMMON, J17, CPU:GOOD, GPU:GK187EGX, FB:2G_ELPIDA, FBA, FBB, SSD:Y, EEEE:FCTC
985-0047	PCBA, MLB, DEV, J17	DEVELOPMENT, J17_DEVEL

CPUs

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
33754532	1	H5W, SR14E, PRQ, C0, 3.20, 84W, 4+2, 1.15, 6M, LGA	CPU	CRITICAL	CPU:GOOD
33754505	1	H5W, QE70, QS, C0, 3.26, 84W, 4+2, 1.15, 6M, LGA	CPU	CRITICAL	CPU:GOOD_TDP

CPU SOCKET

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
511S0080	1	SOCKET, MOLEX, LGA1150, CPU- HF	U0500	CRITICAL	

BOM Groups

BOM GROUP	BOM OPTIONS
J17_COMMON	COMMON, ALTERNATE, J17_COMMON1, J17_PROGPARTS
J17_COMMON1	XDP, SPEAKERID, TBTHV:P12V, CPUVCC:4PHASE, EXT_GPU:YES, VDDQ:P1V5
J17_PROGPARTS	SMC:PROG, BOOTROM:PROG, TBTRM:PROG, CIVROM:PROG, CAMROM:PROG, BLCMCU:PROG
J17_DEVEL	XDP_CONN, LPCPLUS, DDRVREF_DAC
DEVEL_SENSORS	TEMPSNSDEV, AP_ISNS:Y
J17_PRODUCTION	AP_ISNS:N

ASICs

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM_OPTION
33754541	1	1C, SR198, 287, LPT-D, C2, PRQ, 23X22MM, FCBG4708	U1100	CRITICAL	
33754562	1	1C, QEW, LPT-D, C2, SQ5, 23X22MM, FCBG4708	U1100	CRITICAL	PCH_TDP
33851113	1	1C, T87, CR-4C, B1, PRQ, 288 FCBG4, 12X12MM	U2800	CRITICAL	
33550616	1	1C, BCM57766A1, ENET&SD, 8X8	U3900	CRITICAL	

Programmable Parts

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
341S3880	1	IC, PROGRMD, EFI ROM, V0095, J17	U5210	CRITICAL	BOOTROM:PROG
335S0807	1	IC, 64 MBIT SPI SERIAL FLASH	U5210	CRITICAL	BOOTROM:BLANK
341S3874	1	IC, PROG, SMC-A3, V2.14F9, PVT, J17	U5000	CRITICAL	SMC:PROG
338S1159	1	IC, SMC, LM4FS1BH5BBCIGR, SCPL	U5000	CRITICAL	SMC:BLANK
341S3860	1	IC, EEPROM, CR, V23.10, J17	U2890	CRITICAL	TBTROM:PROG
335S0865	1	IC, EEPROM, SERIAL, 8KB, MLP8	U2890	CRITICAL	TBTROM:BLANK
341S3735	1	IC, ENET 1MBIT, SPI, ROM, V1.13 J17	U3990	CRITICAL	CIVROM:PROG
335S0862	1	IC, SERIAL FLASH, 2MBIT, 2.7V, REF F	U3990	CRITICAL	CIVROM:BLANK
341S3778	1	IC, CAMERA FLASH, V7230, J16/17	U4202	CRITICAL	CAMROM:PROG
335S0852	1	IC, FLASH, SPI, 1MBIT, 3V3	U4202	CRITICAL	CAMROM:BLANK
341S3674	1	IC, BLC, MCU, PROPROGRAMMED, V9204, DB	U8100	CRITICAL	BLCMCU:PROG
337S3978	1	IC, BLC MCU LCU2132FBD64/01, LQFP64	U8100	CRITICAL	BLCMCU:BLANK

Bar Code Labels / EEEE #'s

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
825-7896	1	LABEL, MLB, 2D	EEEE_F8YL	CRITICAL	EEEE:F8YL
825-7896	1	LABEL, MLB, 2D	EEEE_FCTC	CRITICAL	EEEE:FCTC

J17 SCHEMATIC / PCB #'S

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
951-9884	1	SCH, MLB, J17	SCH1	CRITICAL	J17
820-3478	1	PCBF, MLB, J17	PCB1	CRITICAL	J17

J17 ALTERNATES

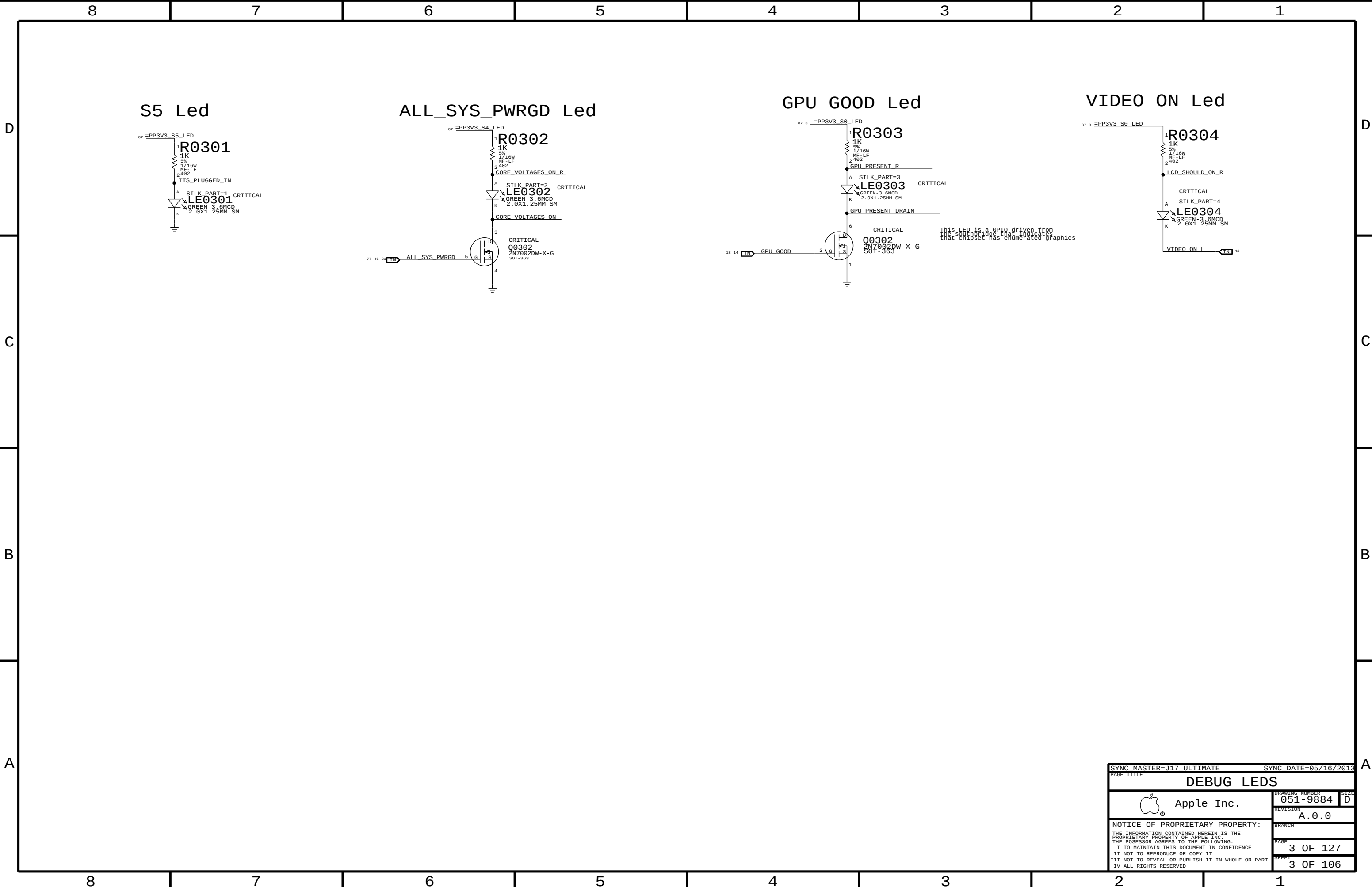
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
377S0147	377S0126		ALL	USB Diodes Array
377S0155	377S0104		ALL	USB Diode
157S0084	157S0058		ALL	Enet Magnetics
376S0975	376S1081		ALL	P/NCH DUAL FET
128S0365	128S0368		ALL	150UF CAPS BLK
138S0803	138S0804		ALL	2.2UF CAPS SOFT
126-0161	126-0160		ALL	56UF CAPS 10X12 THD
197S0481	197S0480		Y1950	25MHz PCH Xtal
197S0479	197S0478		ALL	12MHZ Cam Xtal
372S0186	372S0185		ALL	Alt temp diode
107S0251	107S0249		R5520	Alt 2mOhm sense
107S0254	107S0241		R5450, R5480	Alt 5mOhm sense
107S0255	107S0240		R5450, R5480	Alt 1mOhm sense
341S3747	341S3735		U3990	ENET ROM,ATMEL,V1.13
138S0681	138S0638		ALL	10uF Caps
376S1106	376S0969		ALL	N-Ch FET
138S0860	138S0775		ALL	Single-source 1uF
138S0859	138S0788		ALL	Single-source 10uF

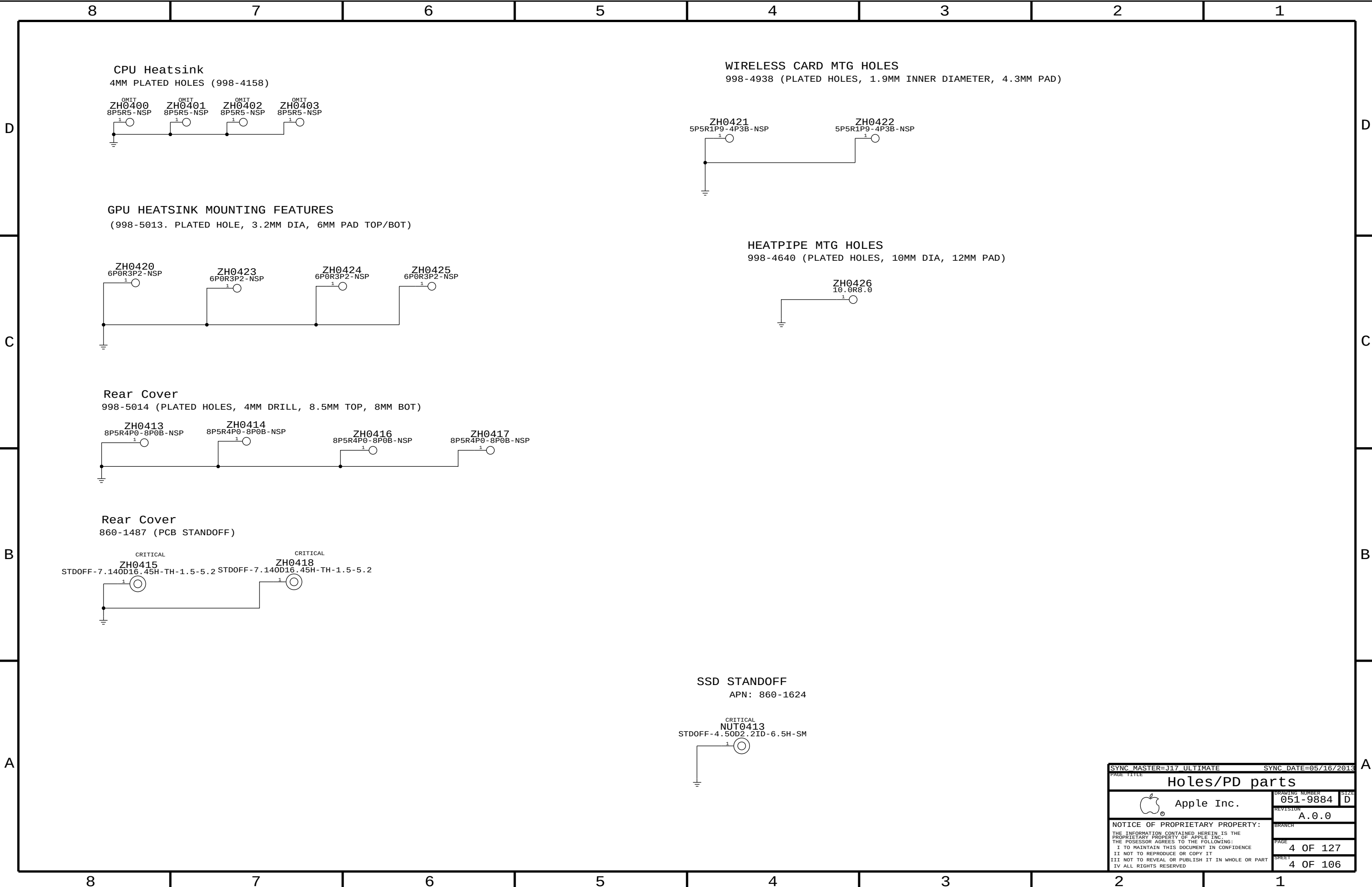
Y4200 and Y8105

GPU and VRAM

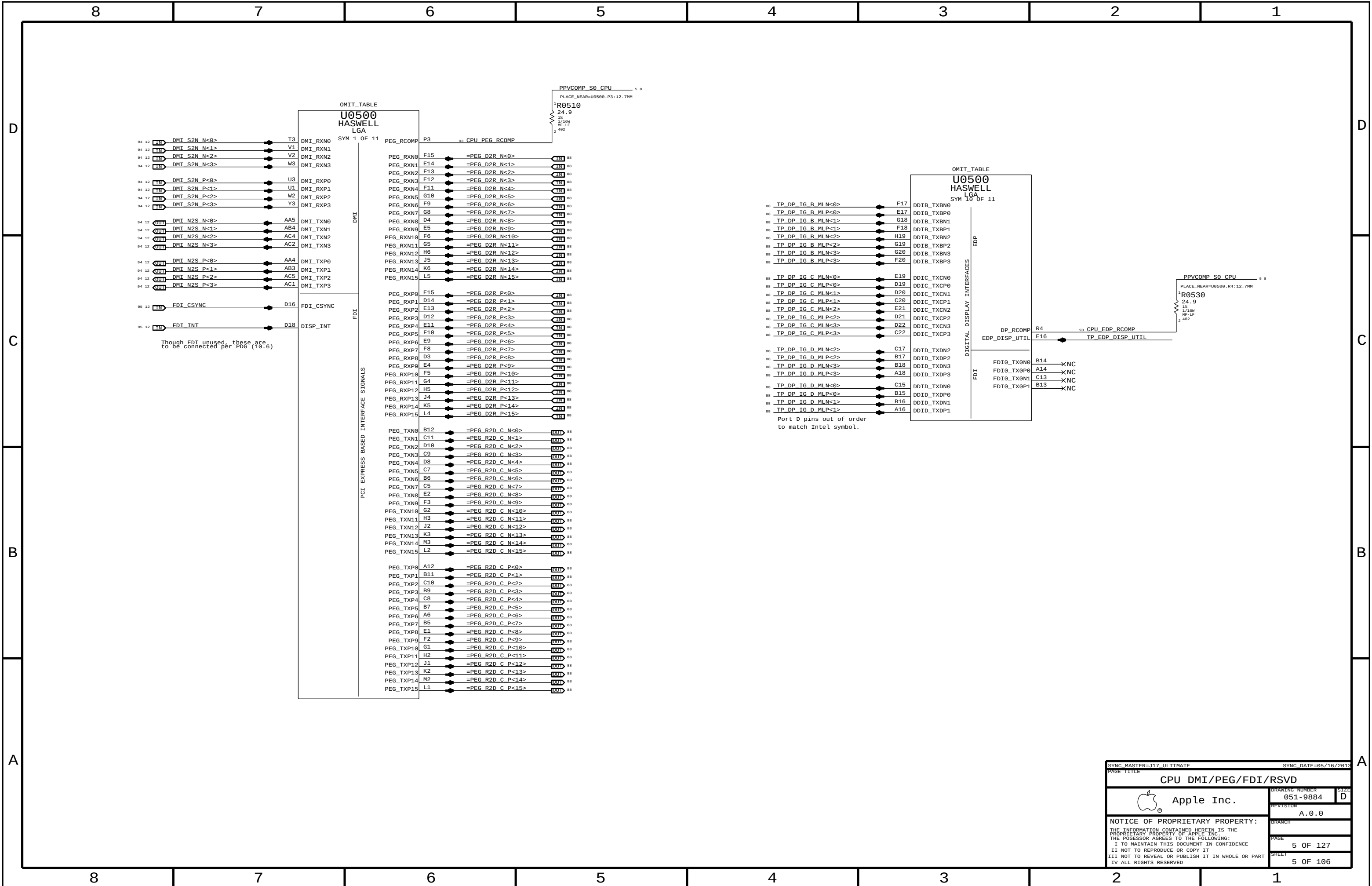
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337S4438	1	IC, GPU, NV, GK107-EGX, PS-A2	U8700	CRITICAL	GPU: GK107EGX
333S0638	4	IC, GDDR5, 2GBIT1, 64KX32, GEMMA-DIE, 1708 FBGA	U9200, U9250, U9300, U9350	CRITICAL	FB: 2G_HYNIX
333S0695	4	IC, GDDR5, 2GBIT1, 64KX32, B-DIE, 1708 FBGA	U9200, U9250, U9300, U9350	CRITICAL	FB: 2G_ELPIDA

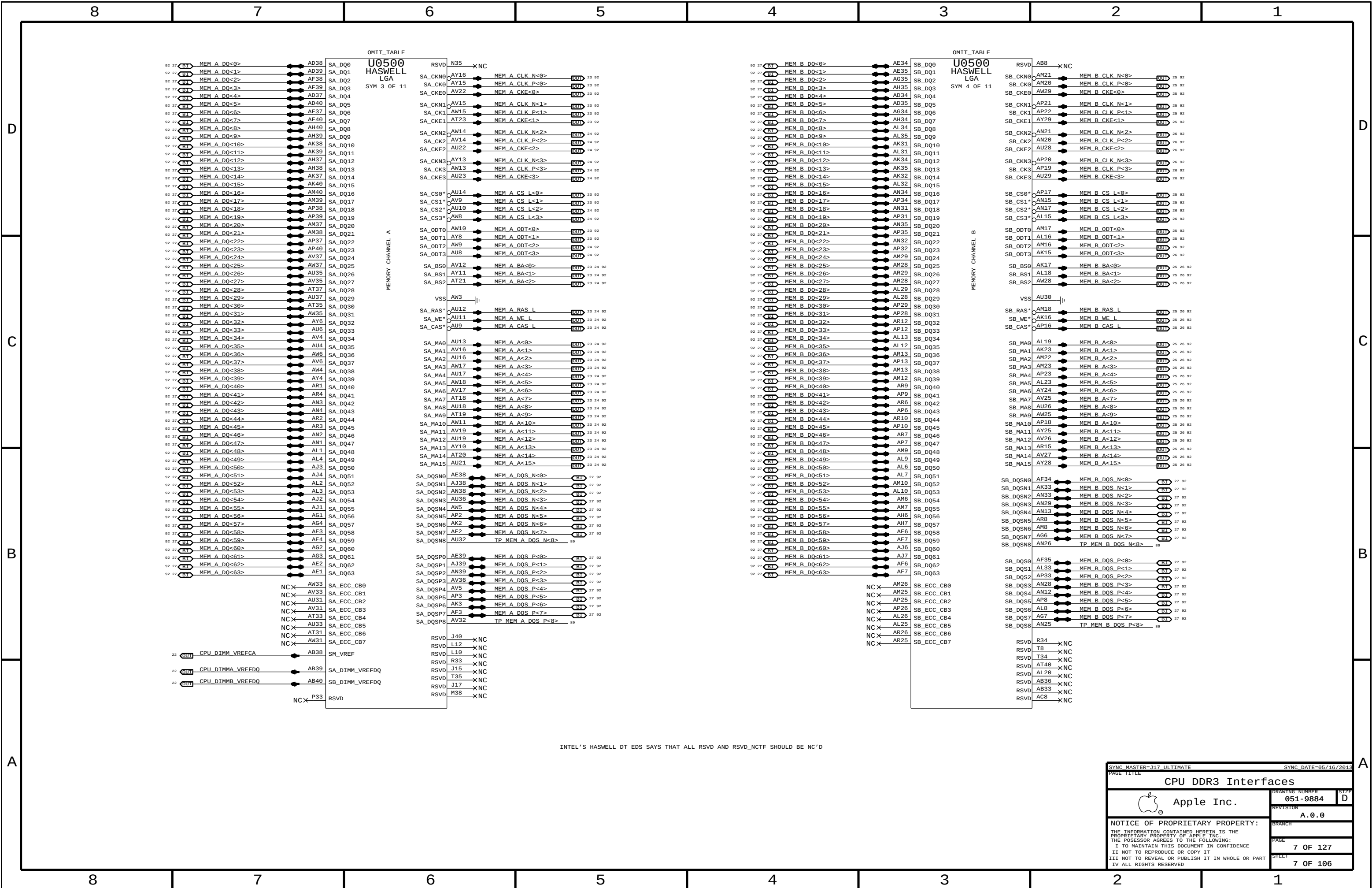
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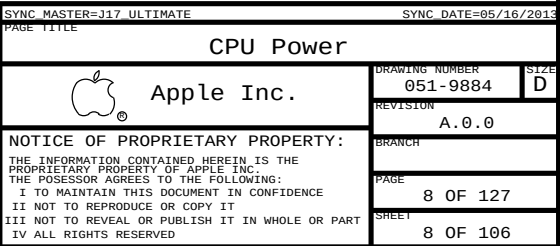


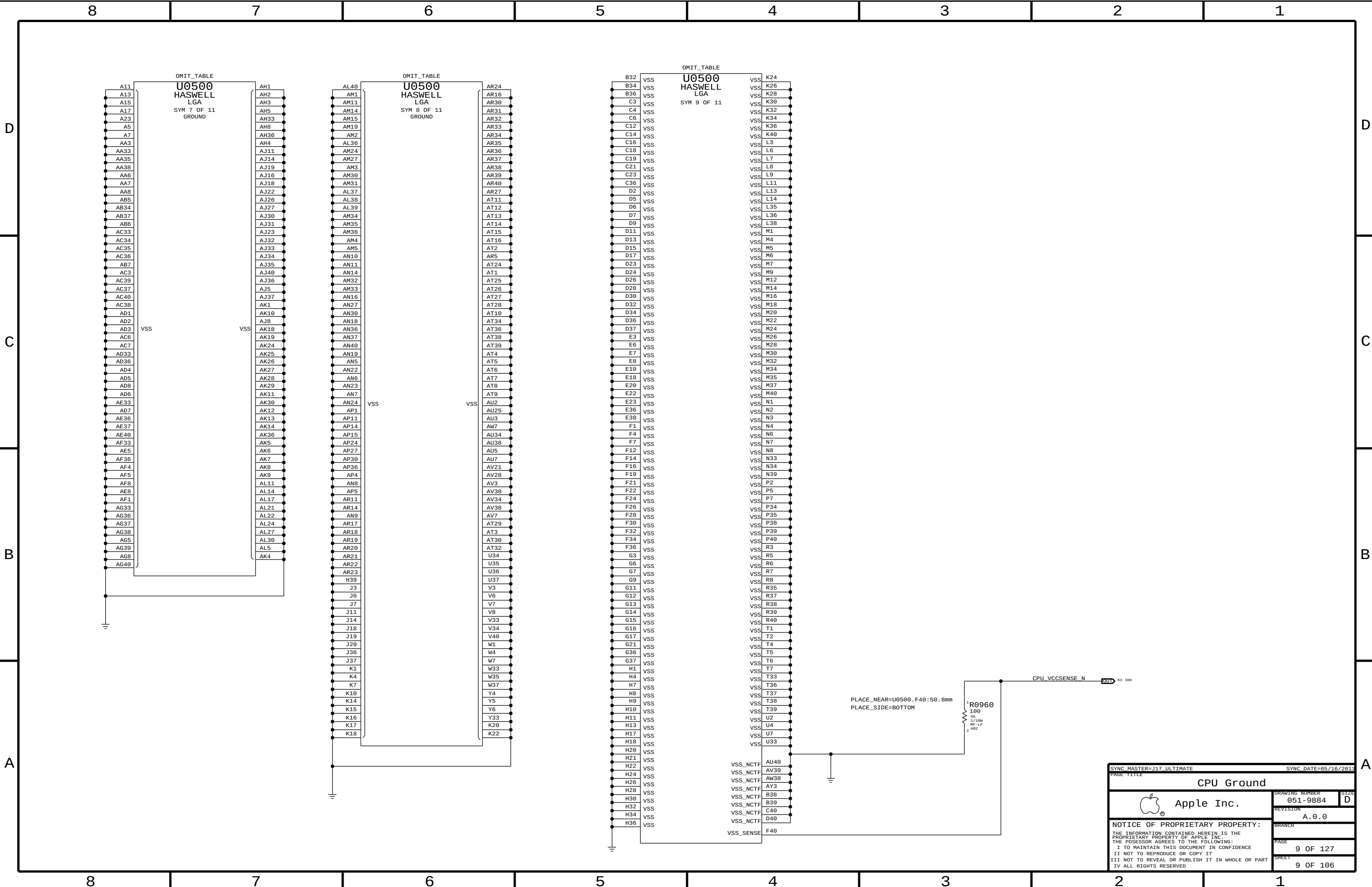
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INTEL'S HASWELL DT EDS SAYS THAT ALL RSVD AND RSVD_NCTF SHOULD BE NC'D

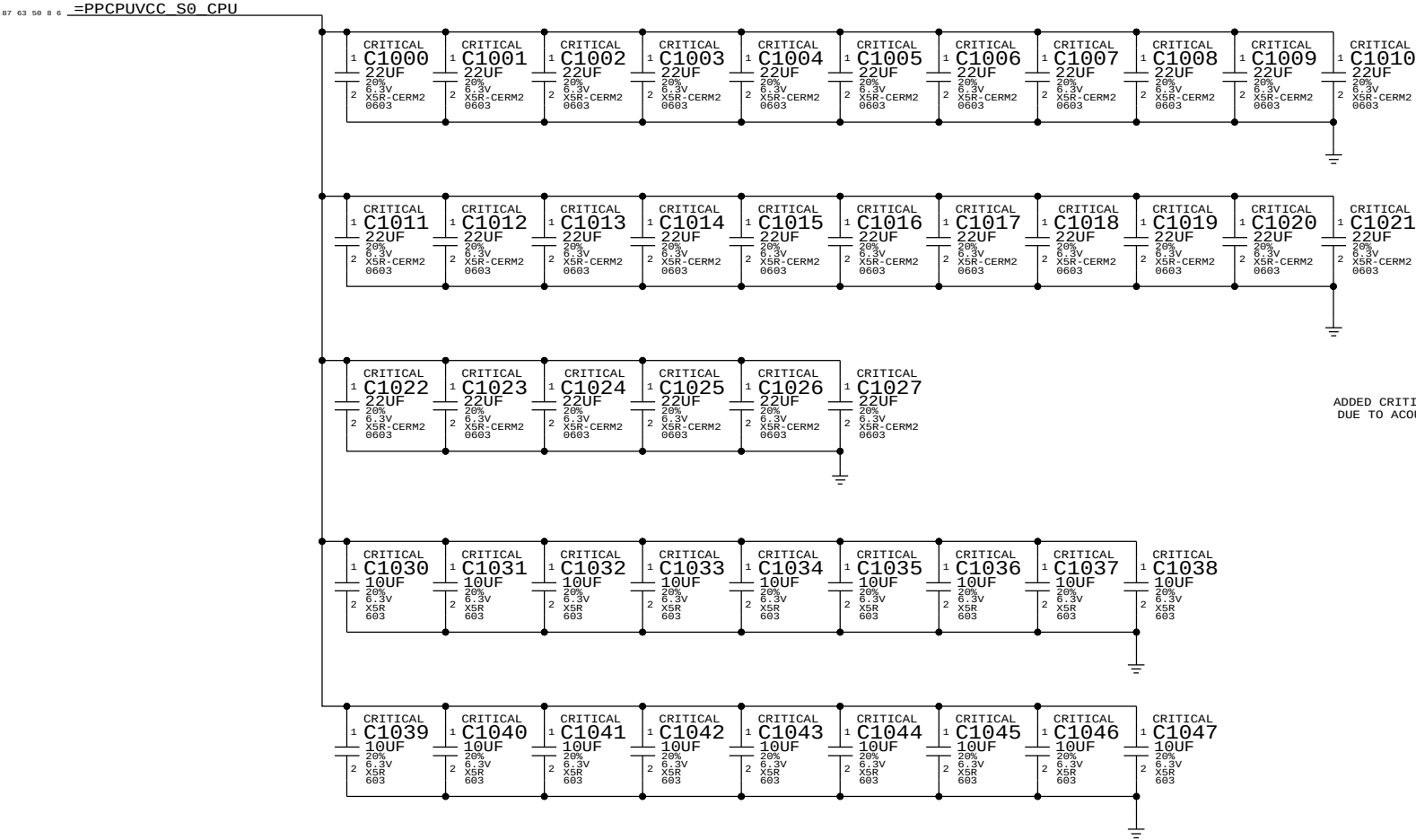




CPU VCORE DECOUPLING

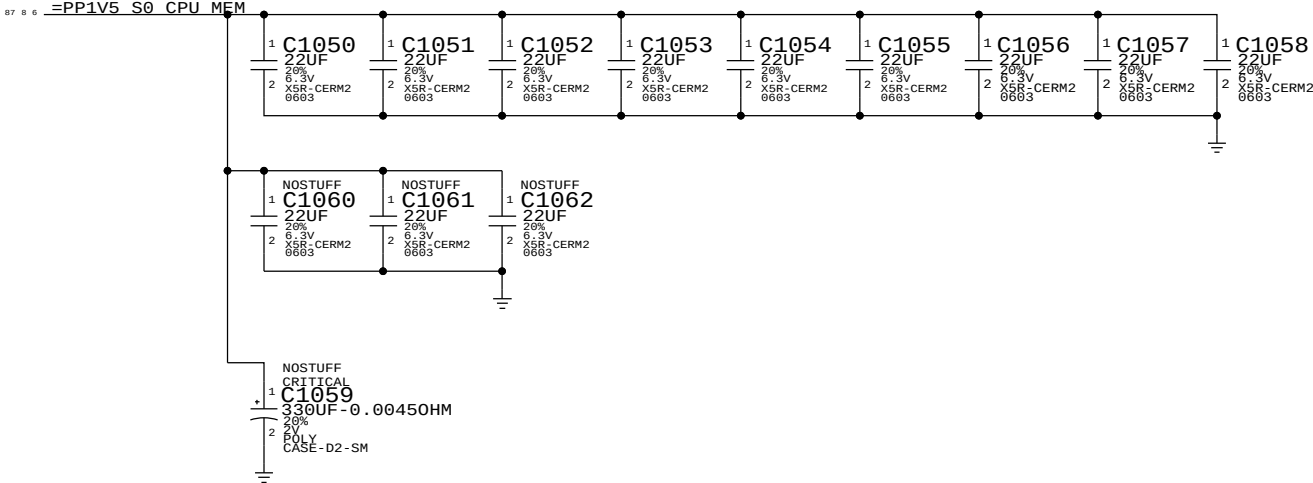
Intel Recommendation:22x 22UF 0805,topside (18 inside cavity, 4 north of processor),8x 470uF bulk caps(5 stuffed,3 no-stuffed)
Apple Implementation:28x 22UF 0603 per Harold
18x 10UF 0603 placed inside socket cavity

Layout Note: These caps should be placed symmetrically on Top and Bottom sides.
BULK CAPS ON CPU VREG PAGE 71



Memory (CPU VCCDDR) DECOUPLING

Intel Recommendation:9x 22UF 0805 near CPU power pins
Apple Implementation:9x 22UF 0603 per Harold
Layout Note: These caps should be placed symmetrically on Top and Bottom sides.



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		DRAWING NUMBER	051-9884
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NOTE: SSC control is ganged on PCIe 0-3 and 4-7 clocks.
PEG-attached (CPU) PCIe devices must use one set,
while PCH-attached PCIe devices use the other set.
If 2 or less devices are attached to PEG the
CLKOUT_PEG outputs can be used for those devices.

SATA Port assignments:
PRIMARY HDD

SSD

R1130¹
7.5K
15
1/20W
MF
201 2

Unused clock terminations for FCIM Mode

SYNC MASTER=117 ULTIMATE

SYNC DATE=05/16/2013

PCH RTC/HDA/JTAG/SATA/CLK

Apple Inc.

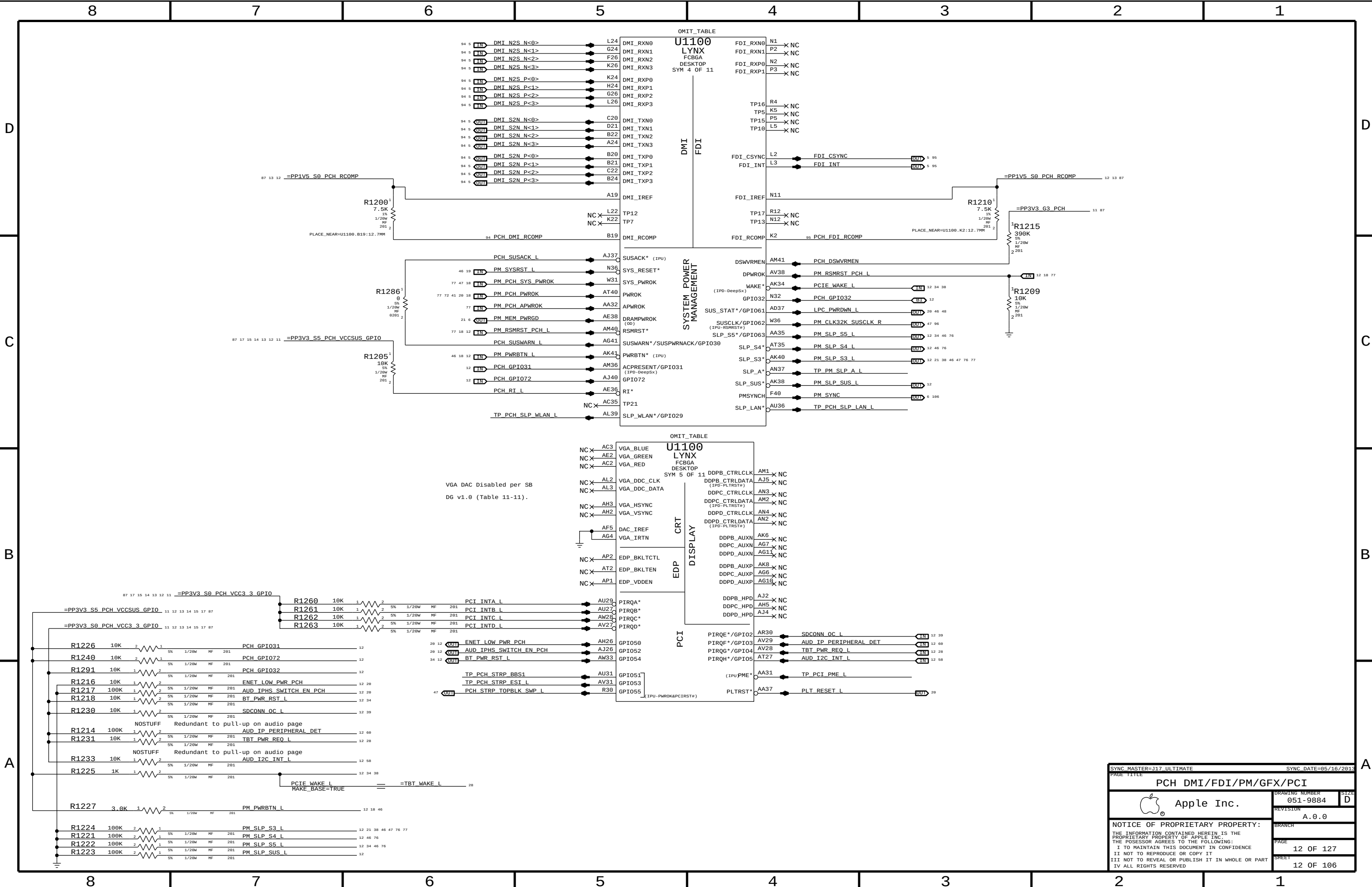
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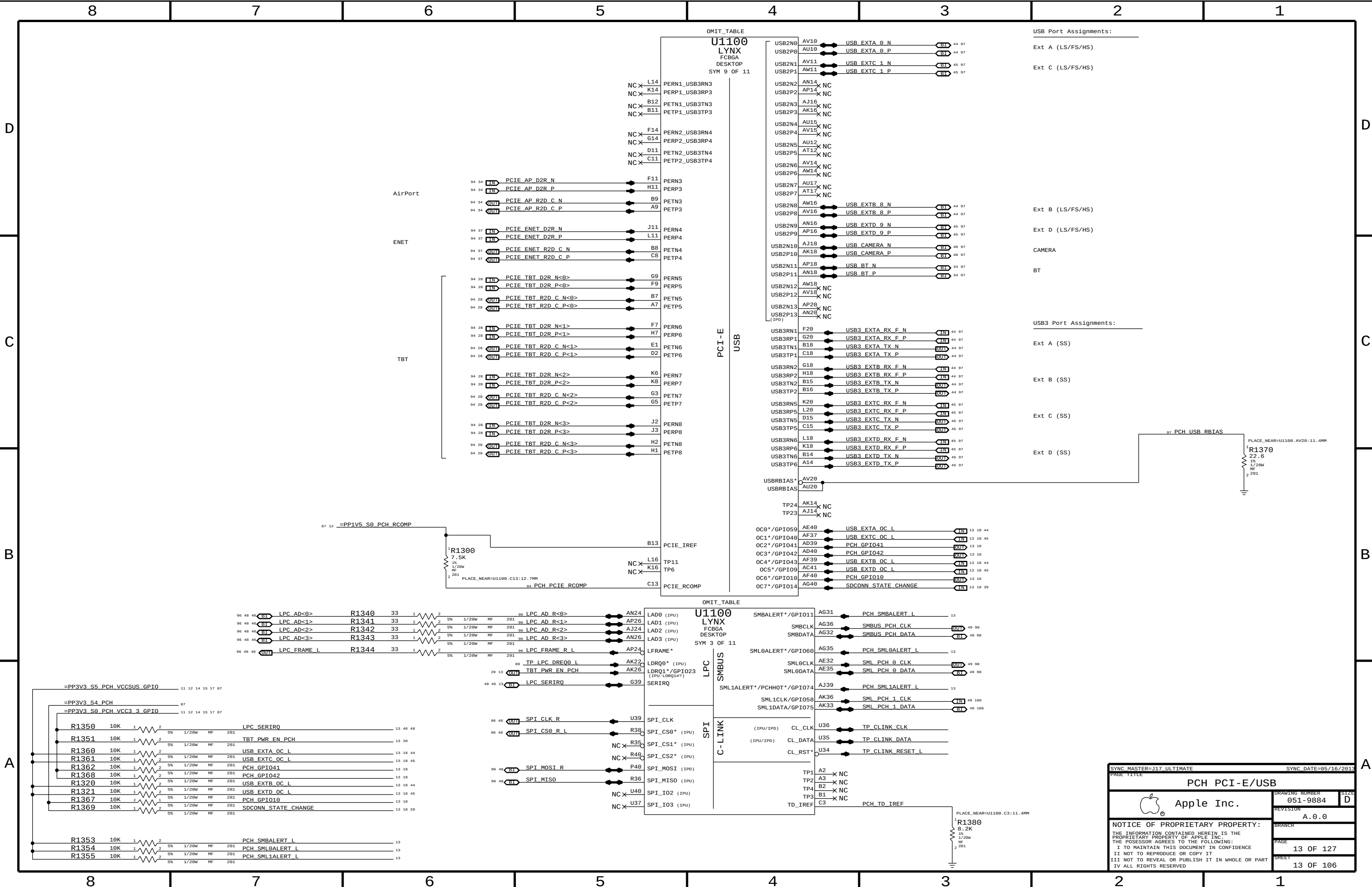
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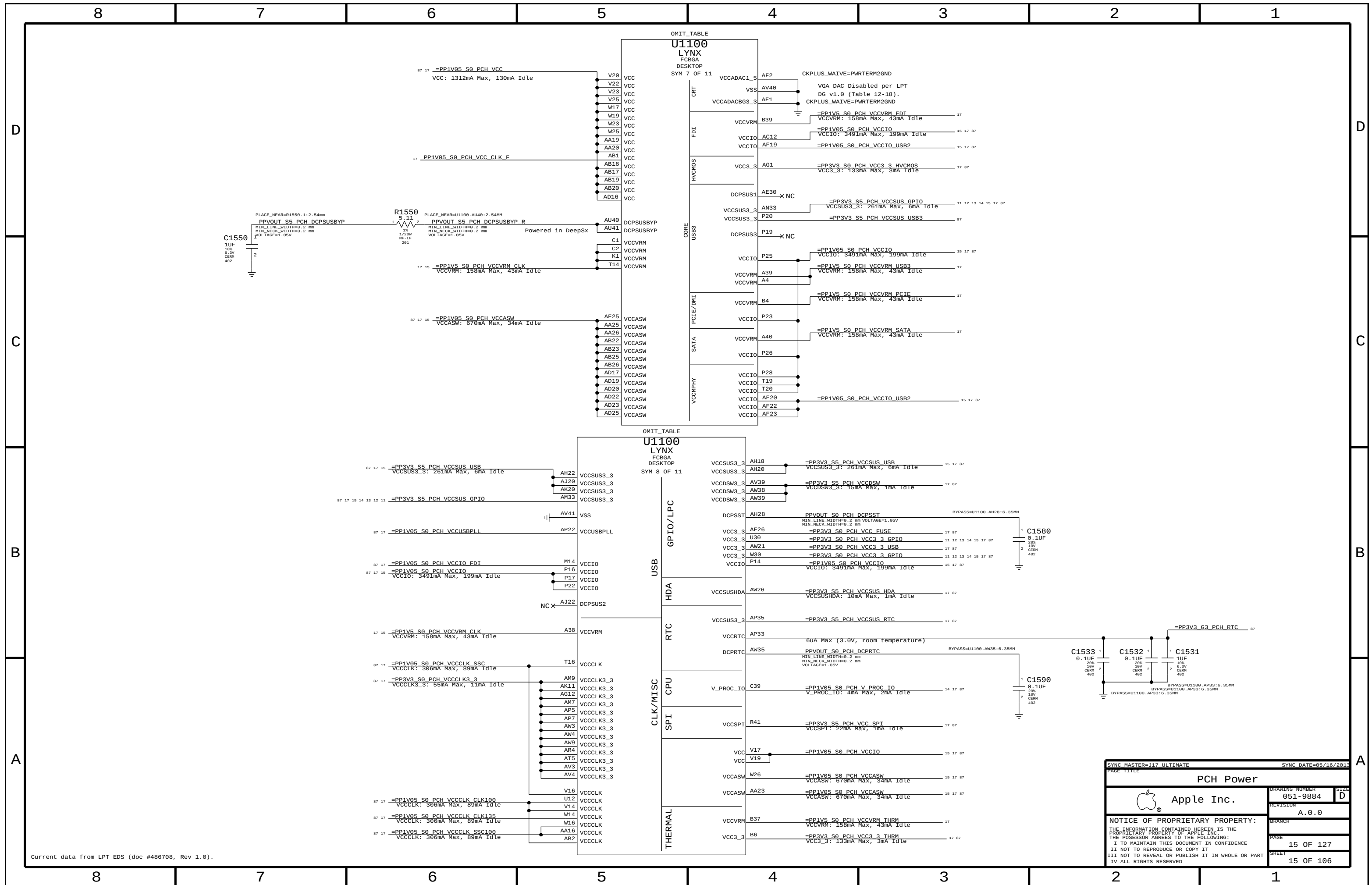
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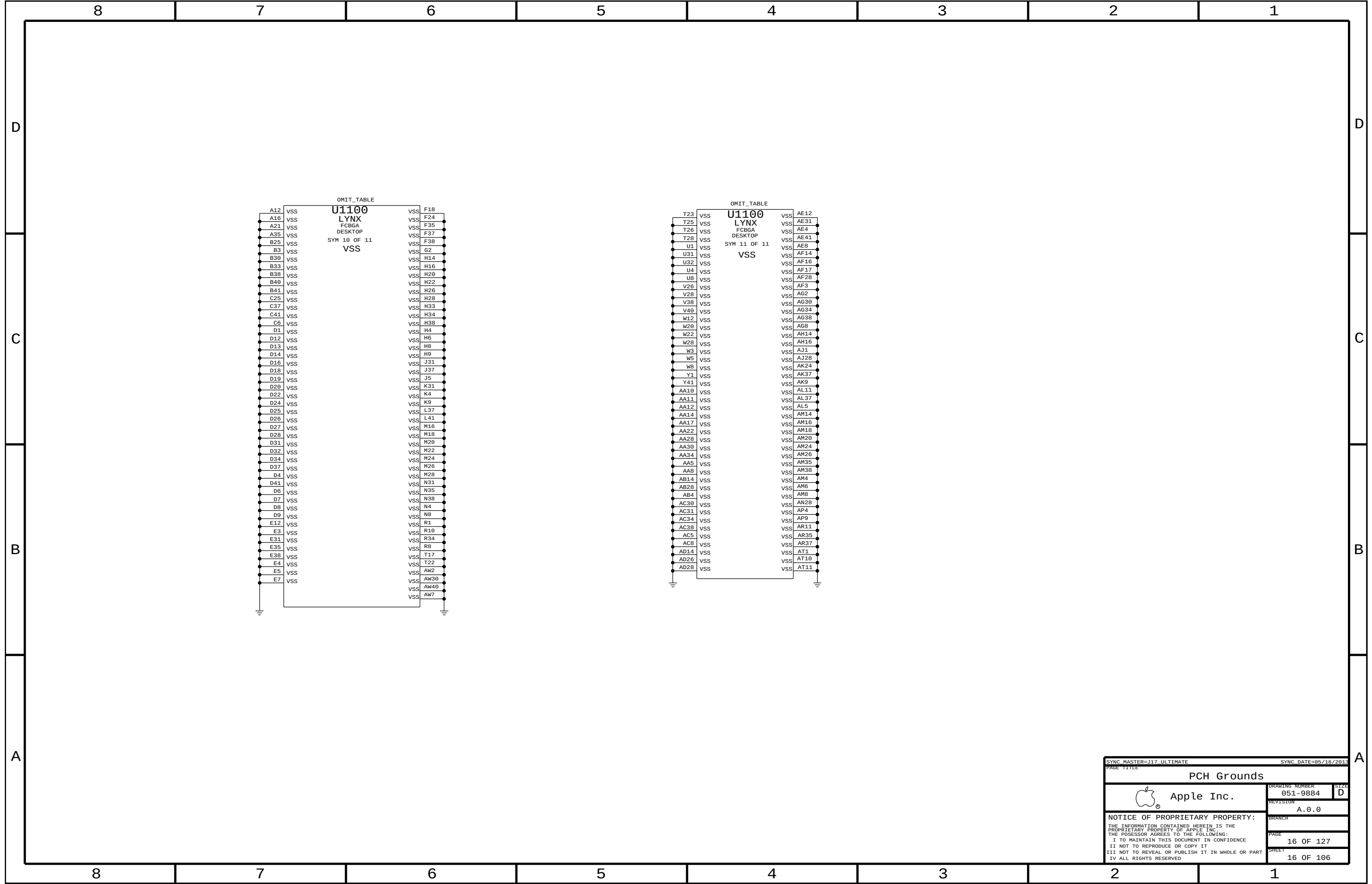
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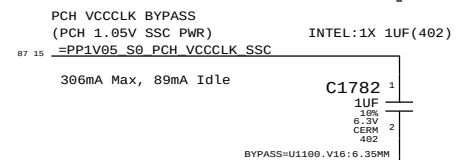
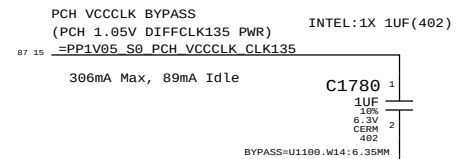
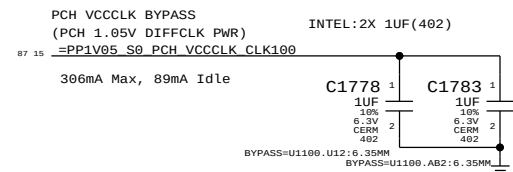
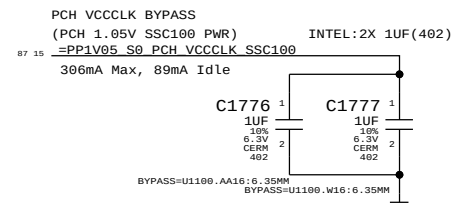
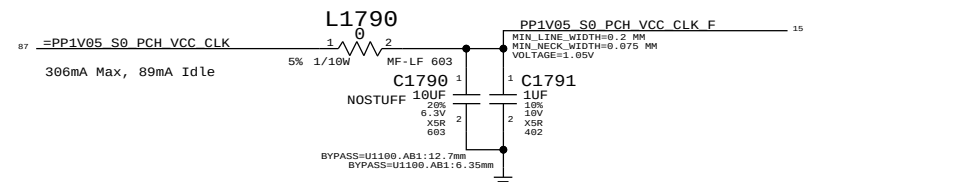
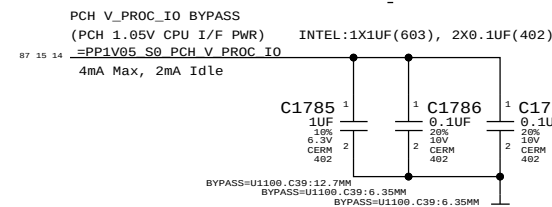
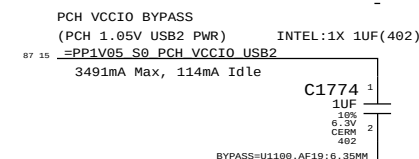
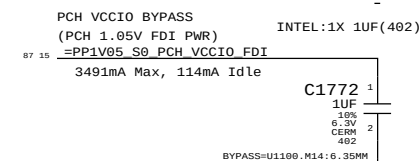
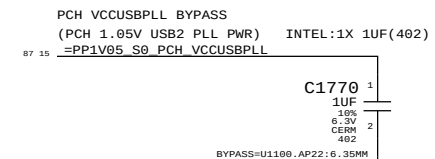
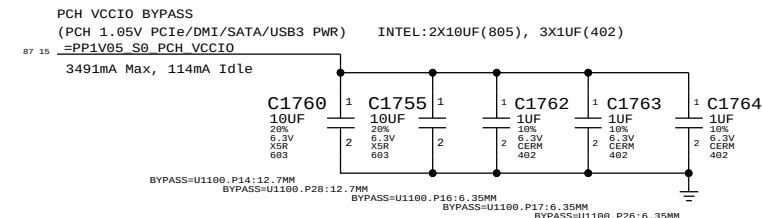
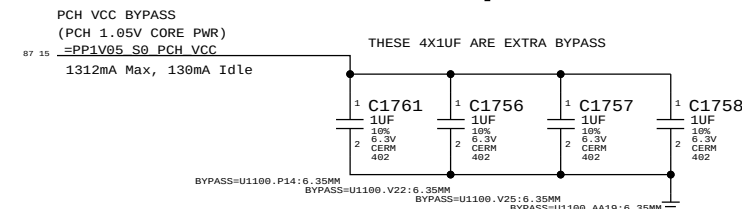
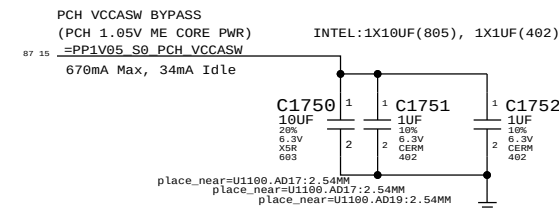
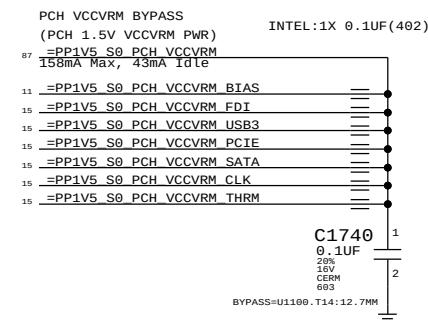
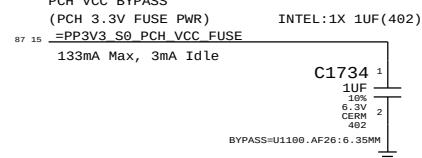
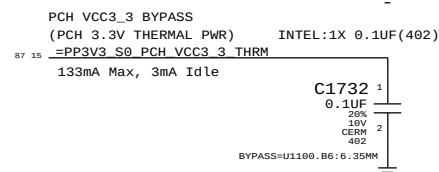
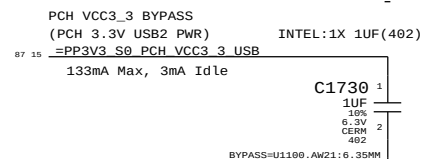
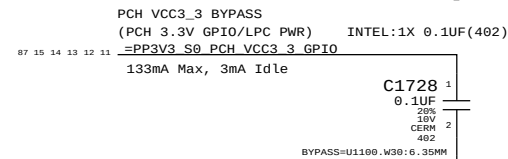
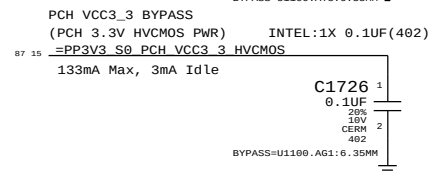
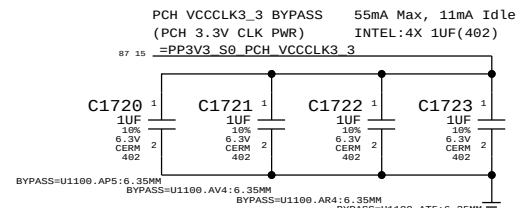
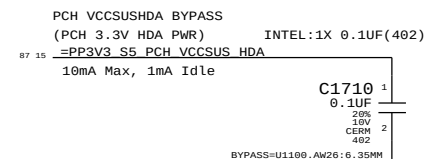
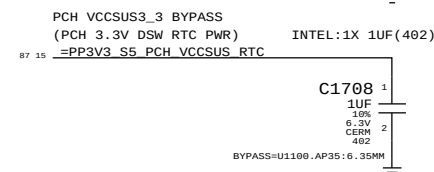
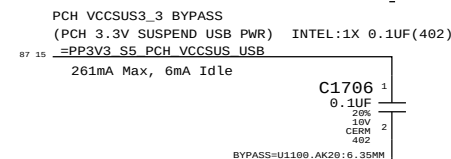
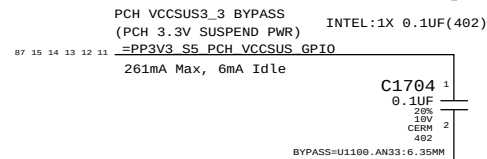
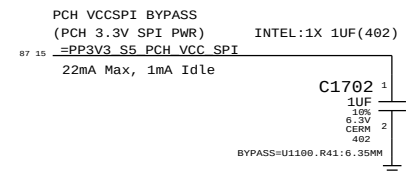
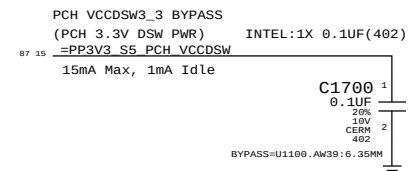
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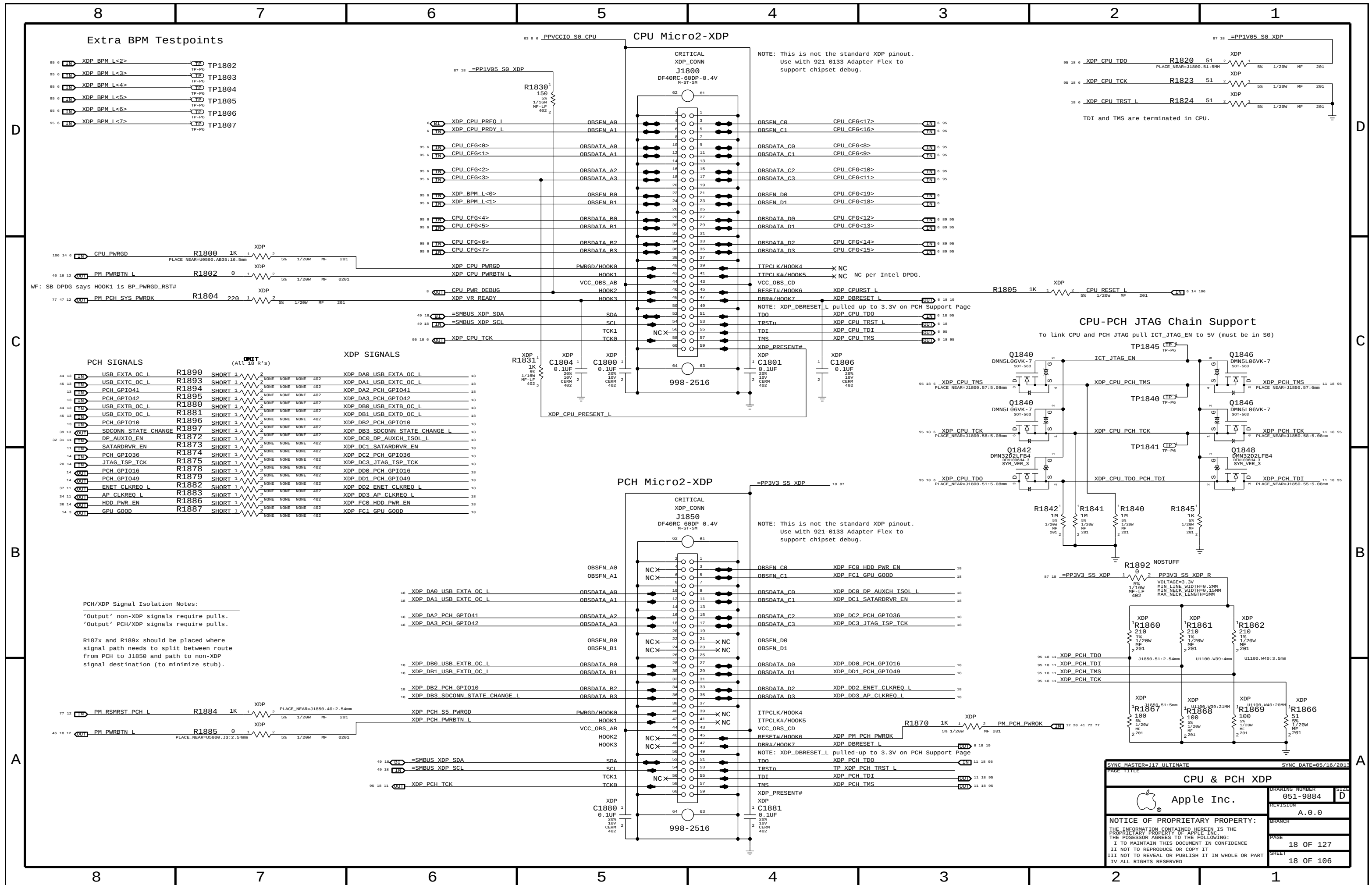




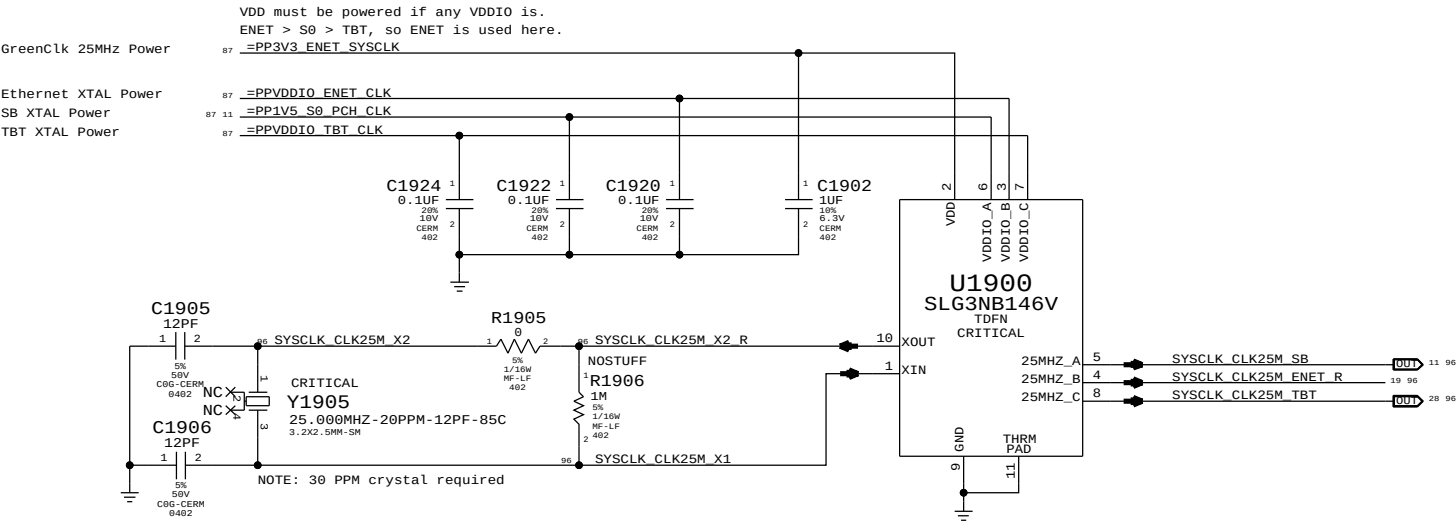




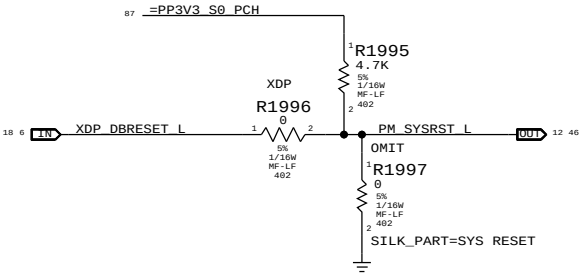




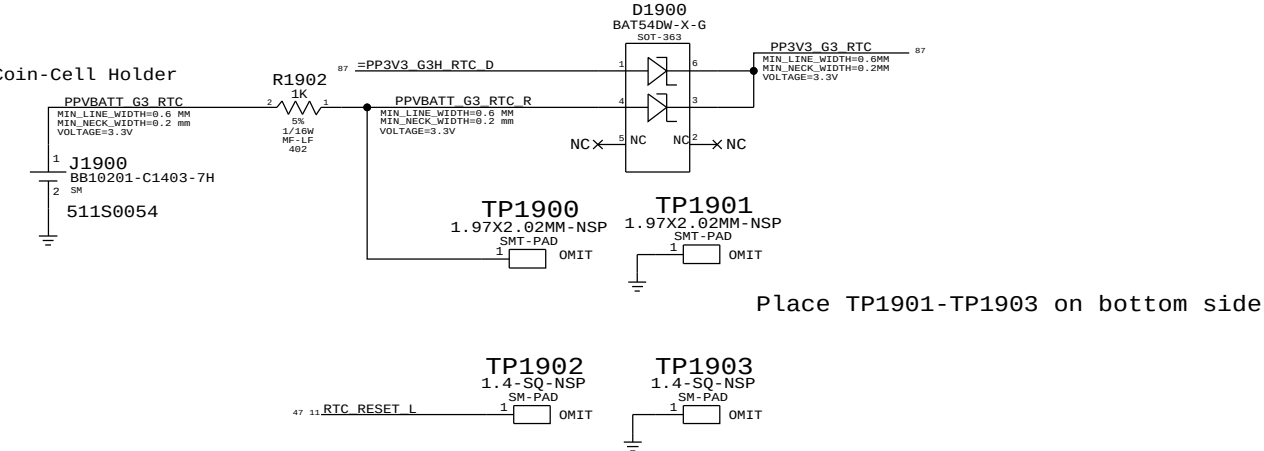
System 25MHz Clock Generator



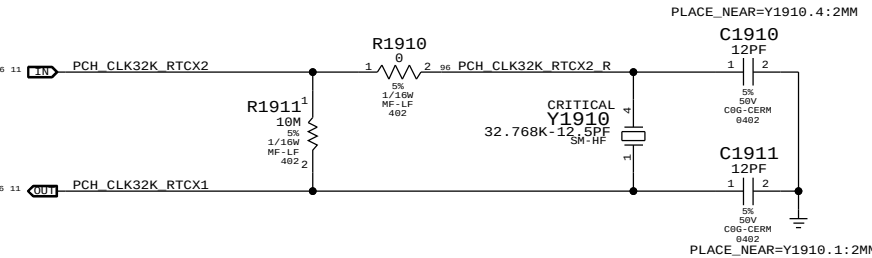
PCH Reset Button



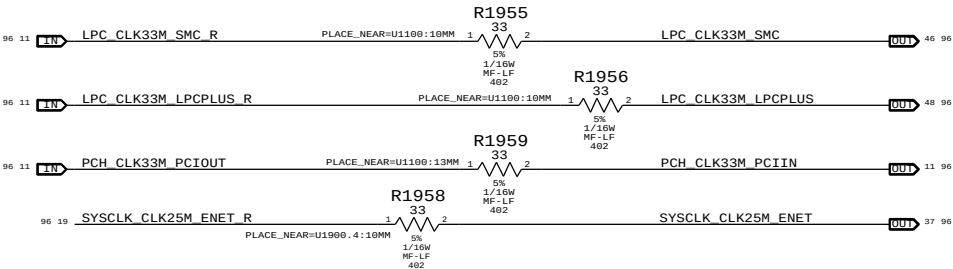
RTC Power Sources



PCH RTC Crystal

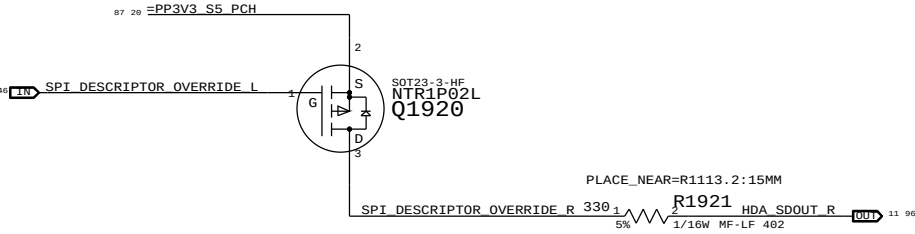


Clock series termination

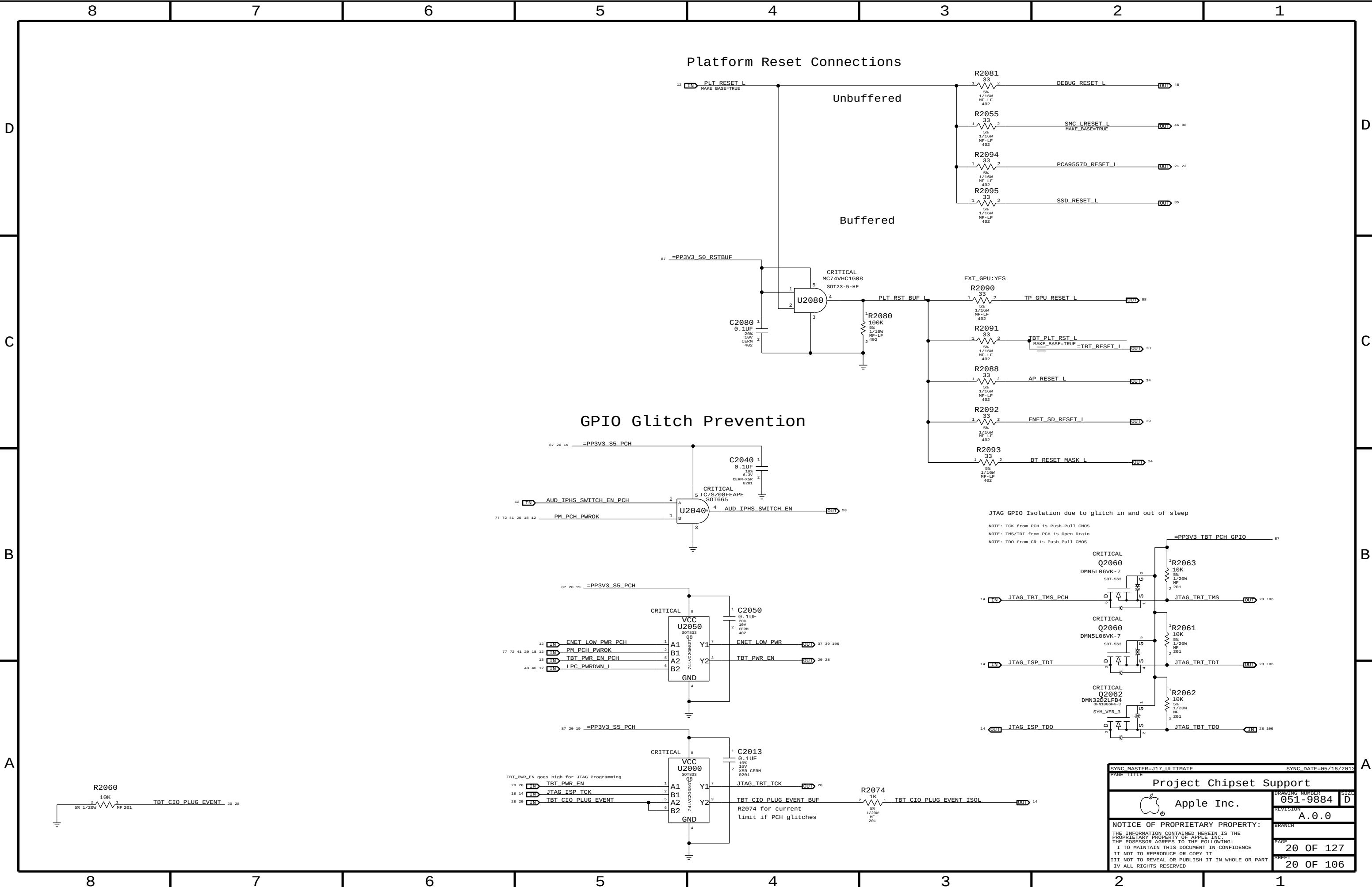


PCH ME Disable Strap

PCH uses HDA_SD0 as a power-up strap. If low, ME functions normally. If high, ME is disabled. This allows for full re-flashing of SPI ROM. SMC controls strap enable to allow in-field control of strap setting.



SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/16/2013	
PAGE TITLE		Chipset Support	
Apple Inc.		DRAWING NUMBER	051-9884
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Project Chipset Support

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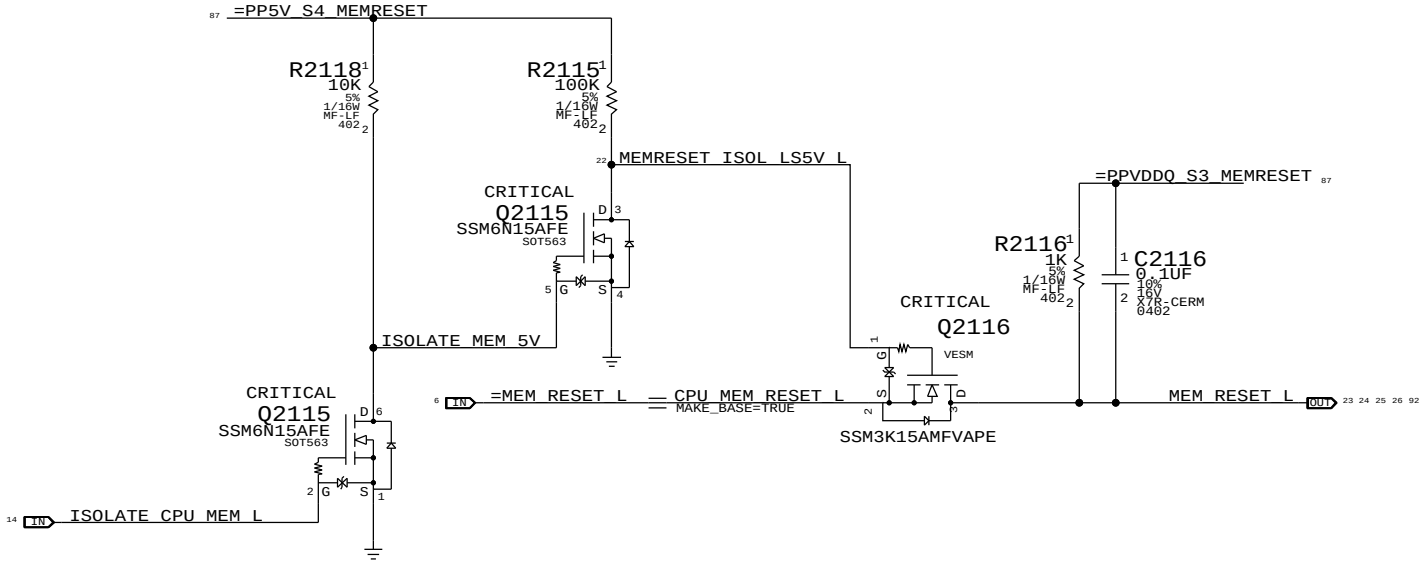
The circuit below handles CPU and VTT power during S0->S3->S0 transitions, as well as isolating the CPU's SM_DRAMRST# output from the S0-DIMMs when necessary.

ISOLATE_CPU_MEM_L GPIO state during S3->S0 transitions determines behavior of signals.

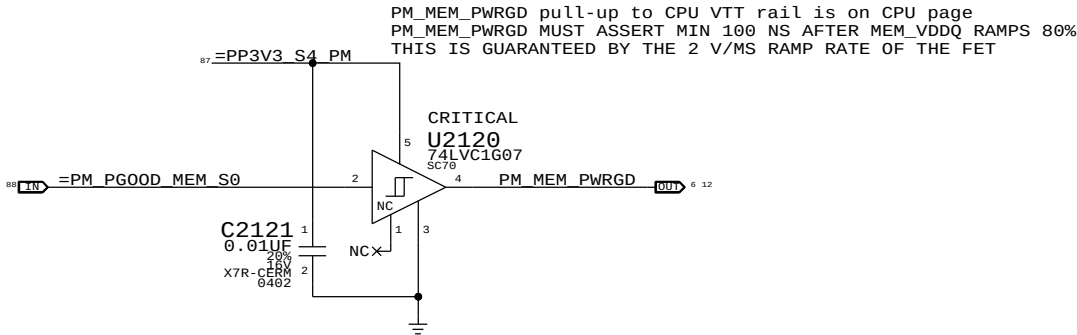
WHEN HIGH: CPU 1.5V remains powered in S3, VTT follows S0 rails, MEM_RESET_L not isolated.

WHEN LOW: CPU 1.5V follows S0 rails, VTT ensures clean CKE transition, MEM_RESET_L isolated.

MEMVTT_EN = PLT_RESET_L * PM_SLP_S3_L
MEM_RESET_L = !ISOLATE_CPU_MEM_L + CPU_MEM_RESET_L

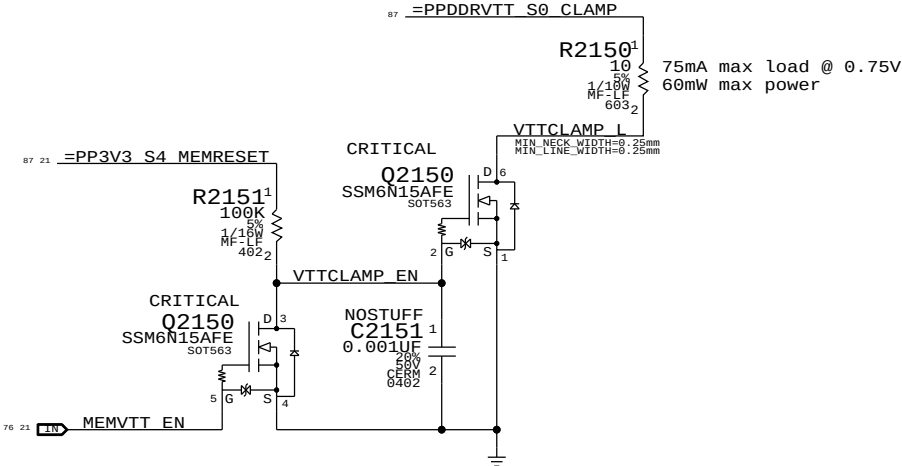
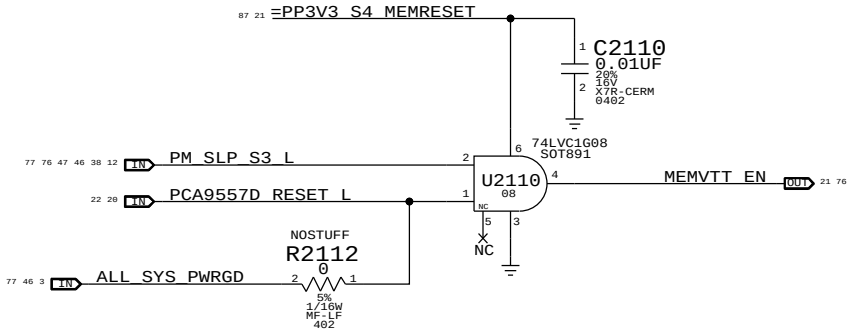


MEM S0 "PGOOD" FOR CPU



MEMVTT Clamp

Ensures CKE signals are held low in S3



Step	ISOLATE_CPU_MEM_L	PLT_RESET_L	PM_SLP_S3_L	CPU_MEM_RESET_L	MEM_RESET_L	MEMVTT_EN
S0	0	1	1	1	CPU_MEM_RESET_L	1
1	1	0	1	1	1	1
2	0	0	1	1	1	0
3	0	0	0	X	1	0
4	0	0	1	X	1	0
5	0	1	1	0 (*)	1	1
6	0	1	1	1	1	1
S0	7	1	1	1	CPU_MEM_RESET_L	1

(*) CPU_MEM_RESET_L asserts due to loss of PM_MEM_PWRGD, must wait for software to clear before deasserting ISOLATE_CPU_MEM_L GPIO.

NOTE: In the event of a S3->S5 transition ISOLATE_CPU_MEM_L will still be asserted on next S5->S0 transition. Rails will power-up as if from S3, but MEM_RESET_L will not properly assert. Software must de-assert ISOLATE_CPU_MEM_L and then generate a valid reset cycle on CPU_MEM_RESET_L.

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SYNC DATE=05/16/2013

CPU Memory S3 Support

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Page Notes

Power aliases required by this page:

- =PP3V3_S3_VREFMRGN
- =PPDDR_S3_MEMVREF

Signal aliases required by this page:

- =I2C_VREFDACS_SCL
- =I2C_VREFDACS_SDA
- =I2C_PCA9557D_SCL
- =I2C_PCA9557D_SDA

BOM options provided by this page:

- DDRVREF_DAC - Stuffs DAC margining circuit.

CPU-Based Margining

FETs for CPU isolation during S3

NOTE: CPU DAC output step sizes:
DDR3 (1.5V) 7.70mV per step
DDR3L (1.35V) 6.99mV per step

NOTE: CPU has single output for
VREFCA. Split into two
signals for independent DAC
margining support. When
DAC margining VREFCA ensure
ISOLATE_CPU_MEM_L is low
to remove short due to CPU.

DAC-Based Margining

DAC sets voltage level, PCA9557 & FETs enable outputs
and disables margining after platform reset.

NOTE: MEMVREG and FRAMEBUF share
a DAC output, cannot enable
both at the same time!

RST* on 'platform reset' so that system
watchdog will disable margining.

NOTE: Margining will be disabled across all
soft-resets and sleep/wake cycles.

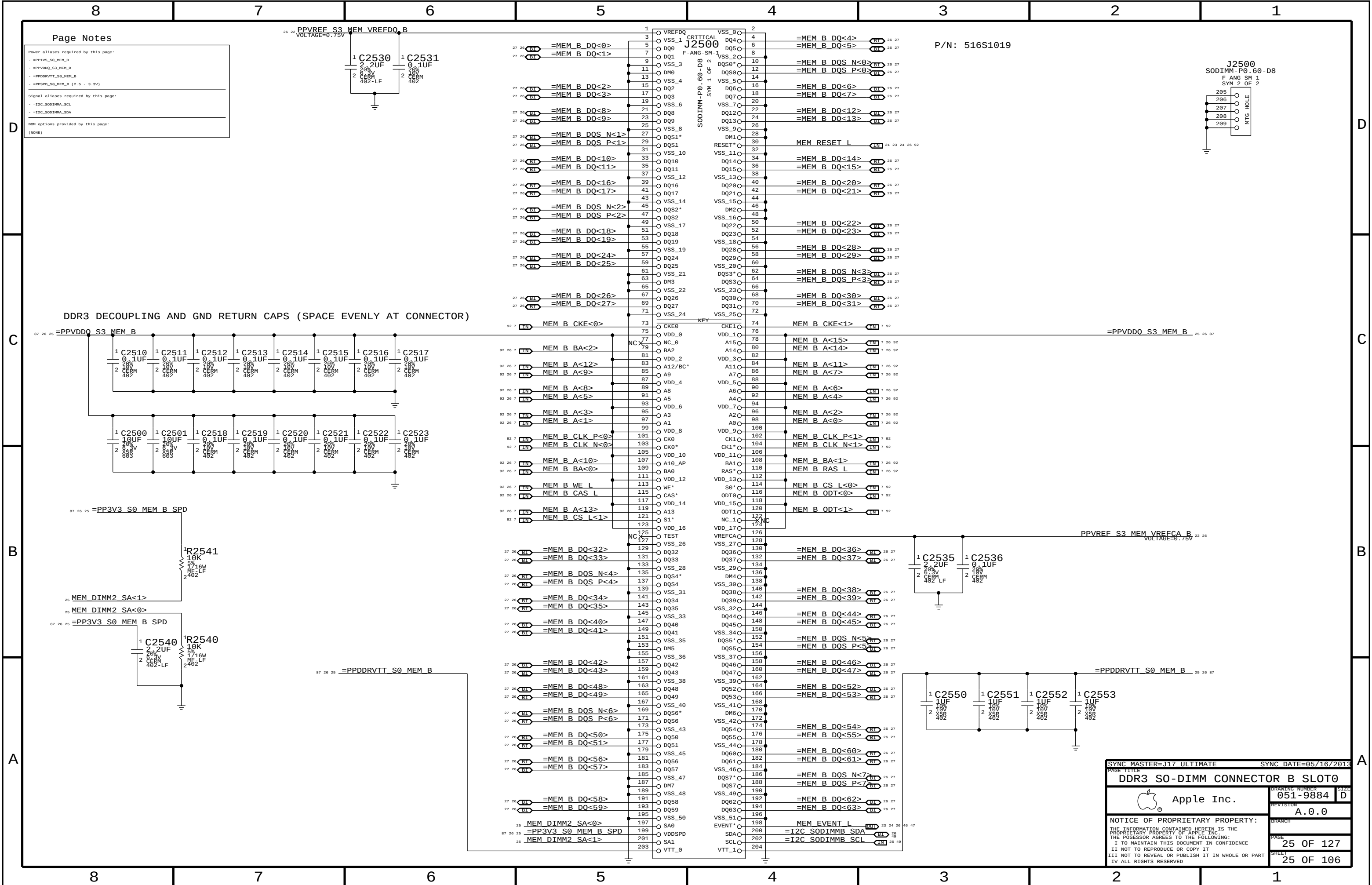
VRef Dividers

Always used, regardless
of margining option.

	MEM A VREF DQ	MEM B VREF DQ	MEM A VREF CA	MEM B VREF CA	MEM VREG
DAC Channel:	A	B	C	C	D
PCA9557D Pin:	1	2	3	4	5
	DDR3 (1.5V)		DDR3L (1.35V)		
Nominal value	0.750V (DAC: 0x3A = 0.747mV)		0.675V (DAC: 0x34 = 0.670mV)		1.500V (DAC: 0x74 = 1.495V)
Margined target:	0.300V - 1.200V (+/- 450mV)		0.275V - 1.075V (+/- 400mV)		1.200V - 1.800V (+/- 300mV)
DAC range:	0.000V - 1.508V (0x00 - 0x75)		0.000V - 1.354V (0x00 - 0x69)		0.000V - 3.004V (0x00 - 0xE9)
Margined range:	0.299V - 1.206V (+/- 453mV)		0.269V - 1.083V (+/- 406mV)		1.199V - 1.801V (+/- 301mV)
VRef current:	+901uA - -911uA (- = sourced)		+811uA - -816uA (- = sourced)		+36uA - -36uA (- = sourced)
DAC step size:	7.68mV / step @ output		7.67mV / step @ output		2.575mV / step @ output

NOTE: DDR3 assumes TPS51916 supply with 10.0k/49.9k divider
DDR3L assumes TPS51916 supply with 19.6k/57.6k divider

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Page Notes

Power aliases required by this page:

- =PP1V5_S0_MEM_B
- =PPVDDQ_S3_MEM_B
- =PPDDRVT_S0_MEM_B
- =PPSPD_S0_MEM_B (2.5 - 3.3V)

Signal aliases required by this page:

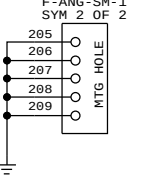
- =I2C_S0DIMM_SCL
- =I2C_S0DIMM_SDA

SOM options provided by this page:

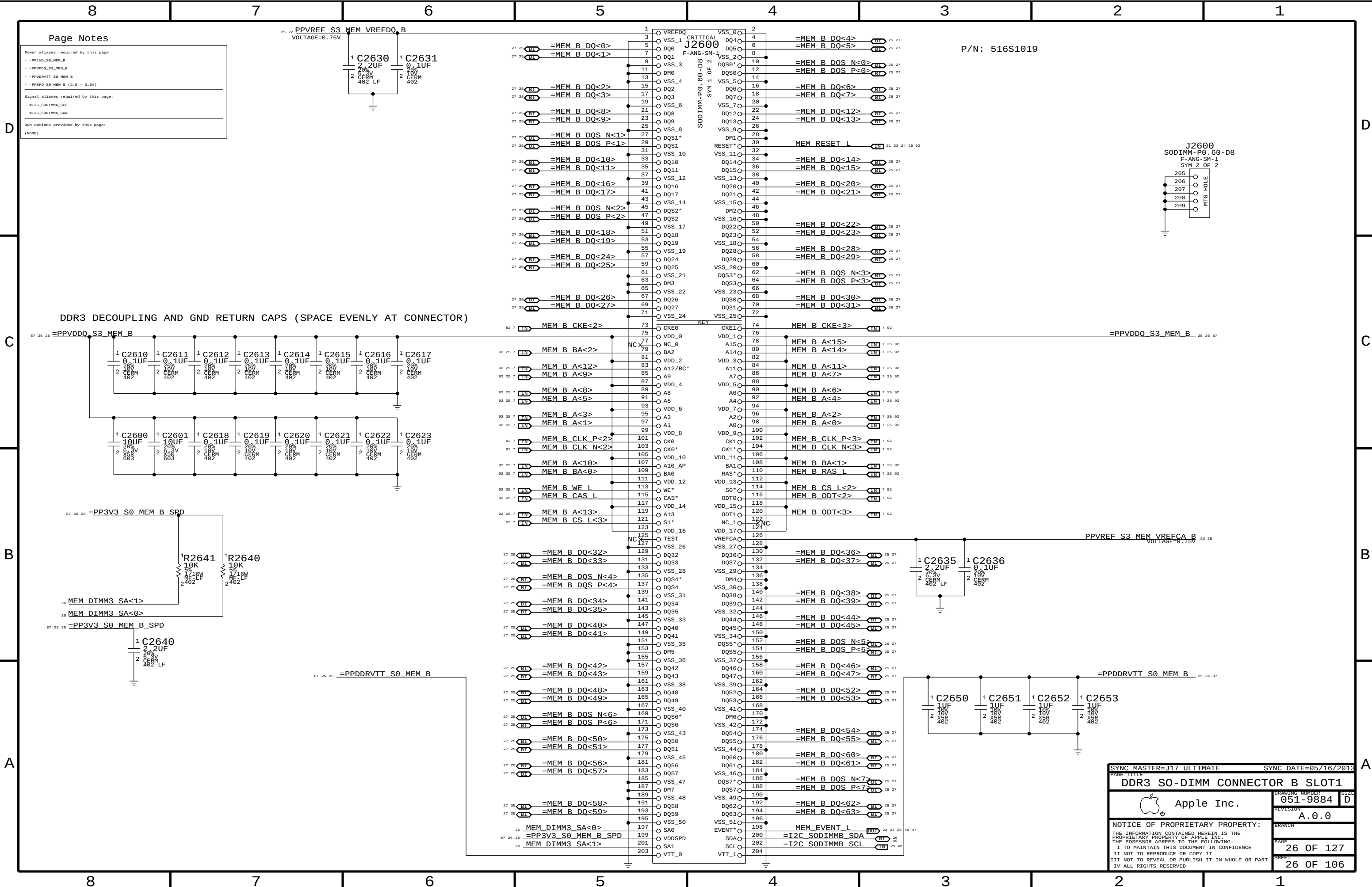
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P/N: 516S1019

J2500
SODIMM-P0.60-D8



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DDR3 SO-DIMM CONNECTOR B SLOT00		051-9884	
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Page Notes

Power aliases required by this page:

- =PP1V5_S0_MEM_B
- =PPVDDQ_S3_MEM_B
- =PPDDRVT_S0_MEM_B
- =PPSPD_S0_MEM_B (2.5 - 3.3V)

Signal aliases required by this page:

- =I2C_S0DIMM_SCL
- =I2C_S0DIMM_SDA

BOM options provided by this page:

(NONE)

P/N: 516S1019

J2600
SODIMM-P0.60-D8
F-ANG-SM-1
SYM 2 OF 2

SYNC MASTER=J17 ULTIMATE SYNC DATE=05/16/2013

DDR3 SO-DIMM CONNECTOR B SLOT1



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Page Notes

Power aliases required by this page:

- =PPVIN_SW_TBTBST (8-13V Boost Input)
- =PP15V_TBT_REG (15V Boost Output)
- =PP3V3_TBT_P3V3TBTFFET (3.3V FET Input)
- =PP3V3_TBT_FET (3.3V FET Output)
- =PP3V3_S0_TBTTPWRCTL
- =PP1V05_TBT_P1V05TBTFFET (1.05V FET Input)
- =PP1V05_TBT_FET (1.05V FET Output)

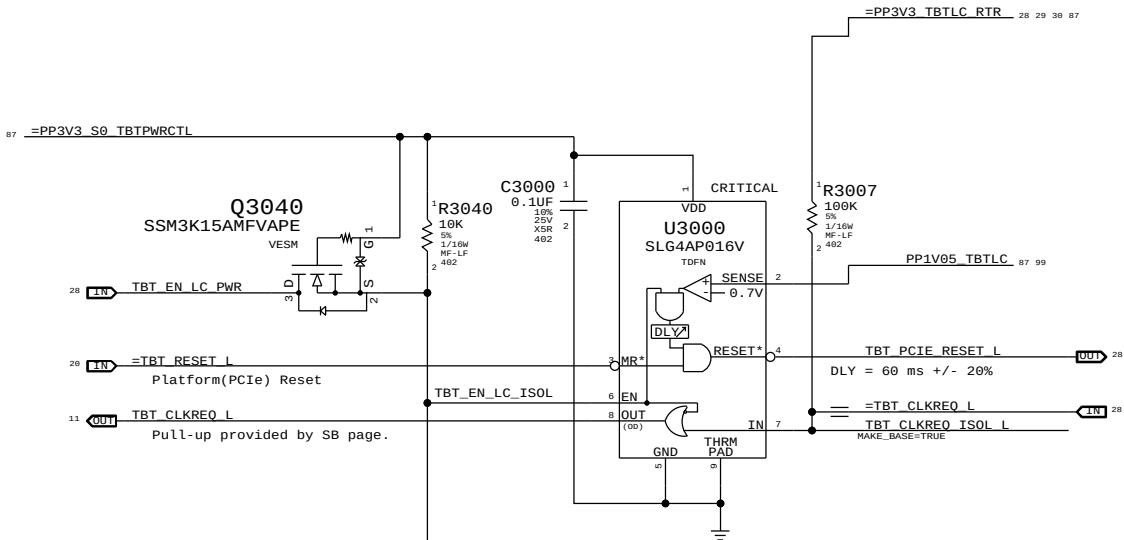
Signal aliases required by this page:

- =TBT_CLKREQ_L
- =TBT_RESET_L

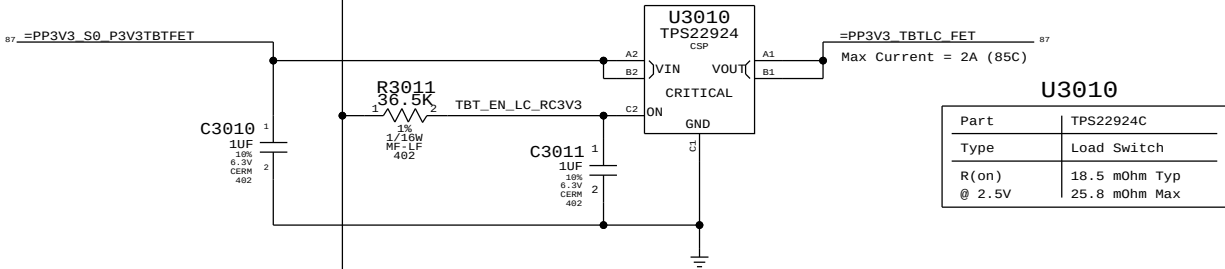
BOM options provided by this page:

TBTBST:Y - Stuffs 15V boost circuitry.

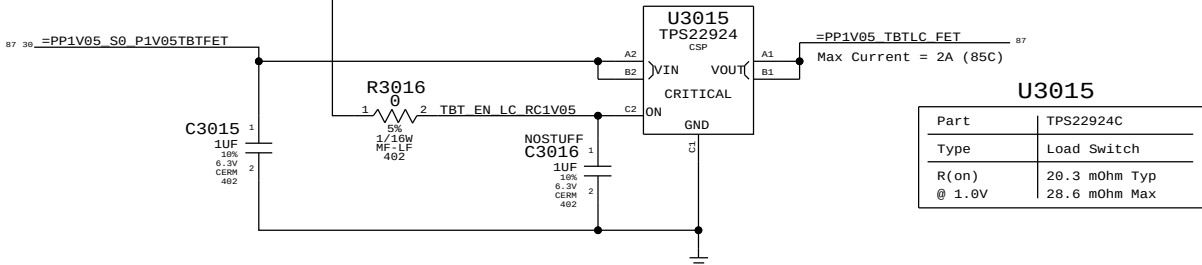
Supervisor & CLKREQ# Isolation



3.3V TBT "LC" Switch

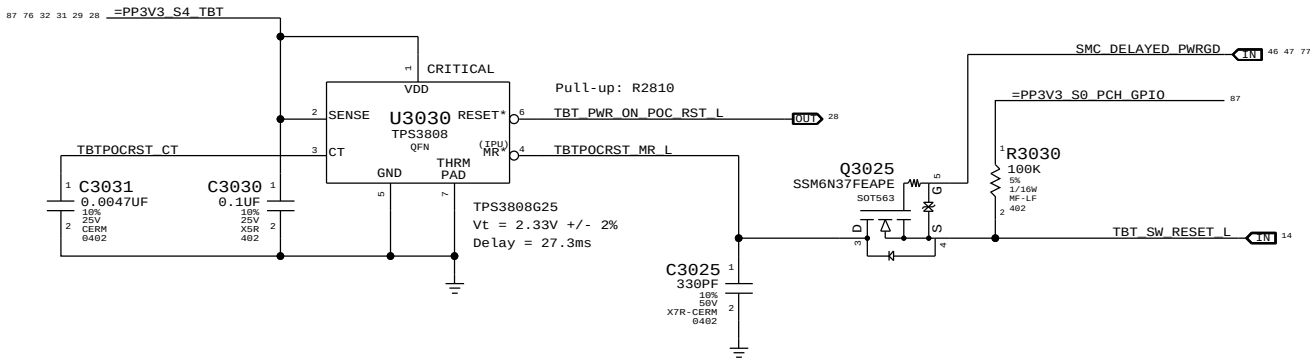


1.05V TBT "LC" Switch

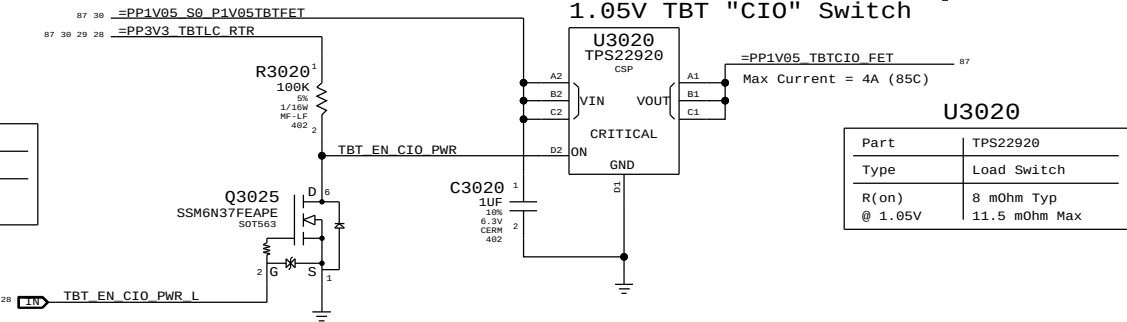


TBT "POC" Power-up Reset

Intel investigating whether RC is sufficient.



1.05V TBT "CIO" Switch



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Thunderbolt Power Support

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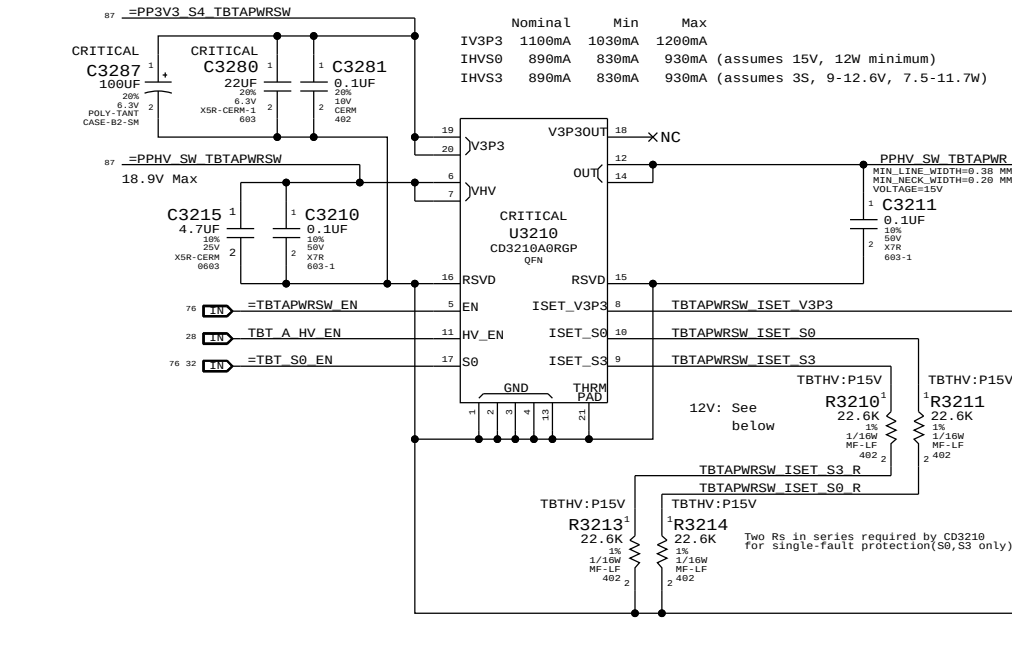
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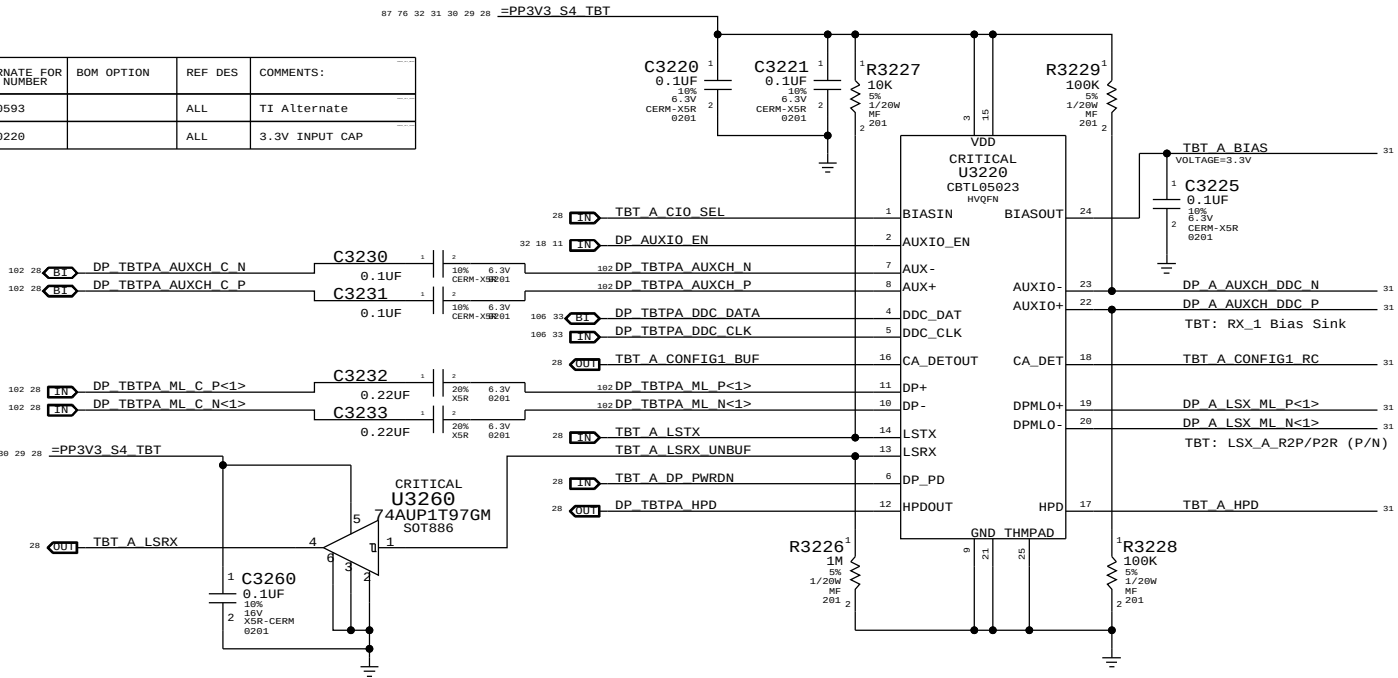
SHEET 30 OF 106

3.3V/HV Power MUX

V3P3 must be S4 to support wake from Thunderbolt devices.



PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
311S0596	311S0593		ALL	T1 Alternate
128S0398	128S0229		ALL	3.3V INPUT CAP

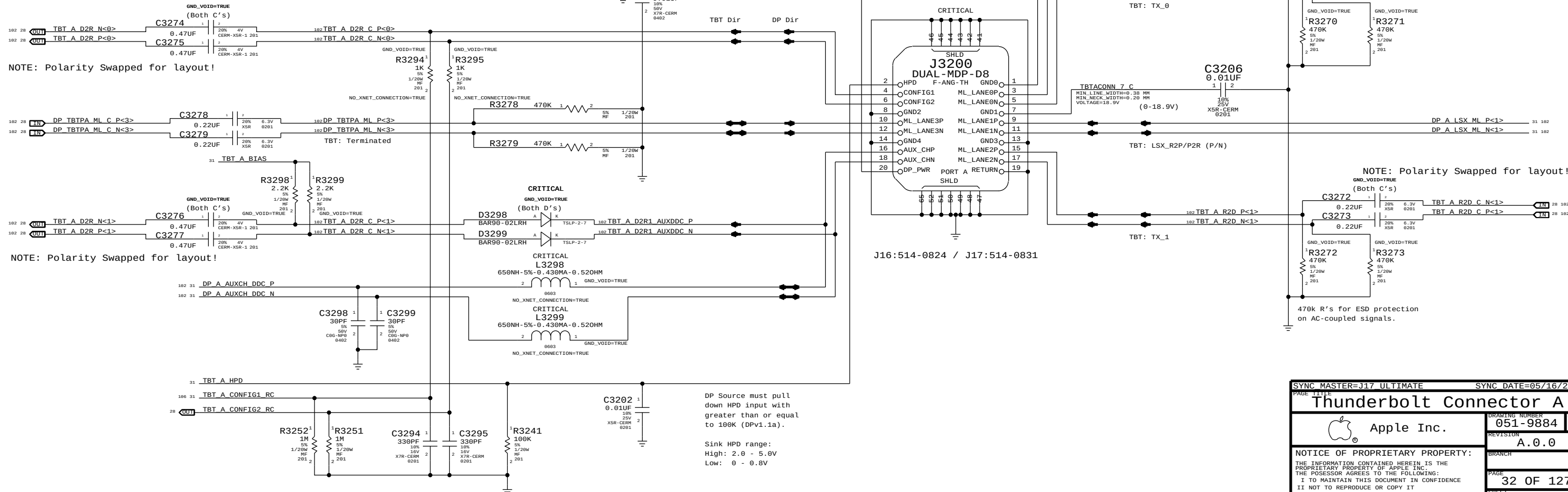


Thunderbolt Connector A

For 12V systems:

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
114S0338	2	RES, MTL FILM, 1/16W, 17.8K, 1, 9402, SMD, LF	R3210, R3213		TBTHV:P12V
114S0338	2	RES, MTL FILM, 1/16W, 17.8K, 1, 9402, SMD, LF	R3211, R3214		TBTHV:P12V

Nominal Min Max
IHVS0/S3 1120mA 1090mA 1170mA (12W minimum)



DP Source must pull down HPD input with greater than or equal to 100K (DPv1.1a).

Sink HPD range:
High: 2.0 - 5.0V
Low: 0 - 0.8V

SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/16/2013	
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8	7	6	5	4	3	2	1
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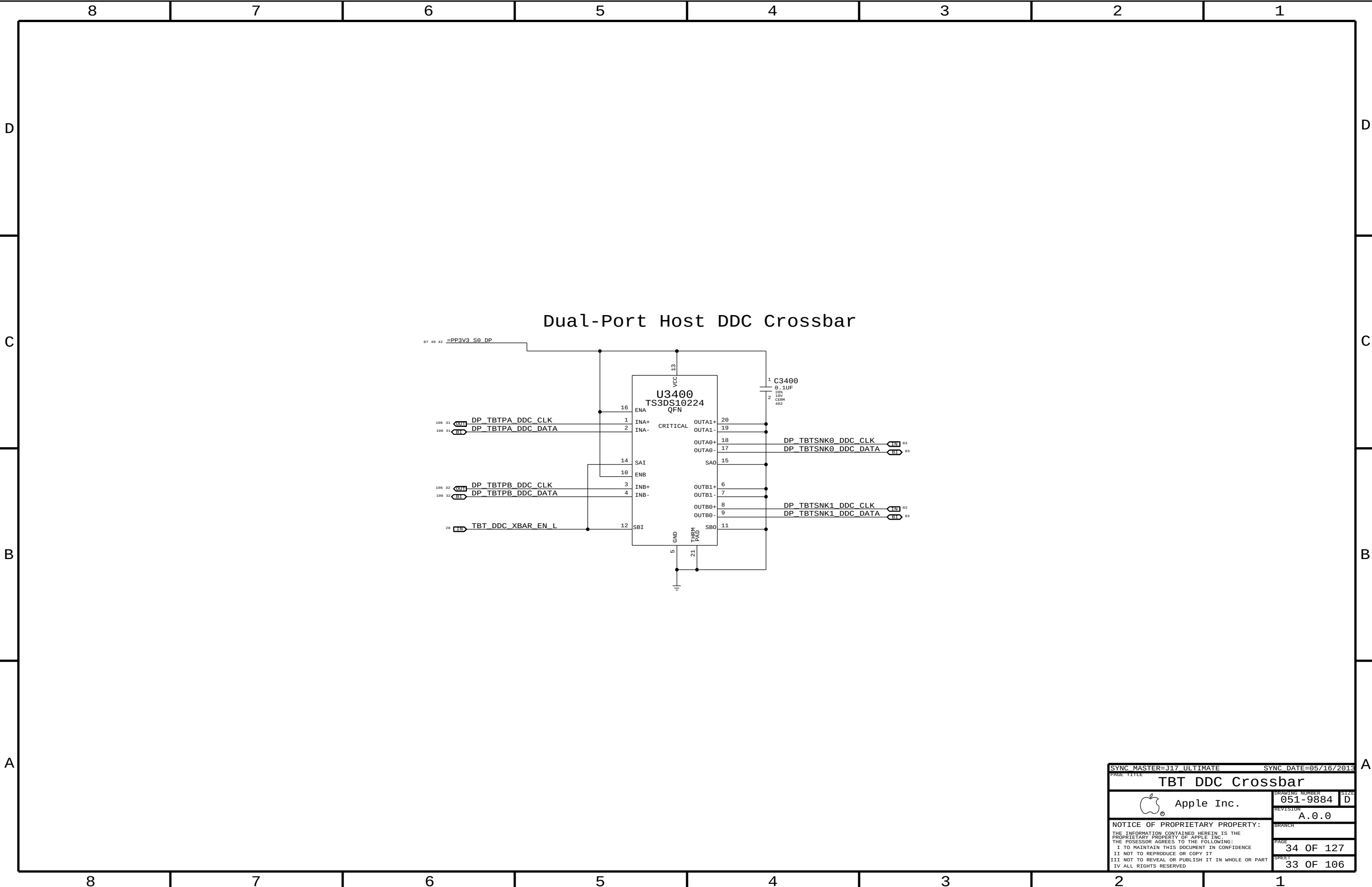


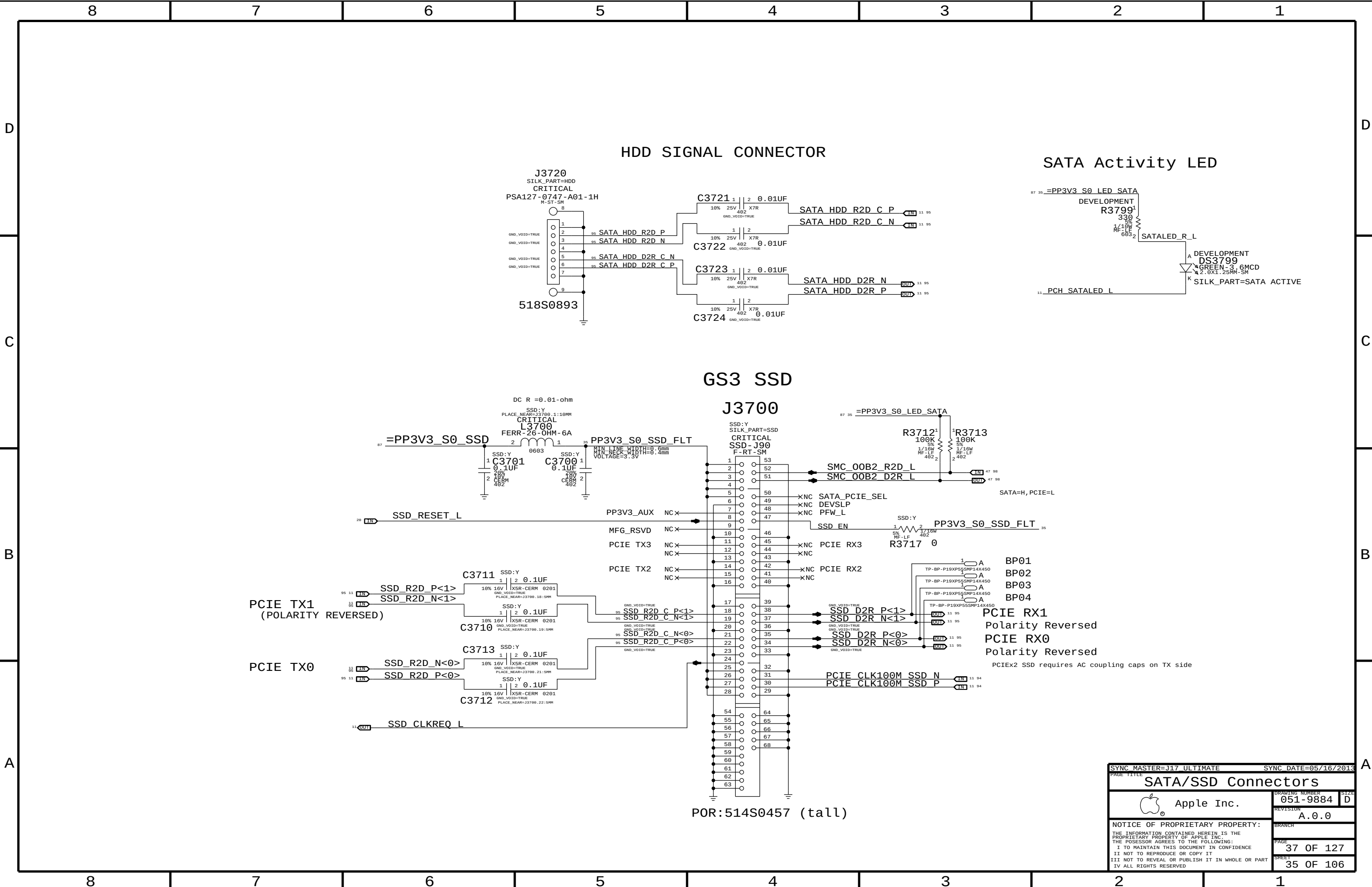
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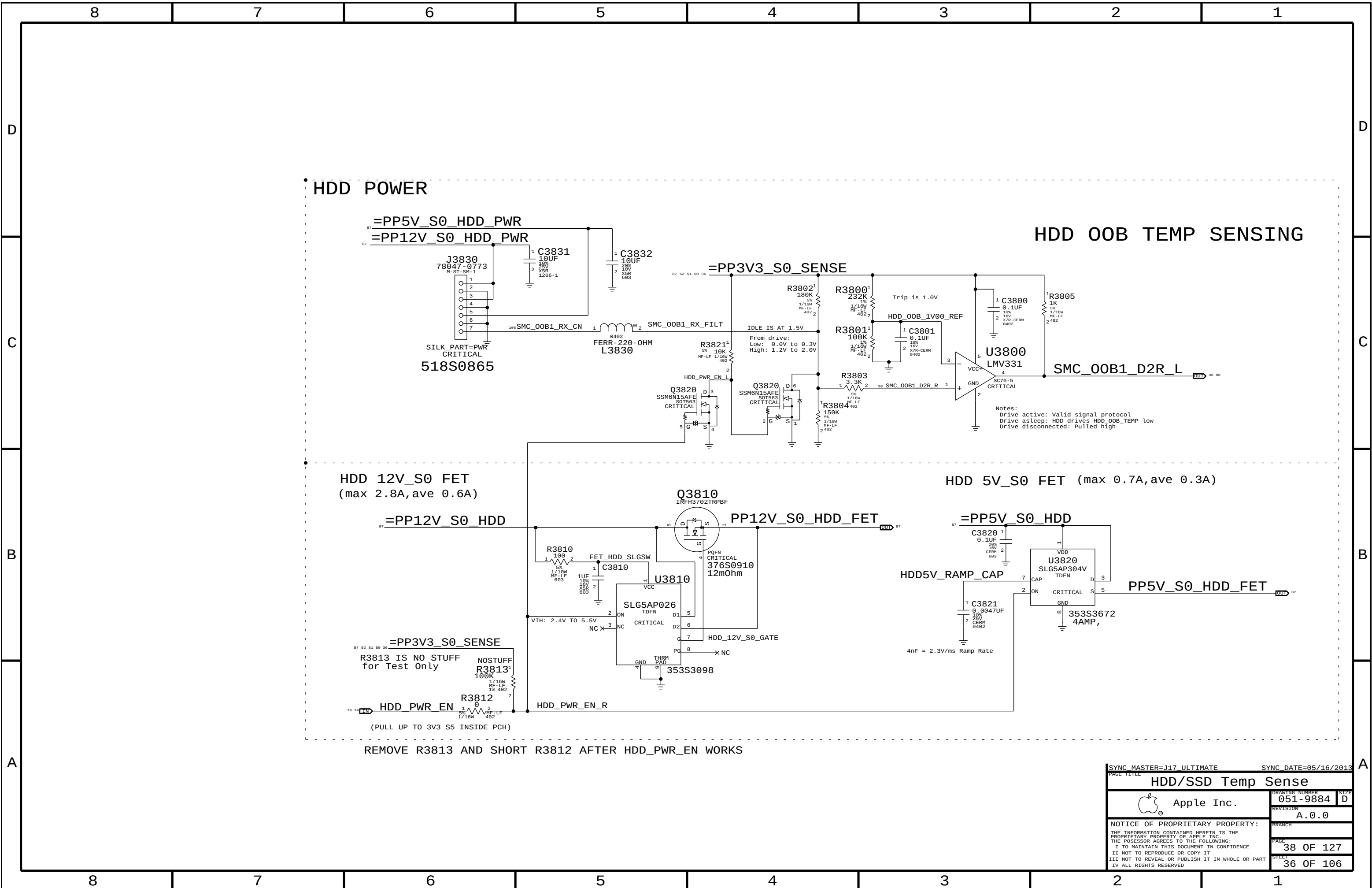
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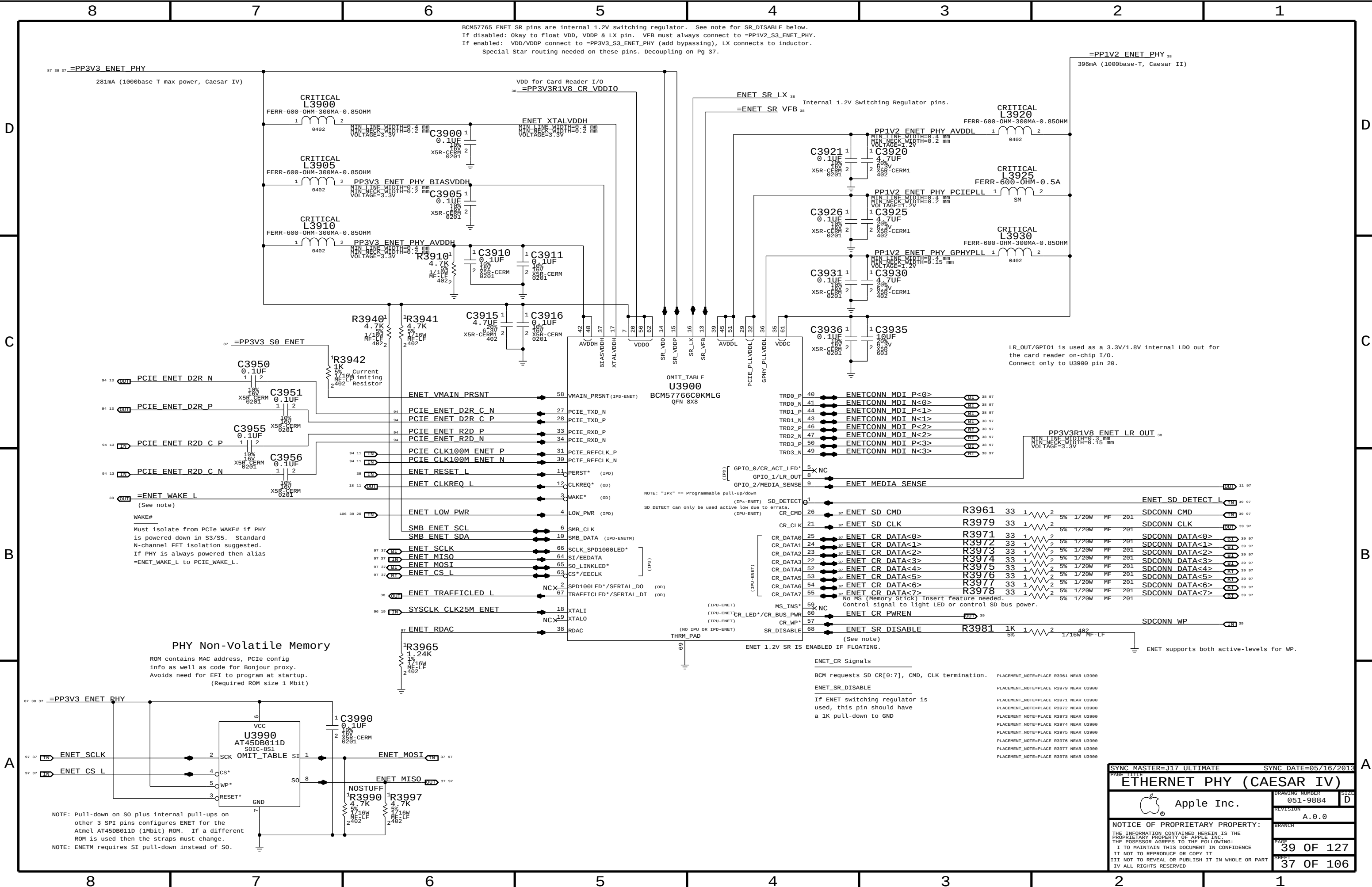
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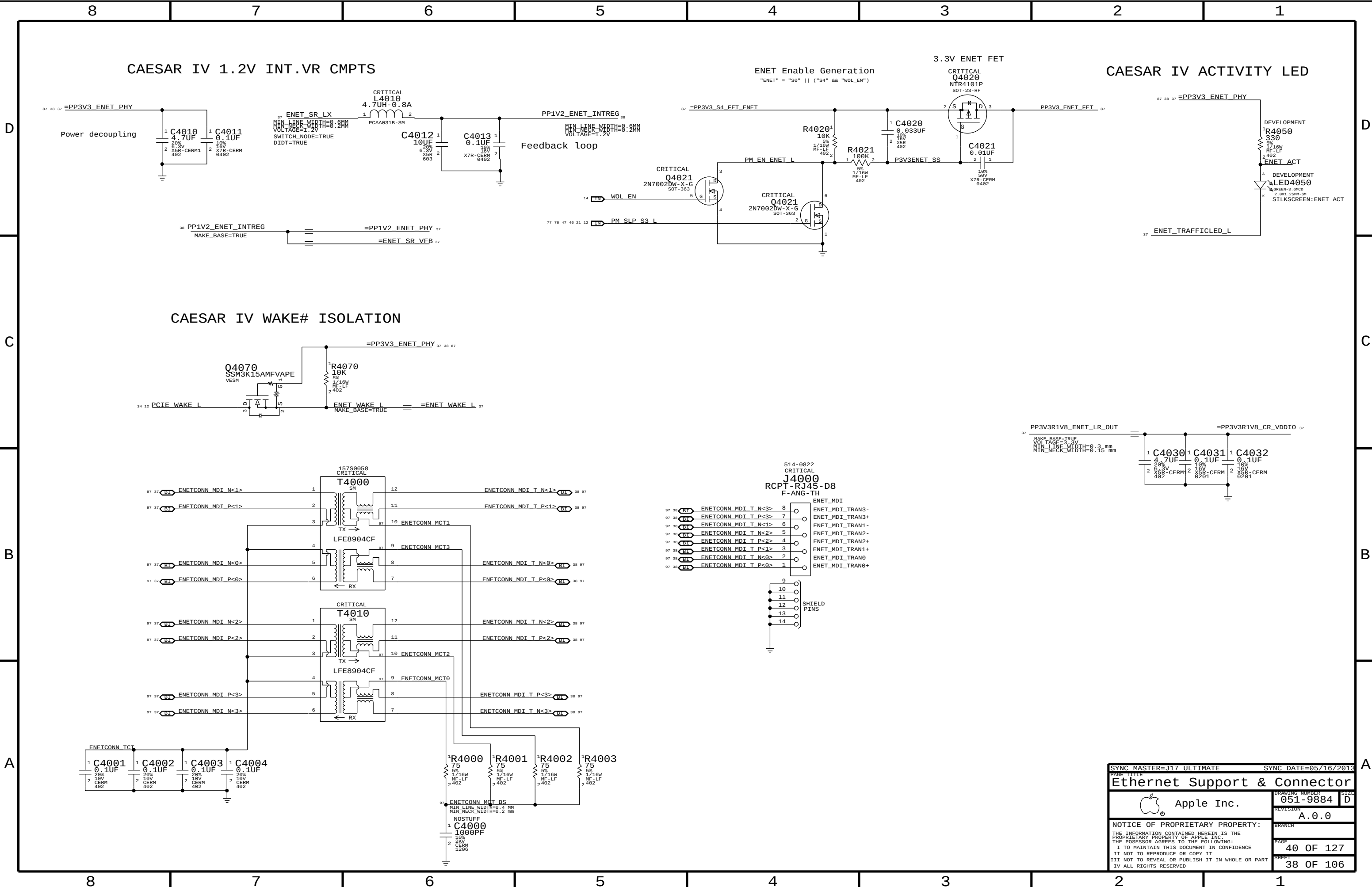


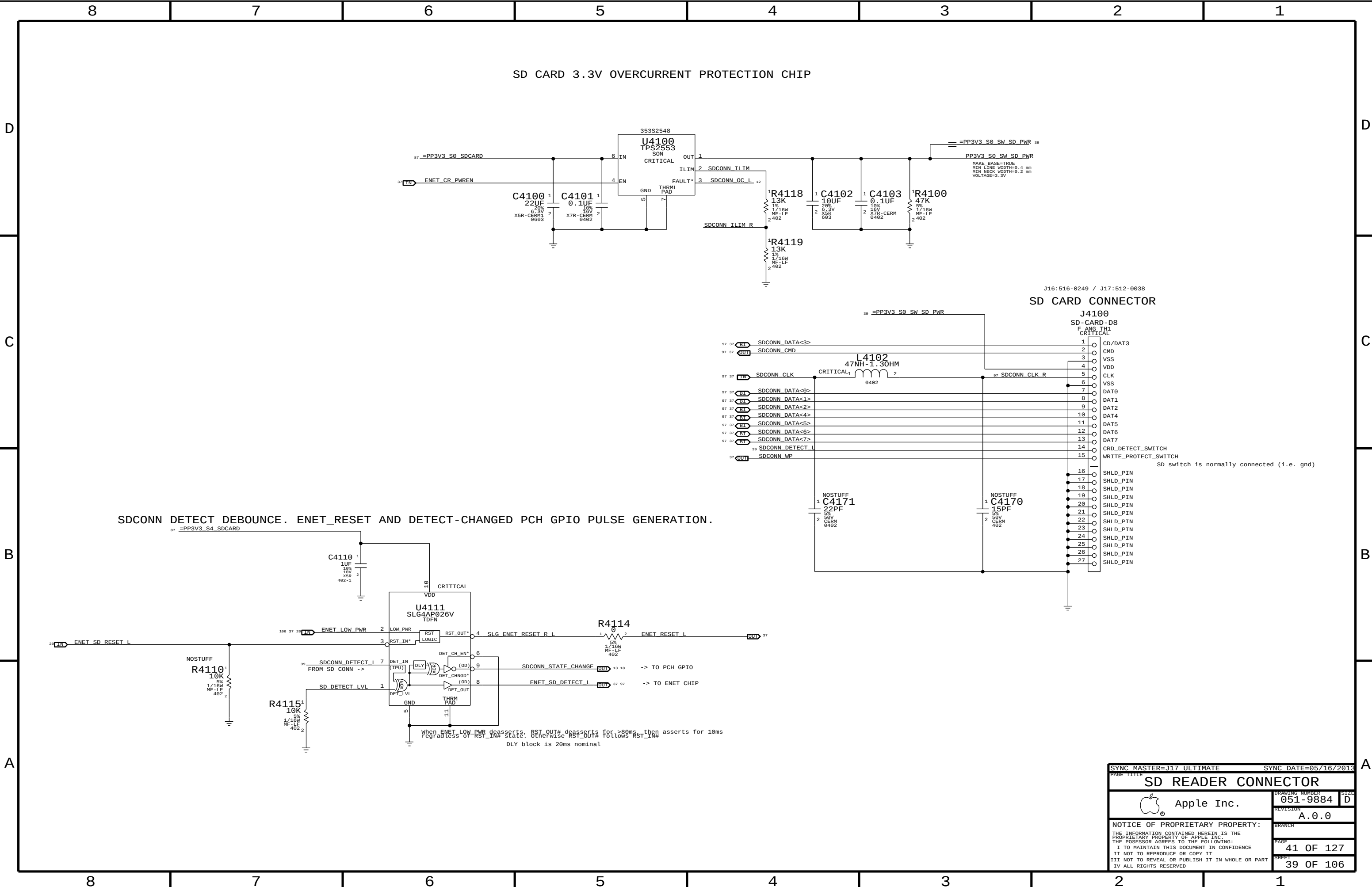


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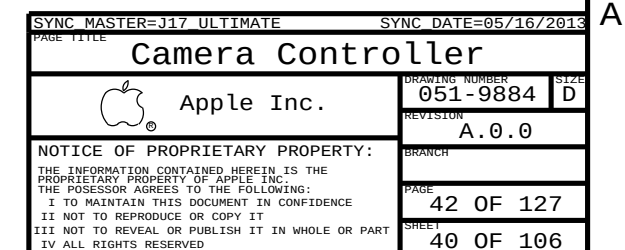


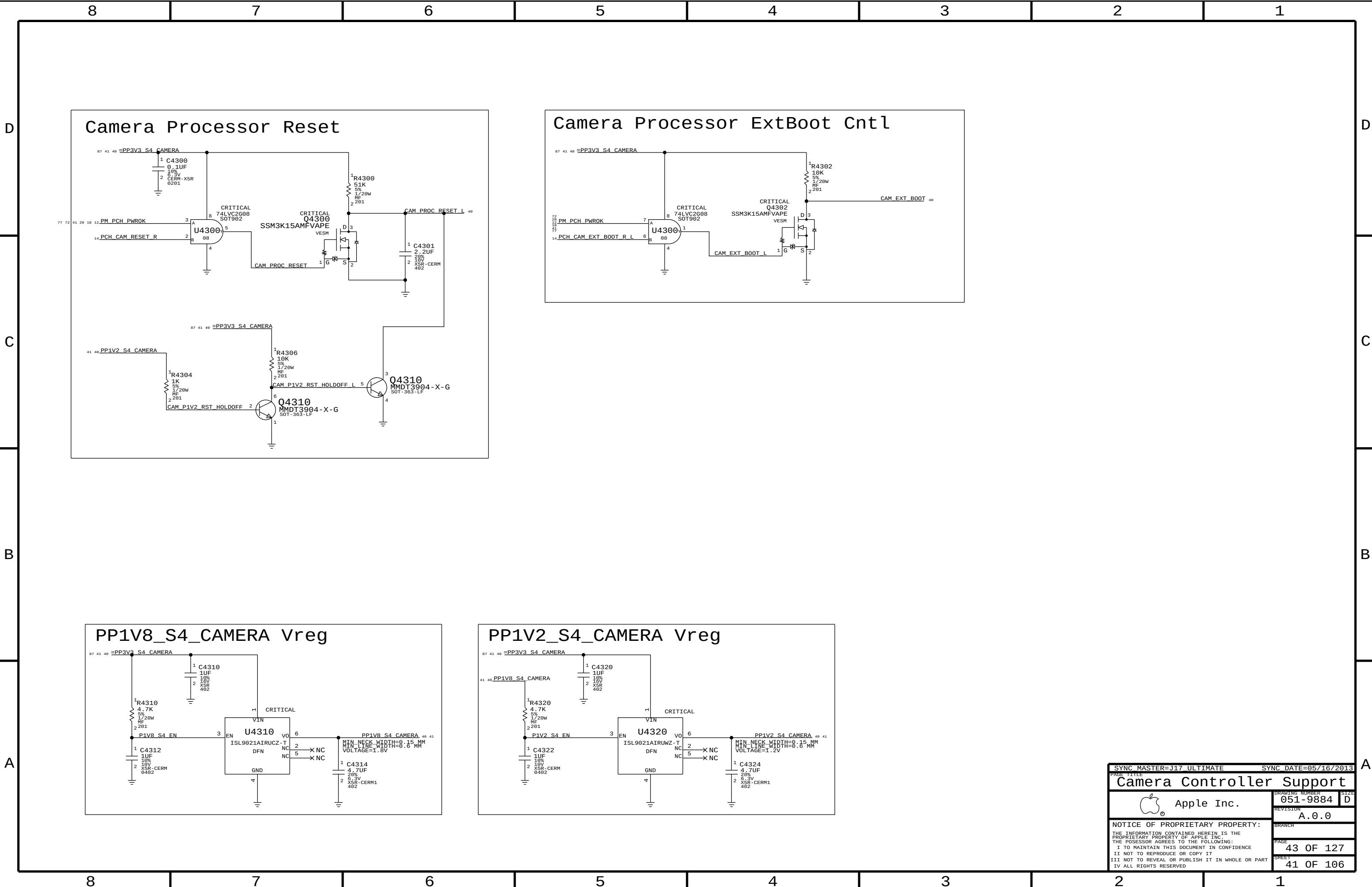


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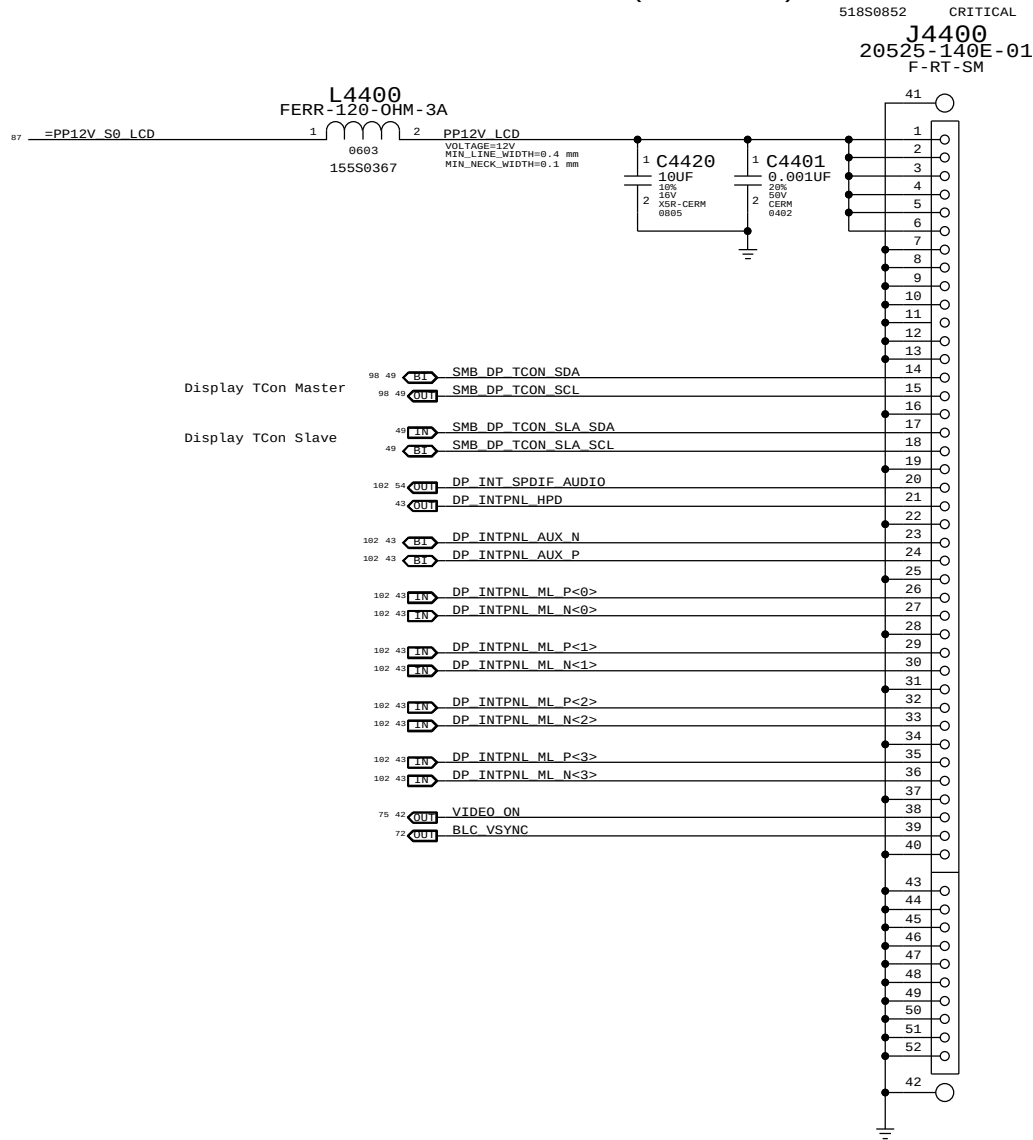


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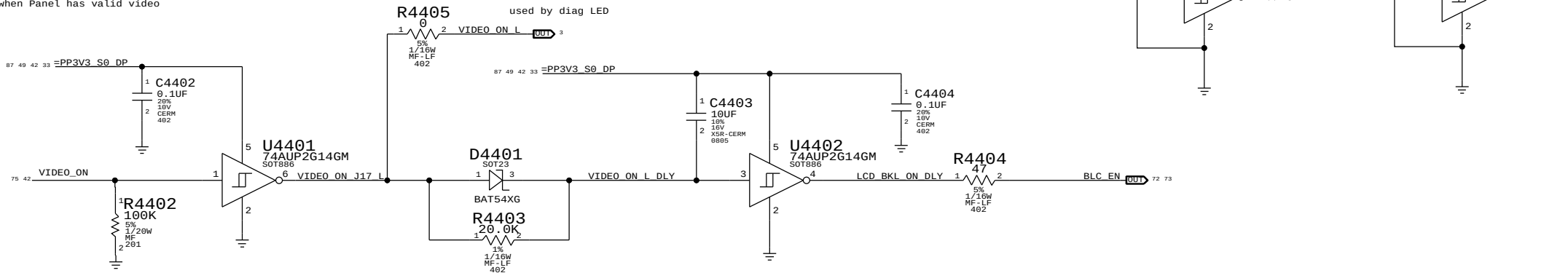


INTERNAL DP (STRAIGHT)

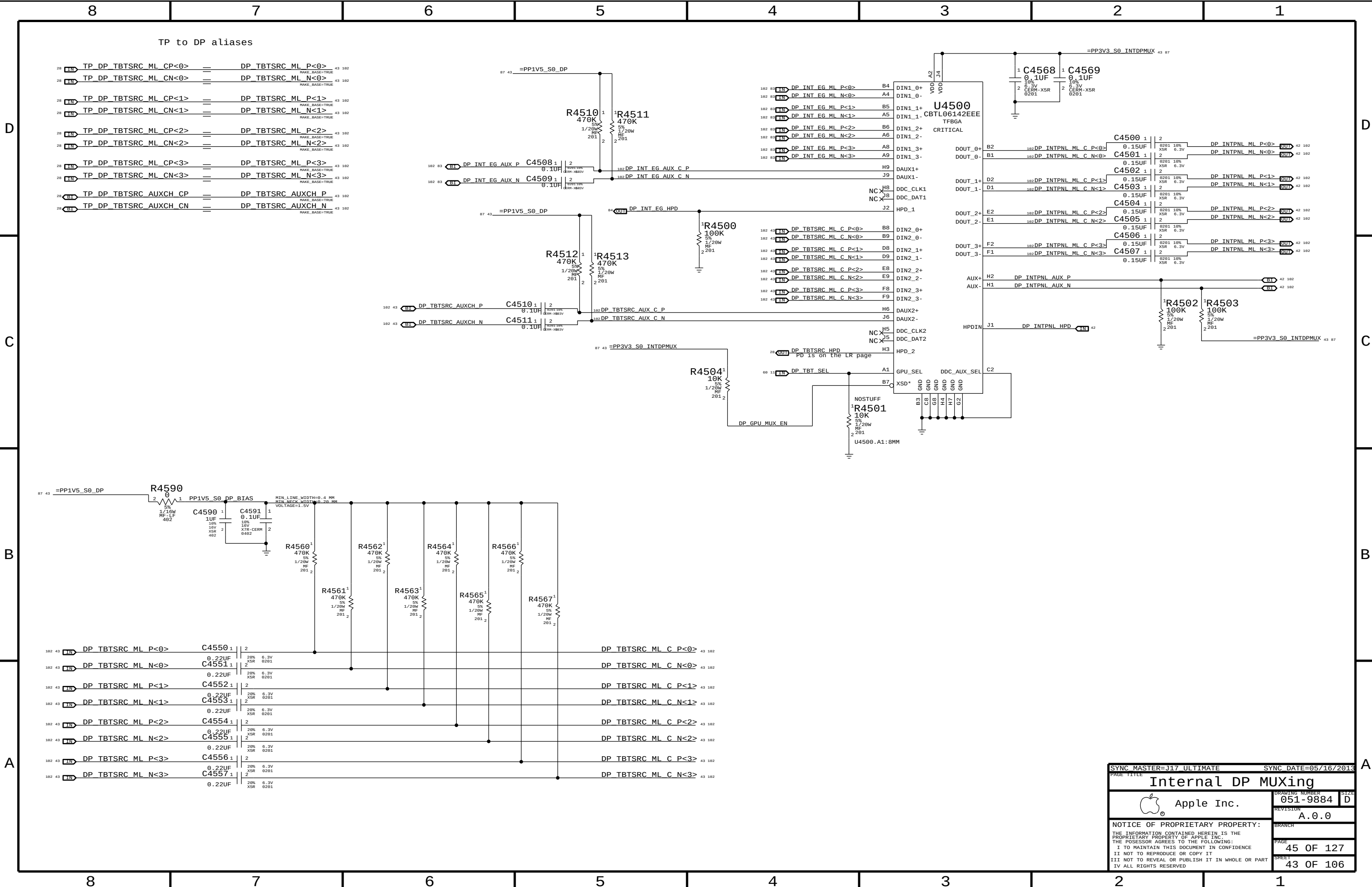


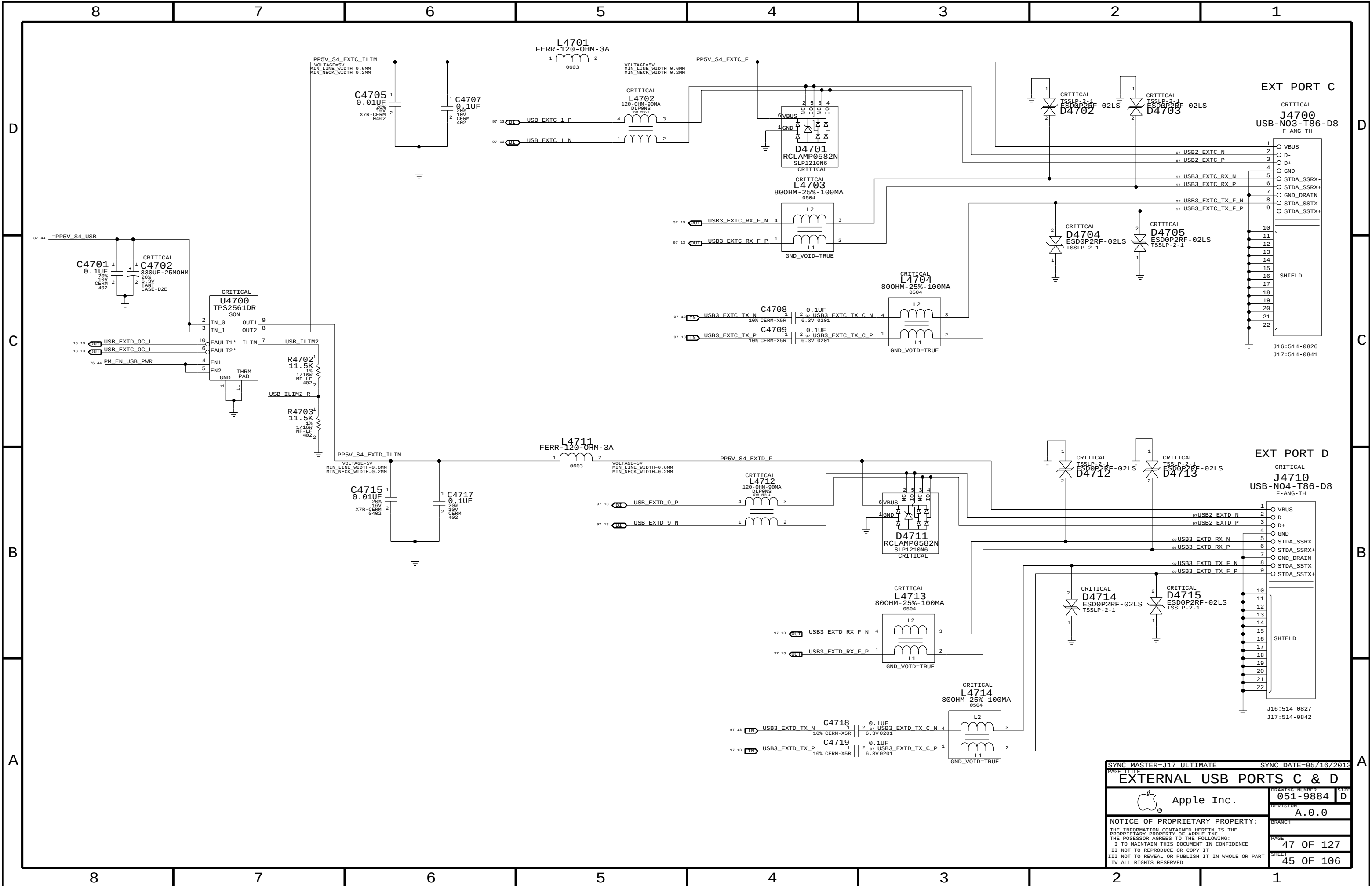
BACKLIGHT CONTROL SUPPORT

guarantee backlight is
only on when Panel has valid video

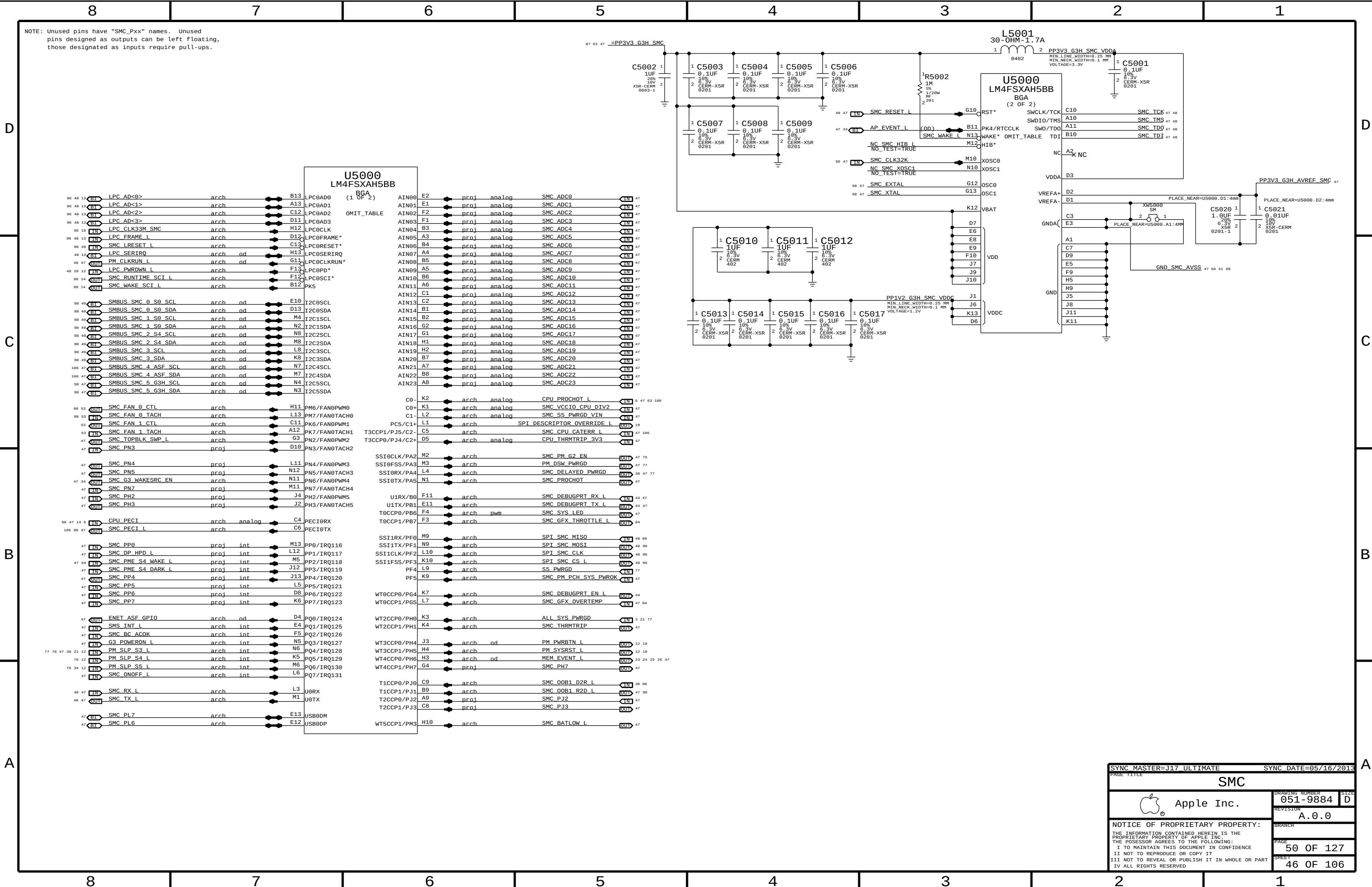


SYNC_MASTER=J17_ULTIMATE		SYNC_DATE=05/16/2013	
PAGE TITLE		Internal DP Support	
 Apple Inc.		DRAWING NUMBER	051-9884
		REVISION	A.0.0
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		PAGE	44 OF 127
		SHEET	42 OF 106

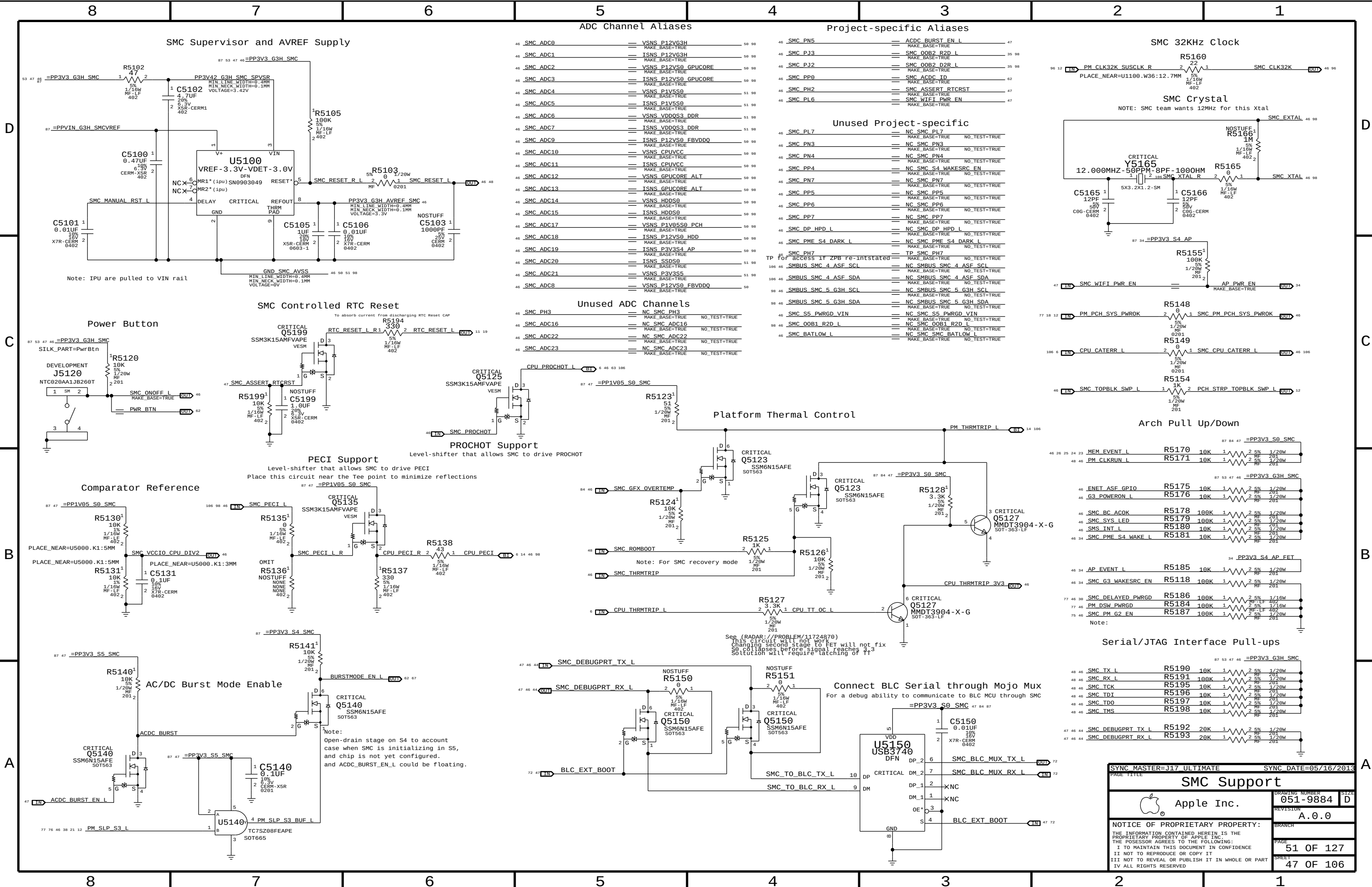




SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/16/2013	
PAGE TITLE		EXTERNAL USB PORTS C & D	
 Apple Inc.		DRAWING NUMBER	051-9884
		REVISION	A.0.0
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		PAGE	47 OF 127
		SHEET	45 OF 106



SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/16/2013	
PAGE TITLE			
SMC			
 Apple Inc.		DRAWING NUMBER	051-9884
		SIZE	D
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		BRANCH	
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		SHEET	46 OF 106



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SYNC DATE=05/16/2013

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051-9884

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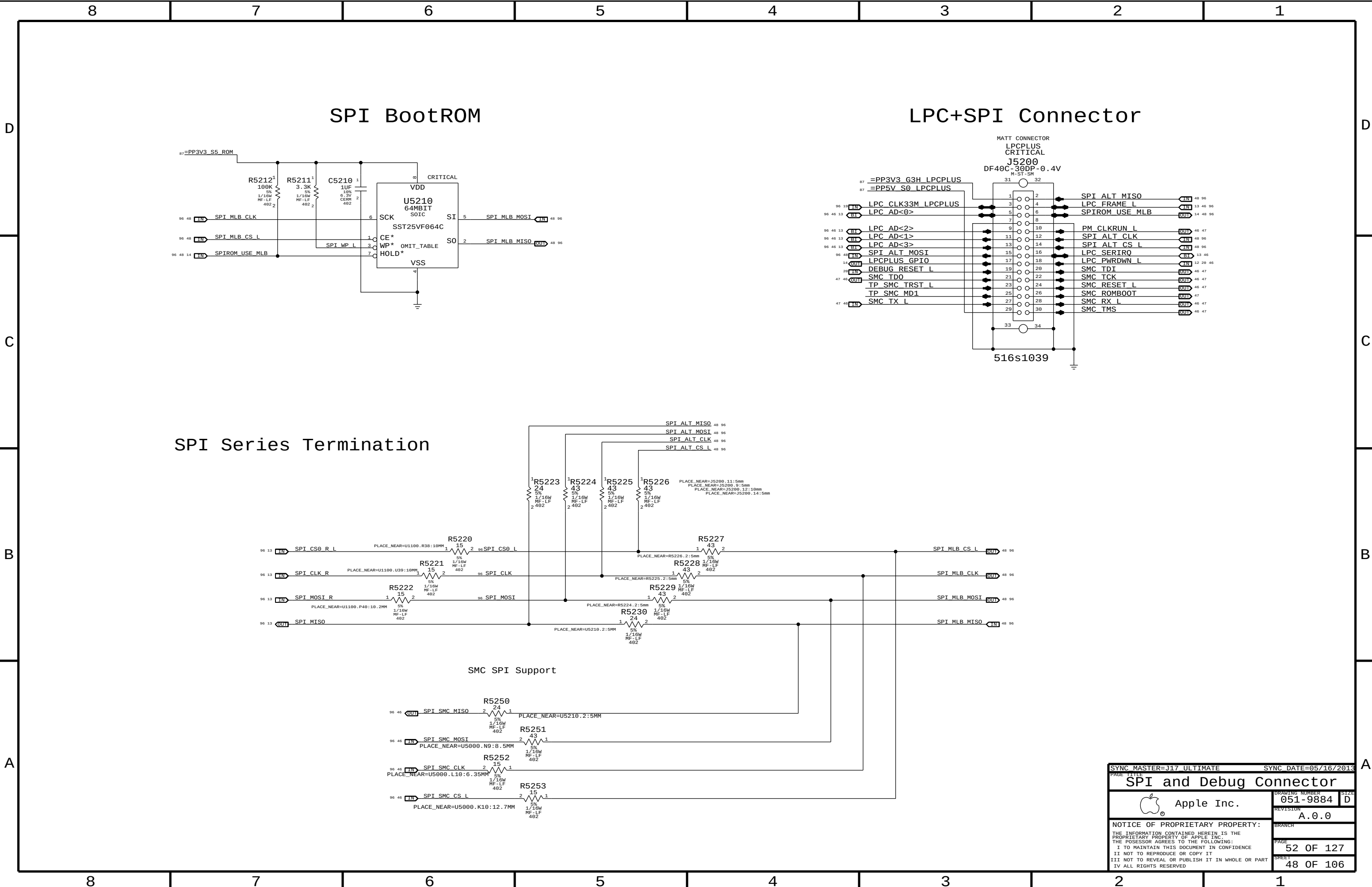
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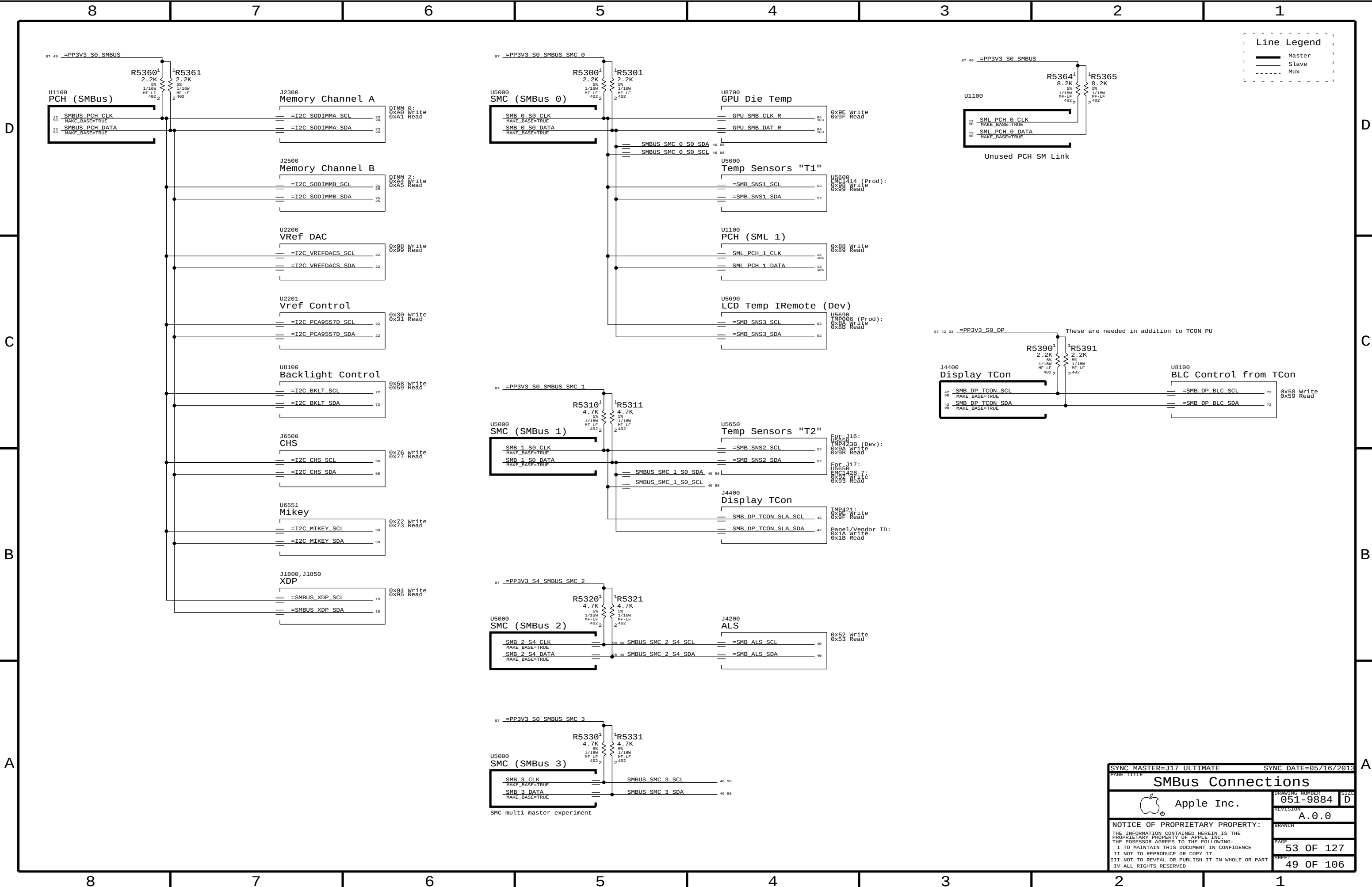
II NOT TO REPRODUCE OR COPY IT

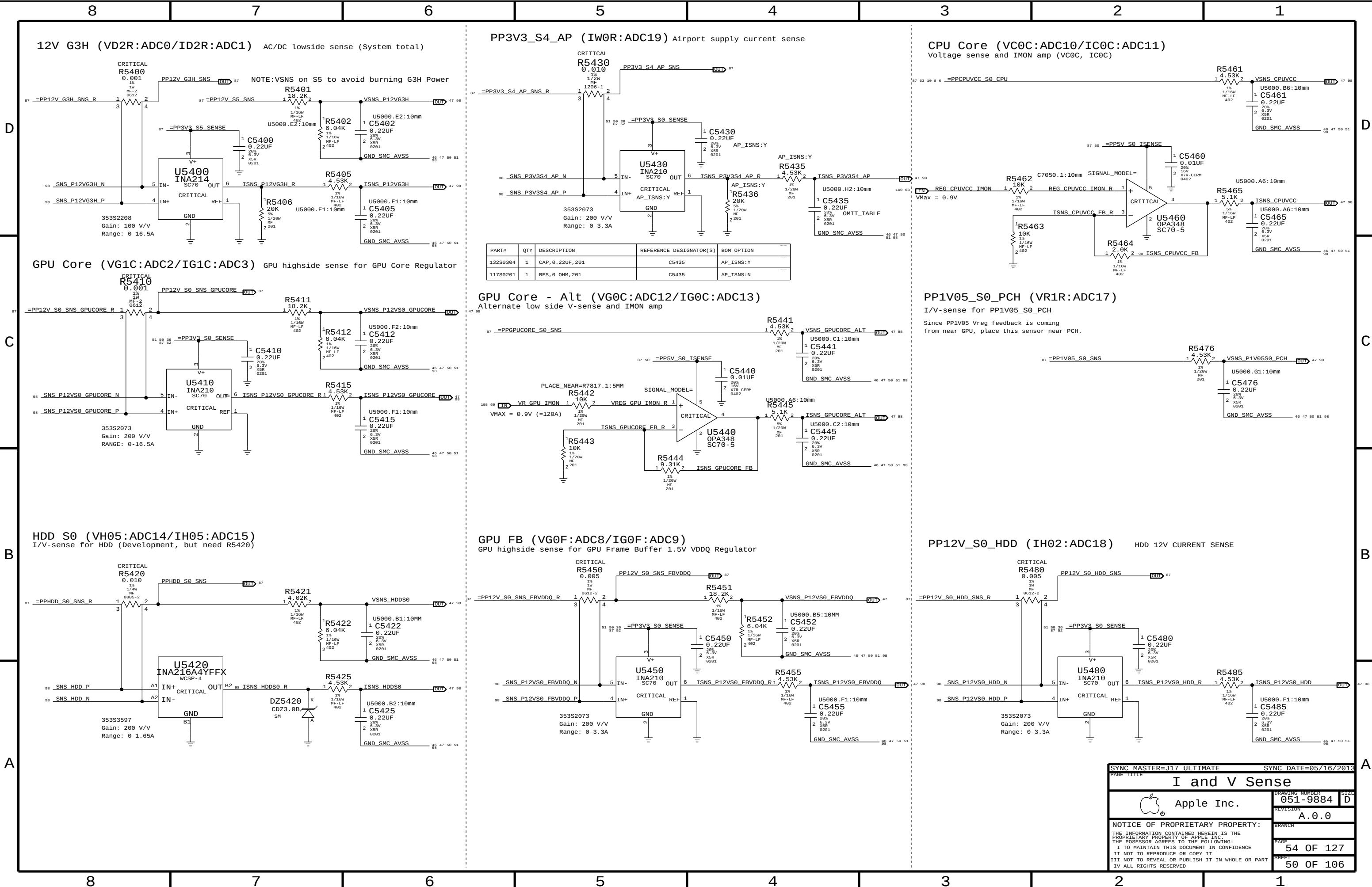
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SPI and Debug Connector		DRAWING NUMBER		051-9884	
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PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
13250304	1	CAP, 0.22UF, 201	C5435	AP_ISNS:Y
11750201	1	RES, 0 OHM, 201	C5435	AP_ISNS:N

SYNC MASTER=J17 ULTIMATE

SYNC DATE=05/16/2013

I and V Sense

Apple Inc.

DRAWING NUMBER051-9884

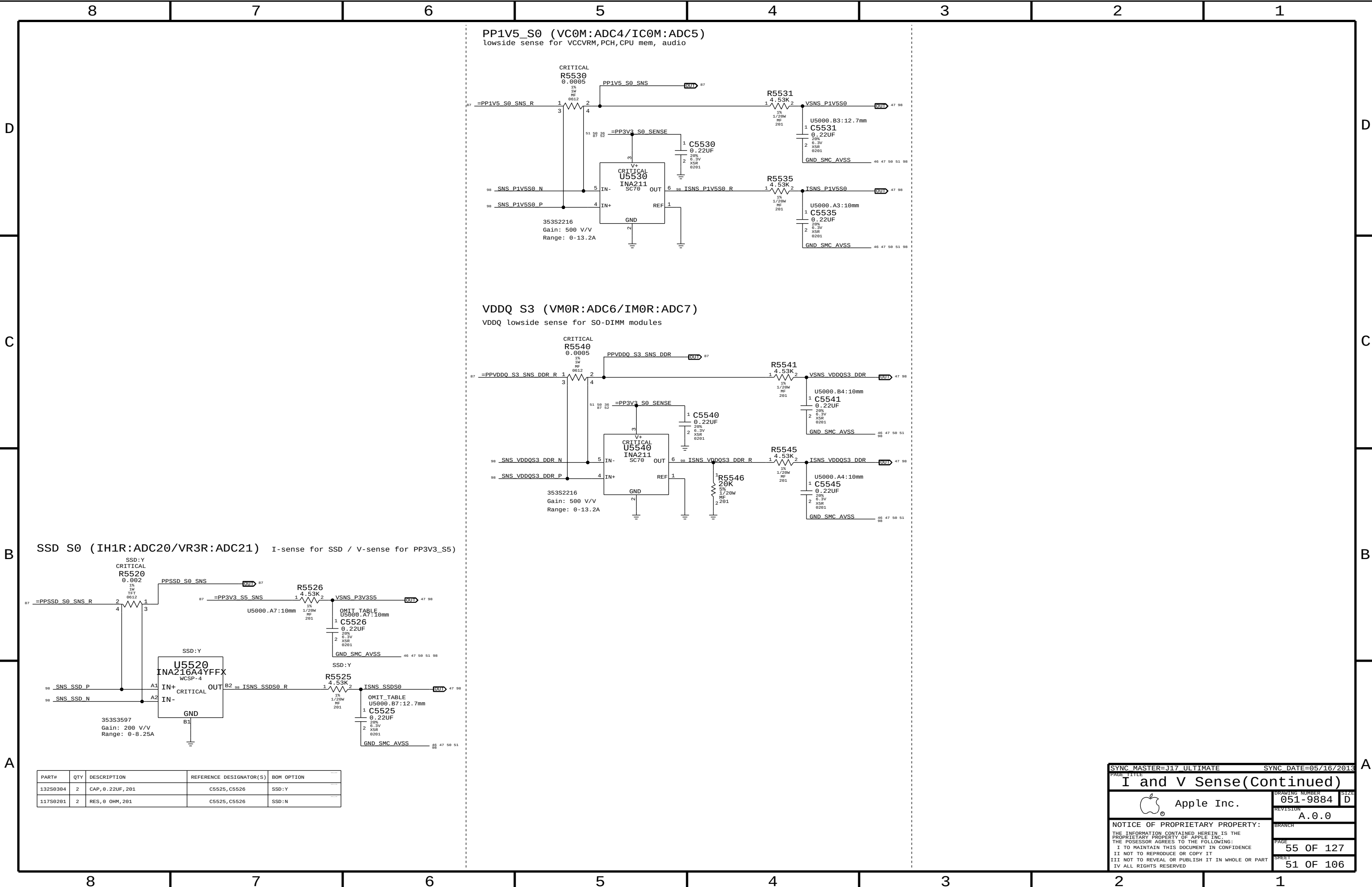
REVISIONA.0.0

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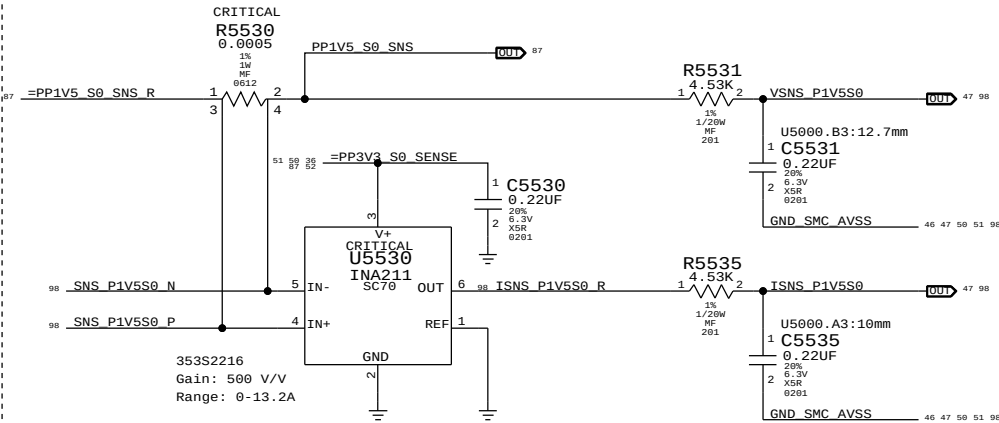
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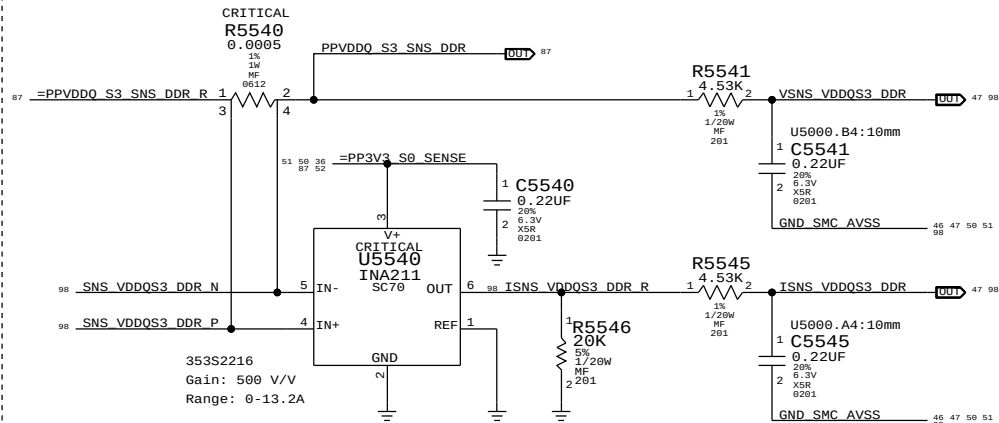
SHEET50 OF 106



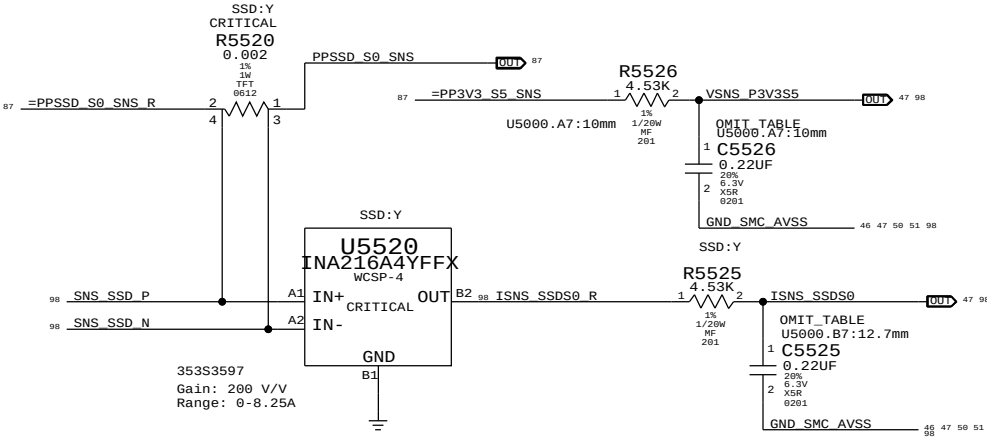
PP1V5_S0 (VC0M:ADC4/IC0M:ADC5)
lowside sense for VCCVRM,PCH,CPU mem, audio



VDDQ_S3 (VM0R:ADC6/IM0R:ADC7)
VDDQ lowside sense for S0-DIMM modules



SSD_S0 (IH1R:ADC20/VR3R:ADC21) I-sense for SSD / V-sense for PP3V3_S5)



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
132S0304	2	CAP, 0.22UF, 201	C5525, C5526	SSD:Y
117S0201	2	RES, 0 OHM, 201	C5525, C5526	SSD:N

SYNC MASTER=J17 ULTIMATE

SYNC DATE=05/16/2013

I and V Sense(Continued)

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PAGE

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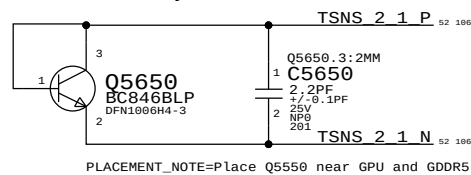
SHEET

51 OF 106

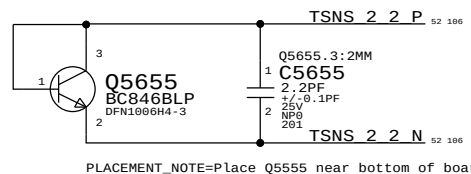
Temperature Sensor T2 EMC1428: Near GPU VR

SNS T2: TEMP SENSOR IC

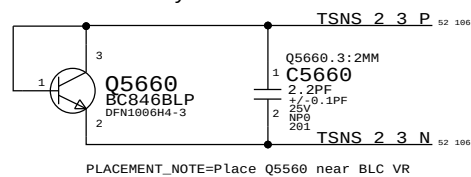
GPU Proximity



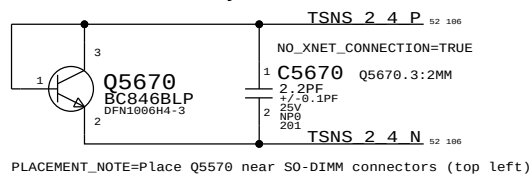
Ambient



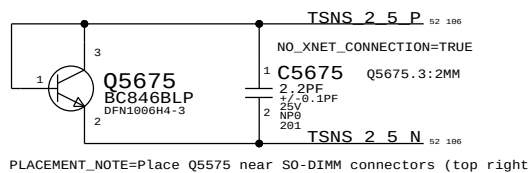
BLC Proximity



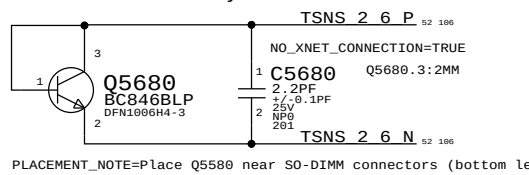
SO-DIMM Proximity 1



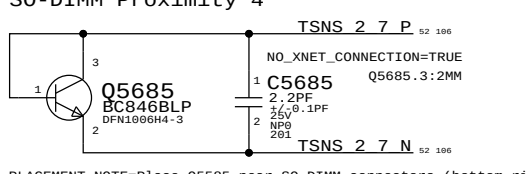
SO-DIMM Proximity 2



SO-DIMM Proximity 3



SO-DIMM Proximity 4



PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
372S0186	372S0185		ALL	Alternate Temp Diode

GPU Prox (TG0p)

TSNS 2 1 P

TSNS 2 1 N

Ambient (TA0p)

TSNS 2 2 P

TSNS 2 2 N

SO-DIMM Prox 1 (TM0p)

TSNS 2 4 P

TSNS 2 4 N

SO-DIMM Prox 3 (TM2p)

TSNS 2 6 P

TSNS 2 6 N

SoDIMM Prox 4 (TM3p)

TSNS 2 7 P

TSNS 2 7 N

SoDIMM Prox 2 (TM1p)

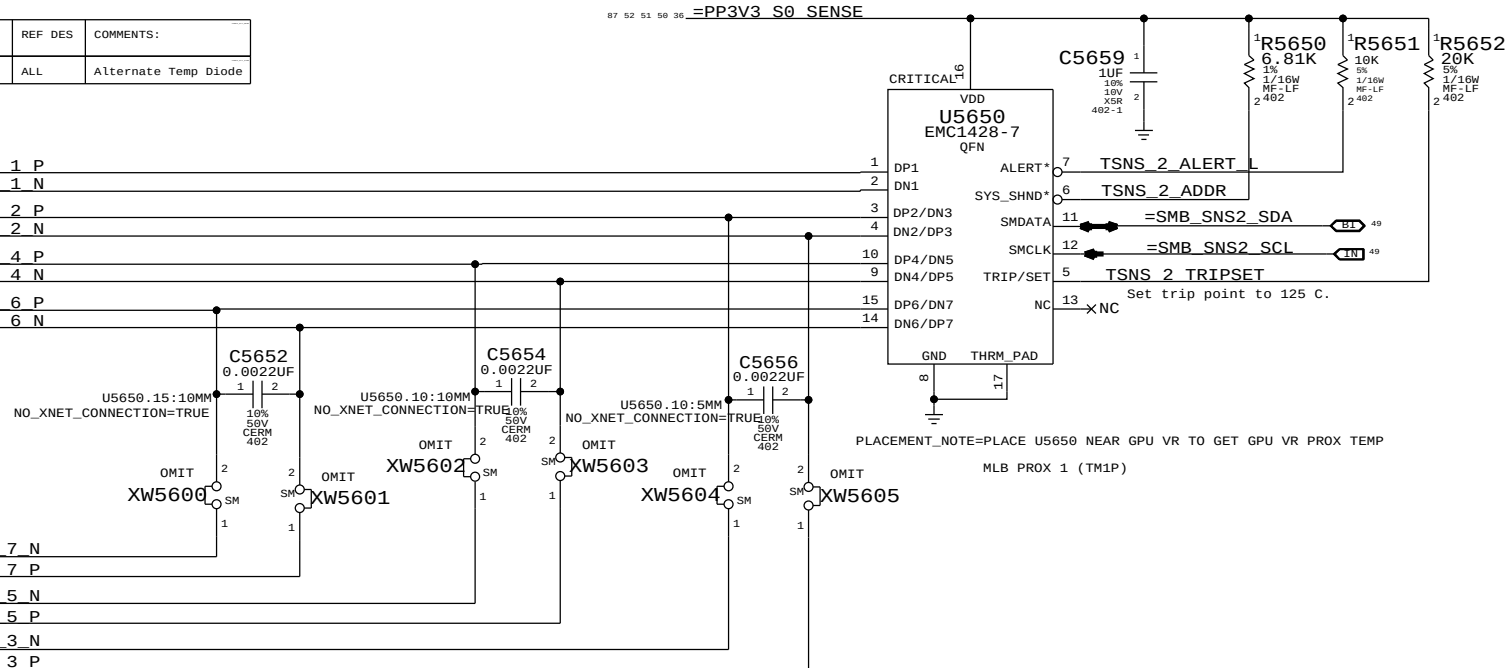
TSNS 2 5 P

TSNS 2 5 N

BLC PROX (Tb0p)

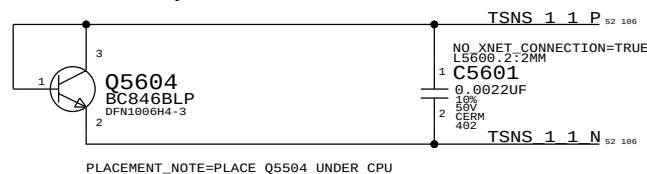
TSNS 2 3 P

TSNS 2 3 N

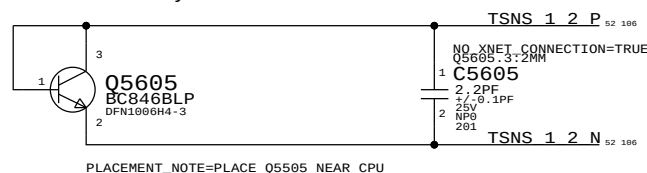


Temperature Sensor T1 EMC1414: Near PSU Conn

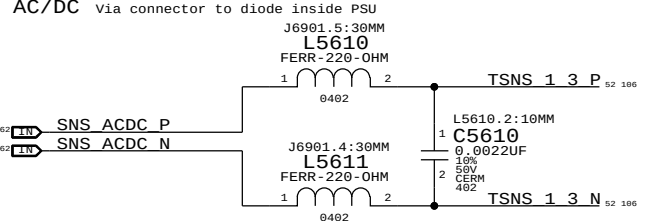
MLB Proximity



CPU Proximity



AC/DC



MLB PROX 2 (Tm2p)

TSNS 1 1 P

TSNS 1 1 N

CPU Prox (TC0p)

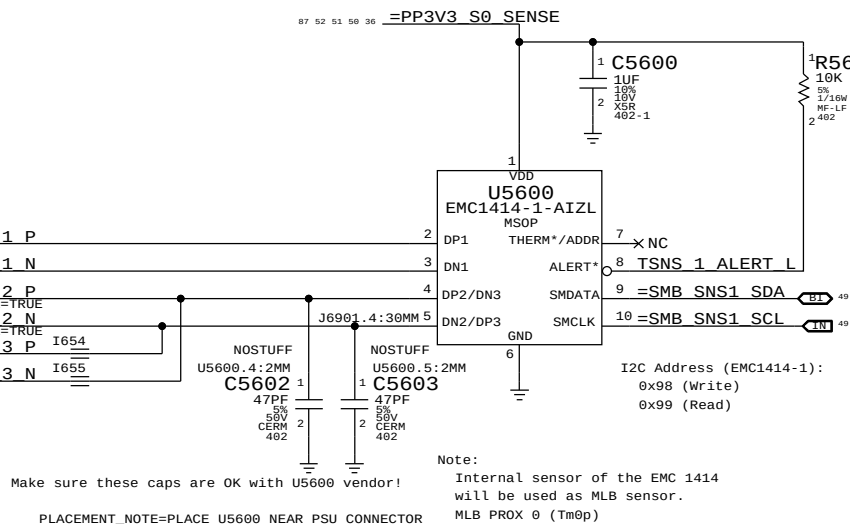
TSNS 1 2 P

TSNS 1 2 N

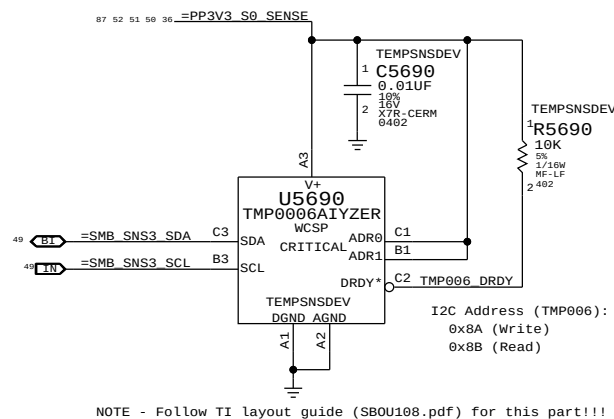
AC/DC (Tp2h)

TSNS 1 3 P

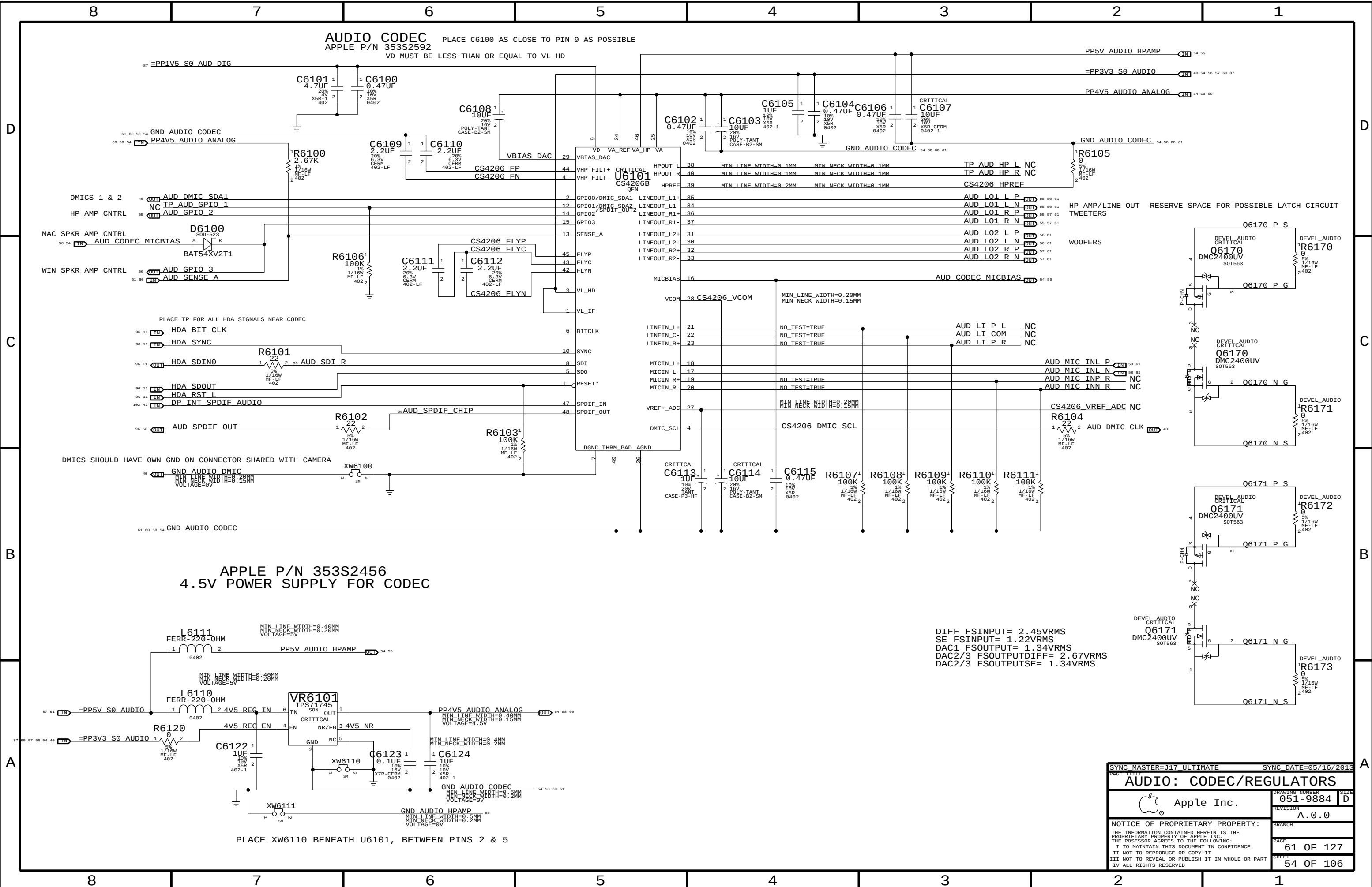
TSNS 1 3 N



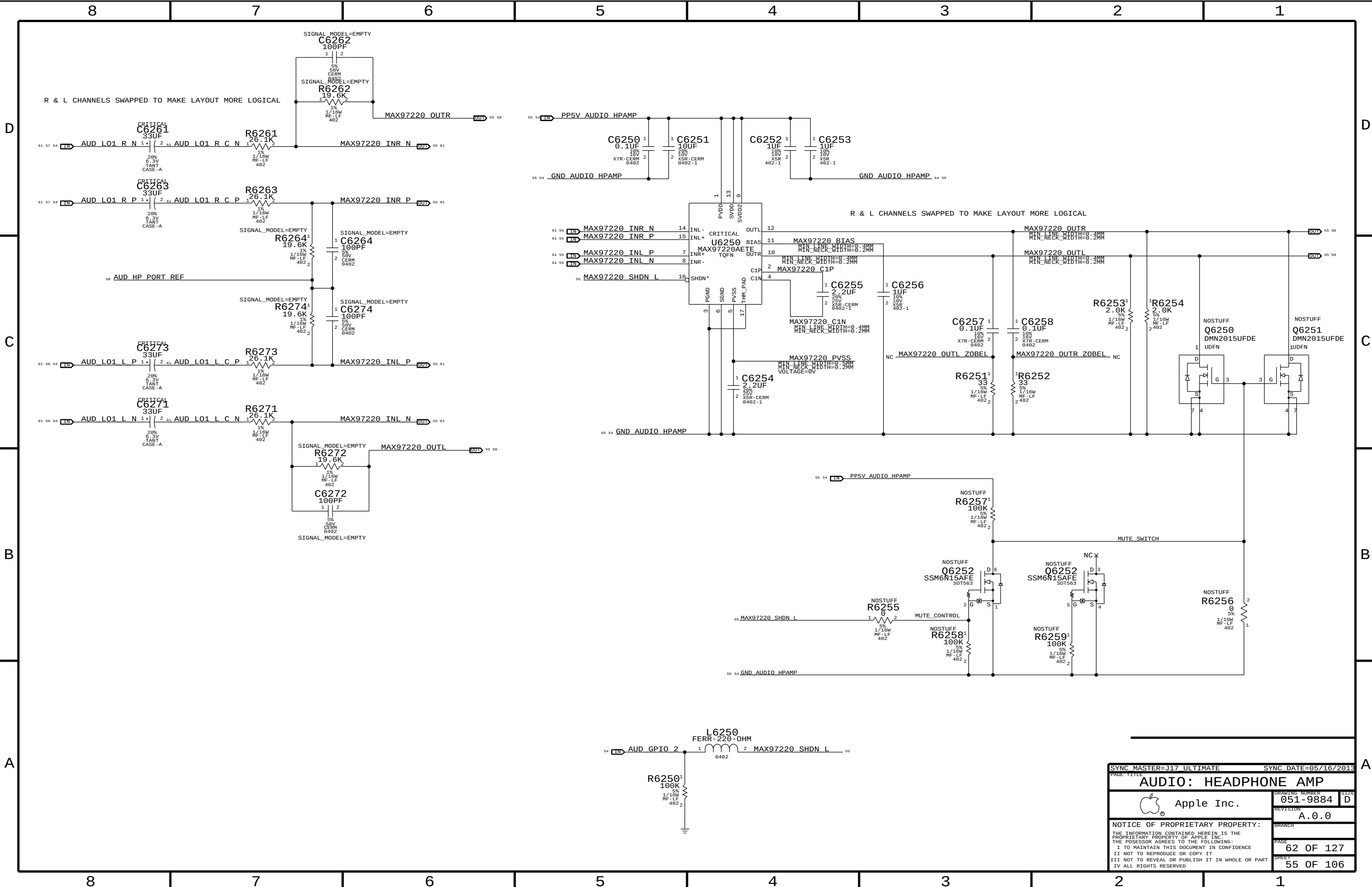
Temperature Sensor T3: LCD Remote Sensor(Dev Only)



SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/16/2013	
PAGE TITLE		Temperature Sensors	
Apple Inc.		DRAWING NUMBER	051-9884
		REVISION	A.0.0
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AUDIO: CODEC/REGULATORS		DRAWING NUMBER	
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SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/16/2013	
PAGE TITLE		AUDIO: HEADPHONE AMP	
Apple Inc.		DRAWING NUMBER	051-9884
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LEFT CH SPEAKER AMP
APPLE P/N 353S3163

SPEAKER AMP GAIN = +12 DB
SPEAKER AMP RIN = 40K NOMINAL
FC_HPF, TWEETERS = ~847 HZ (4700 PF)
FC_HPF, WOOFERS = ~4 HZ (1.0 UF)

D

D

C

C

B

B

A

A

INPUT POLARITY FLIP OK -- TRUE DIFF INPUTS

OUTPUT POLARITY FLIP TO
MAKE LAYOUT MORE LOGICAL

PINS 14 & 15 ARE TEST PINS AND
SHOULD BE TIED TO GND

EDGE RATE
CONTROL
ON
OFF

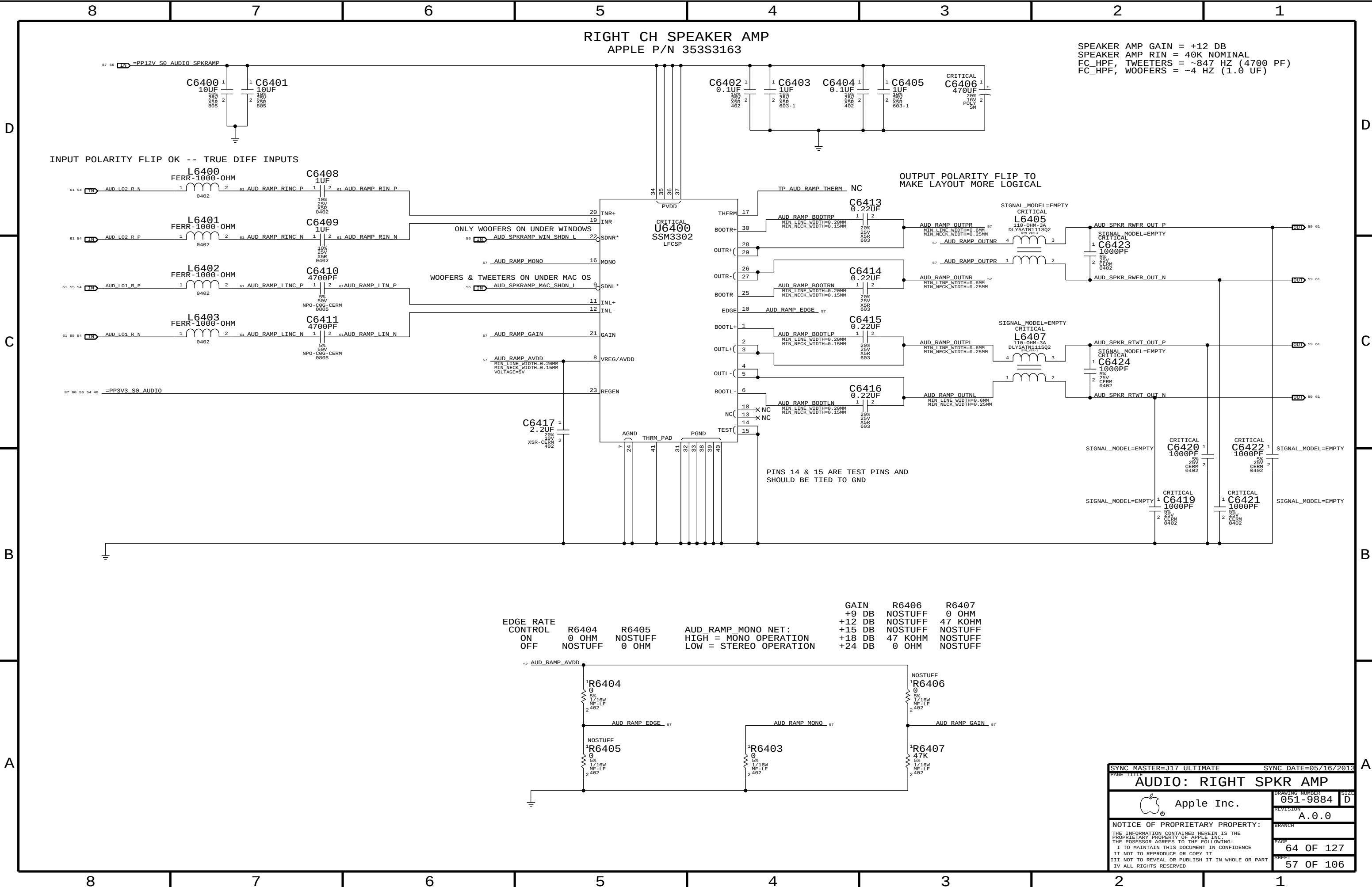
R6304
0 OHM
NOSTUFF

R6305
0 OHM
NOSTUFF

AUD_RAMP_MONO NET:
HIGH = MONO OPERATION
LOW = STEREO OPERATION

GAIN	R6306	R6307
+9 DB	NOSTUFF	0 OHM
+12 DB	NOSTUFF	47 KOHM
+15 DB	NOSTUFF	NOSTUFF
+18 DB	47 KOHM	NOSTUFF
+24 DB	0 OHM	NOSTUFF

SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/16/2013	
PAGE TITLE		AUDIO: LEFT SPKR AMP	
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MIKEY RECEIVER CKT

WRITE: 0X72 READ: 0X73 APN 353S3231

I2C ADDRESSES

MIKEY	U6551	READ	0111 0011	0X73
MIKEY	U6551	WRITE	0111 0010	0X72

D

D

C

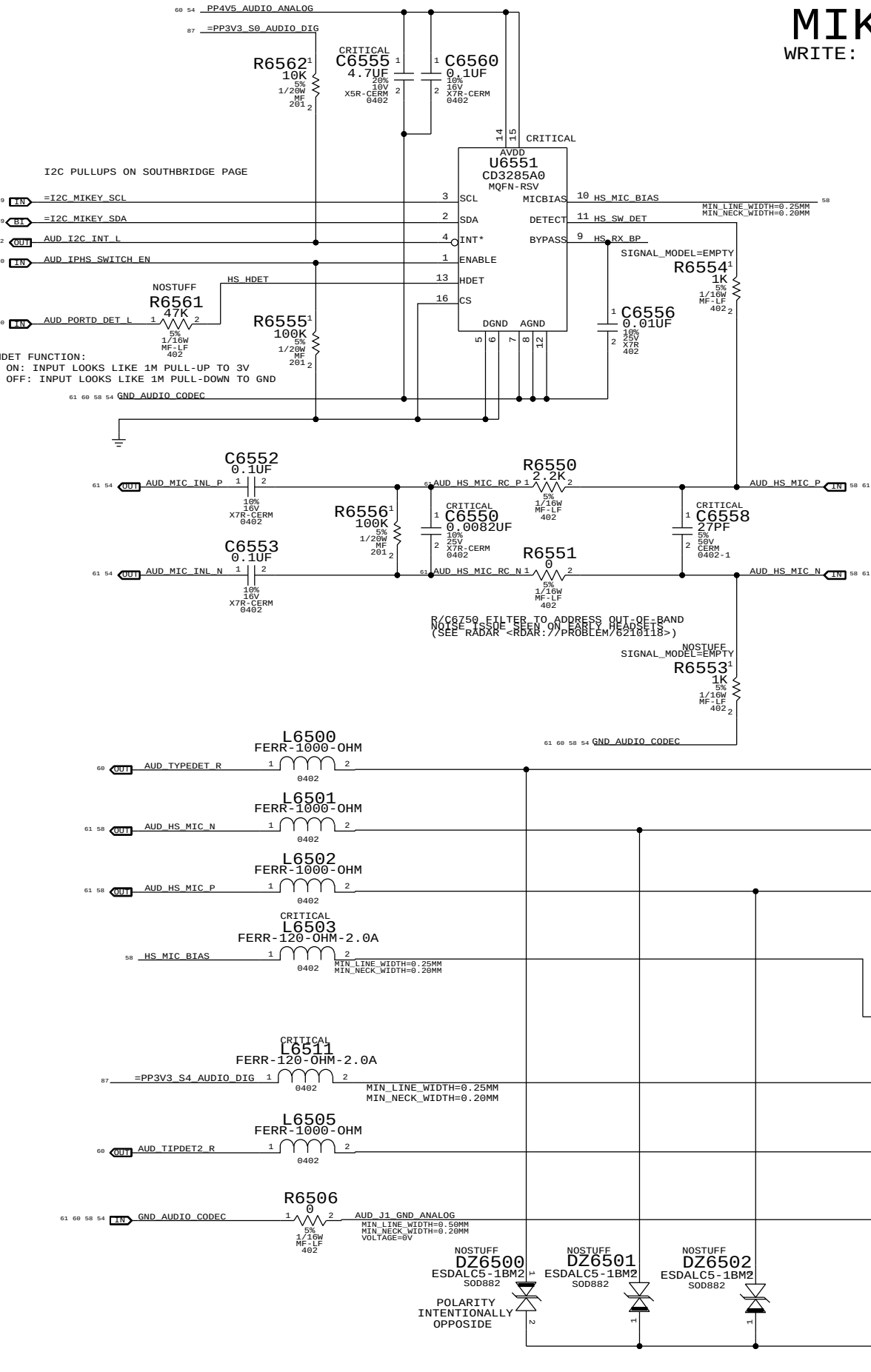
C

B

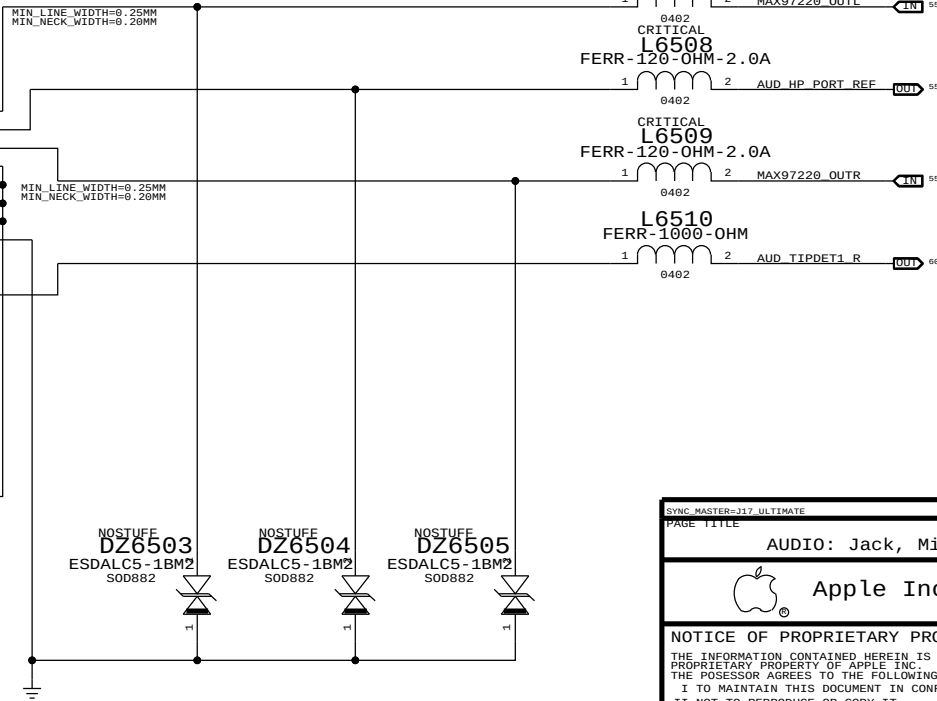
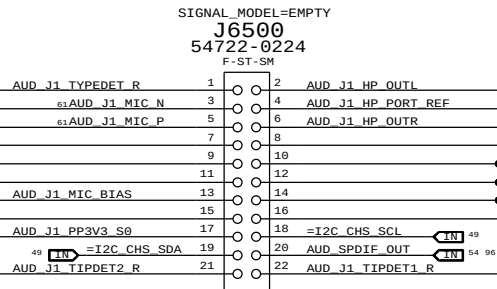
B

A

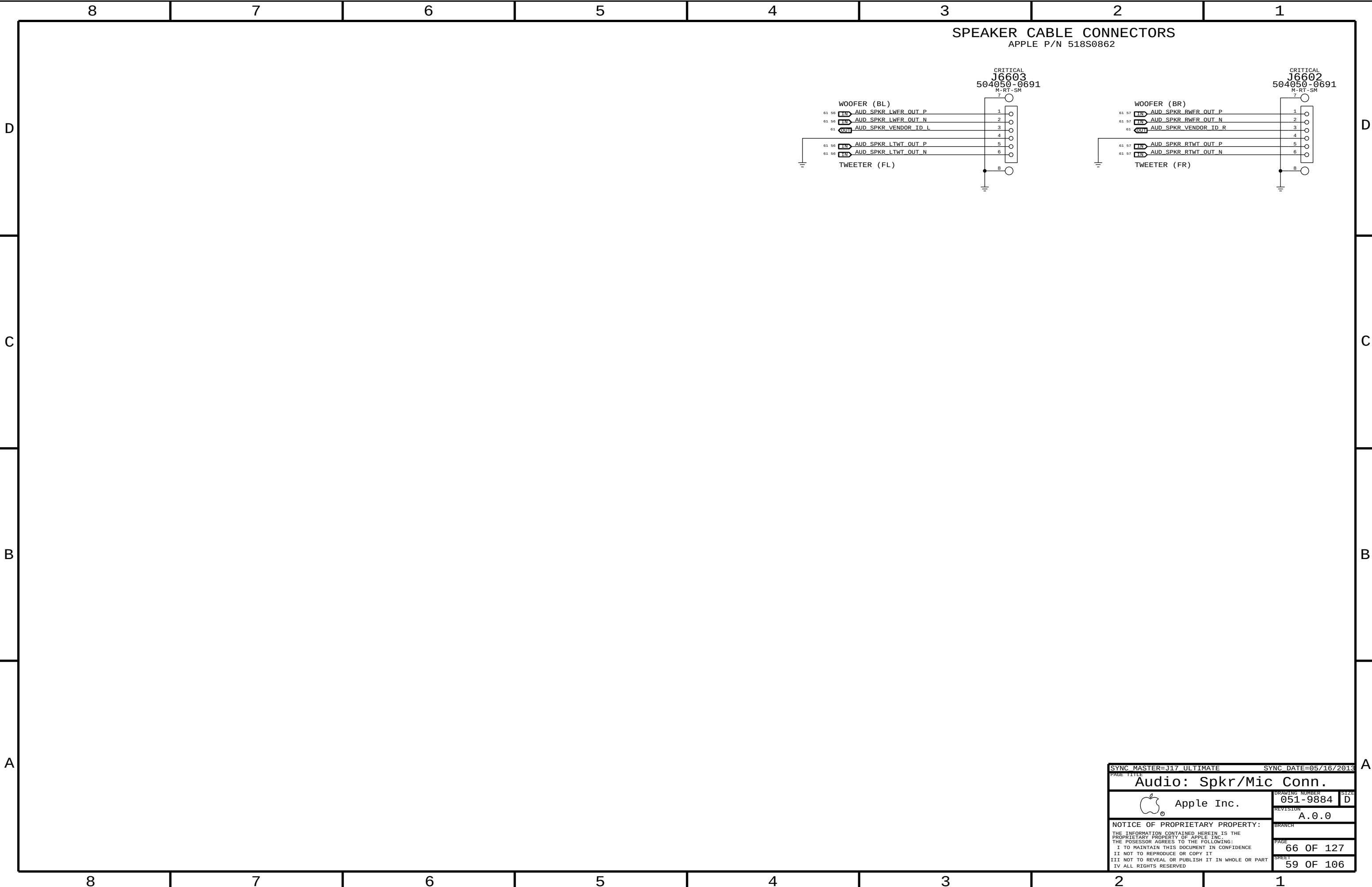
A



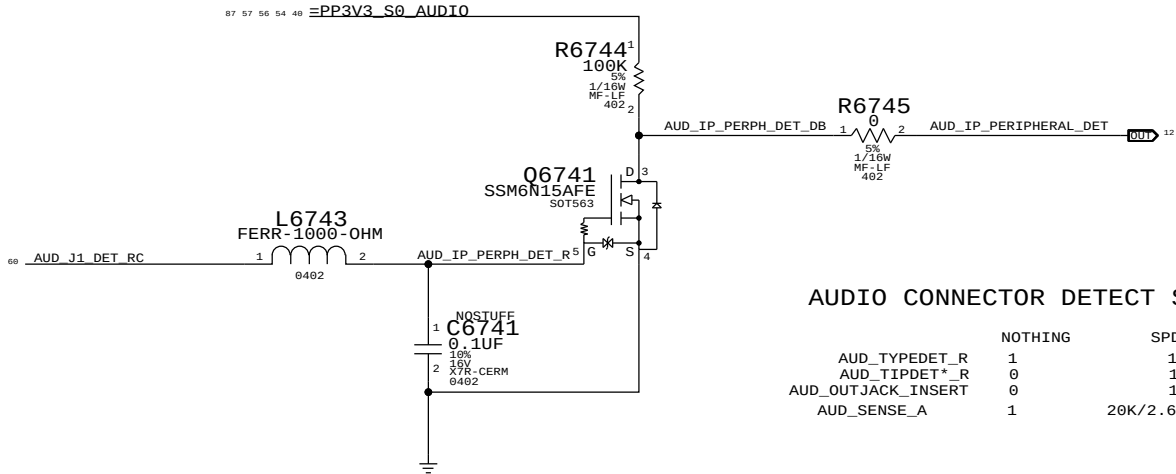
APPLE P/N 518S0687



PAGE TITLE		PAGE TITLE	
AUDIO: Jack, Mikey, CHS Switch		DRAWING NUMBER	
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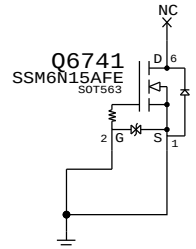


IPHS HS Detect Debounce CKT

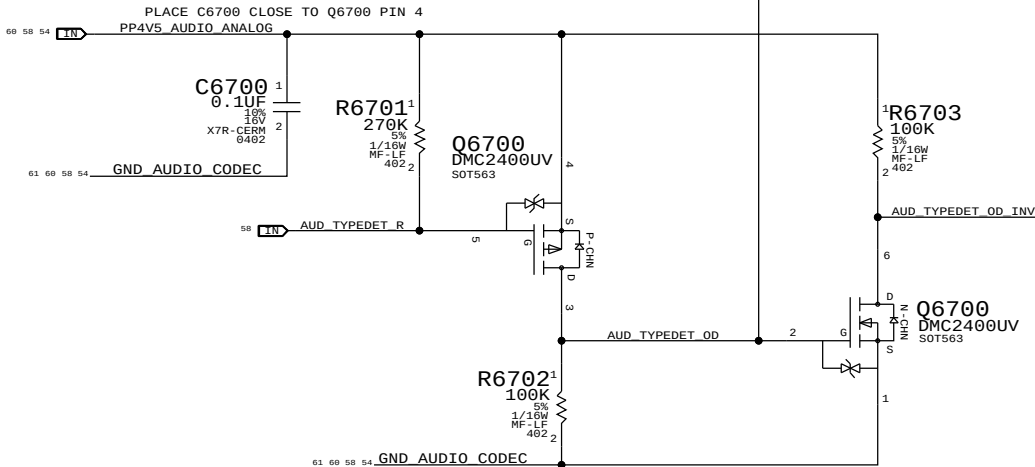
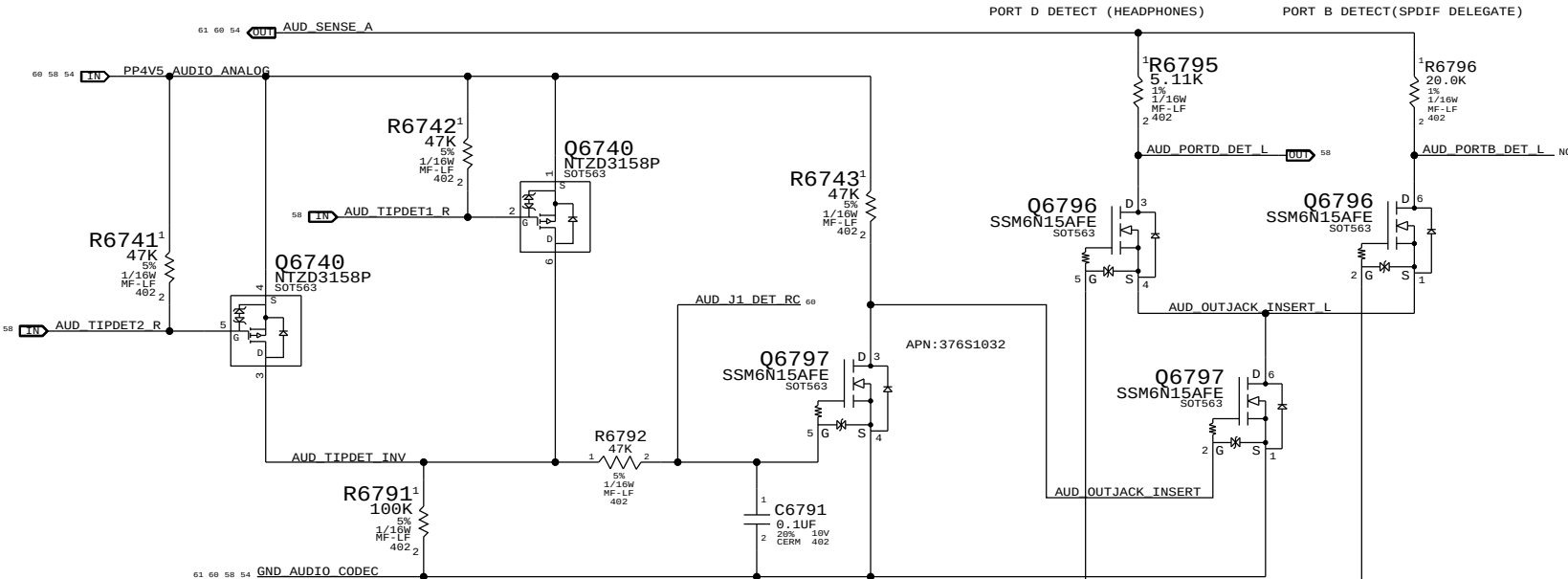
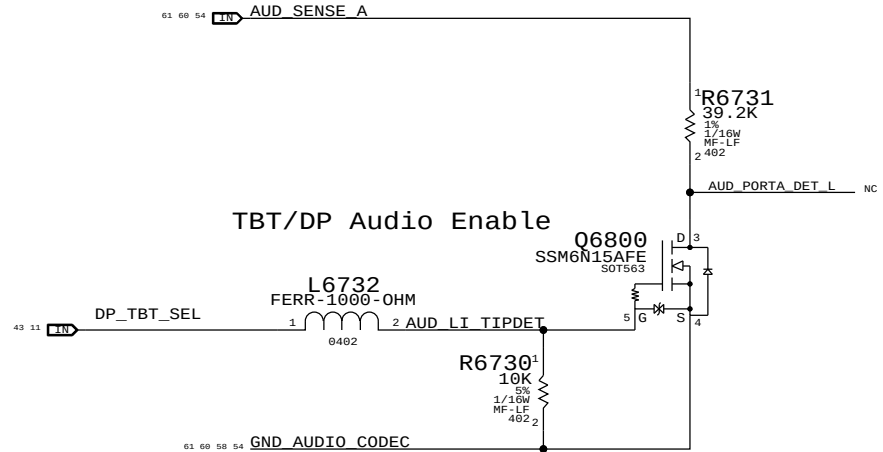


AUDIO CONNECTOR DETECT STATES

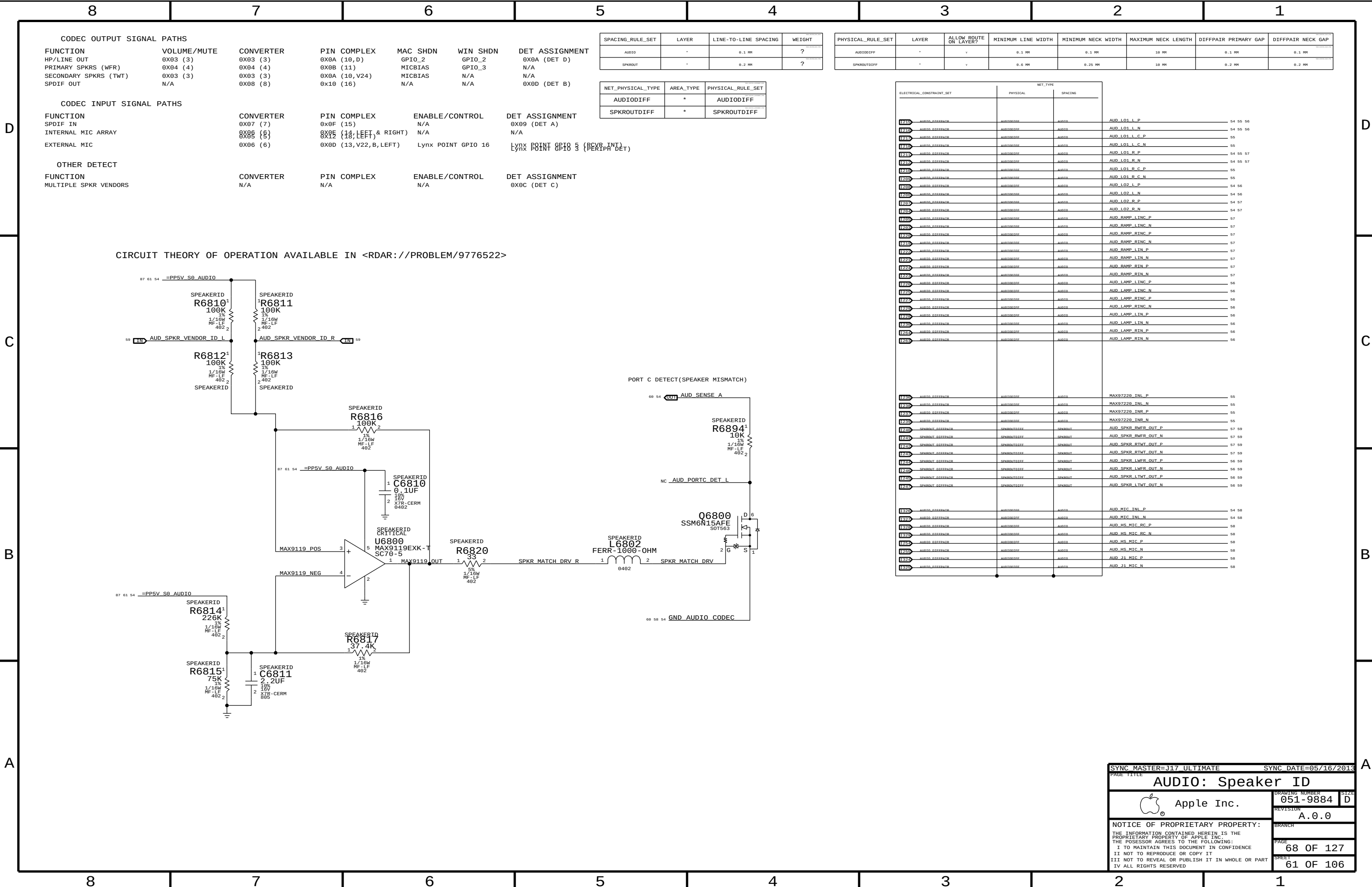
	NOTHING	SPDIF	HEADPHONE
AUD_TYPEDET_R	1	1	0
AUD_TIPDET*_R	0	1	1
AUD_OUTJACK_INSERT	0	1	1
AUD_SENSE_A	1	20K/2.67K RDIV	5.11K/2.67K RDIV



Target Display Mode Detect



PAGE TITLE		SYNC DATE=05/16/2013	
AUDIO: Detects/Grounding		DRAWING NUMBER	
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		60 OF 106	

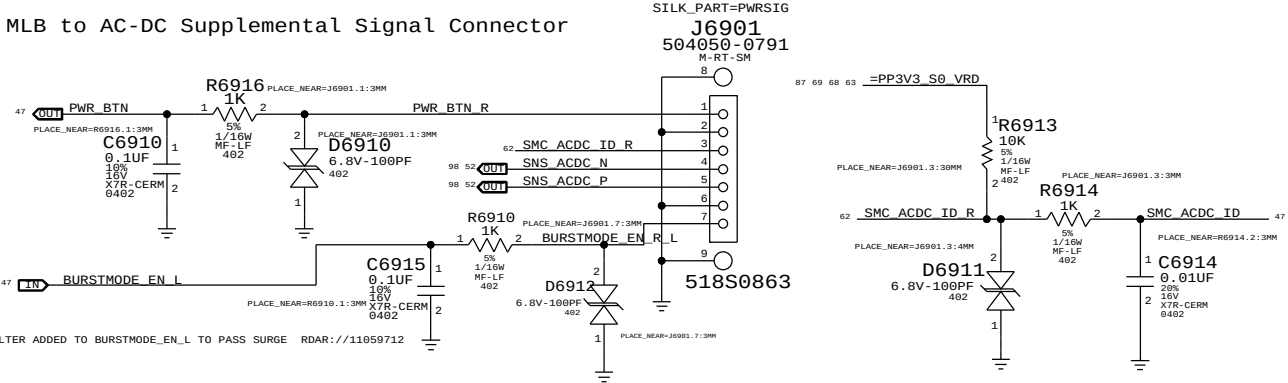
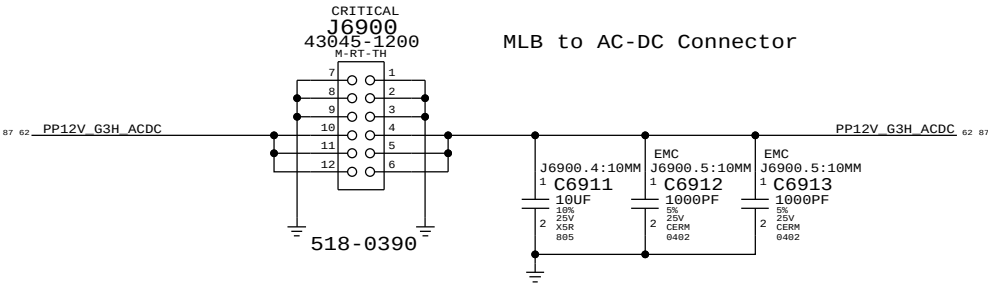
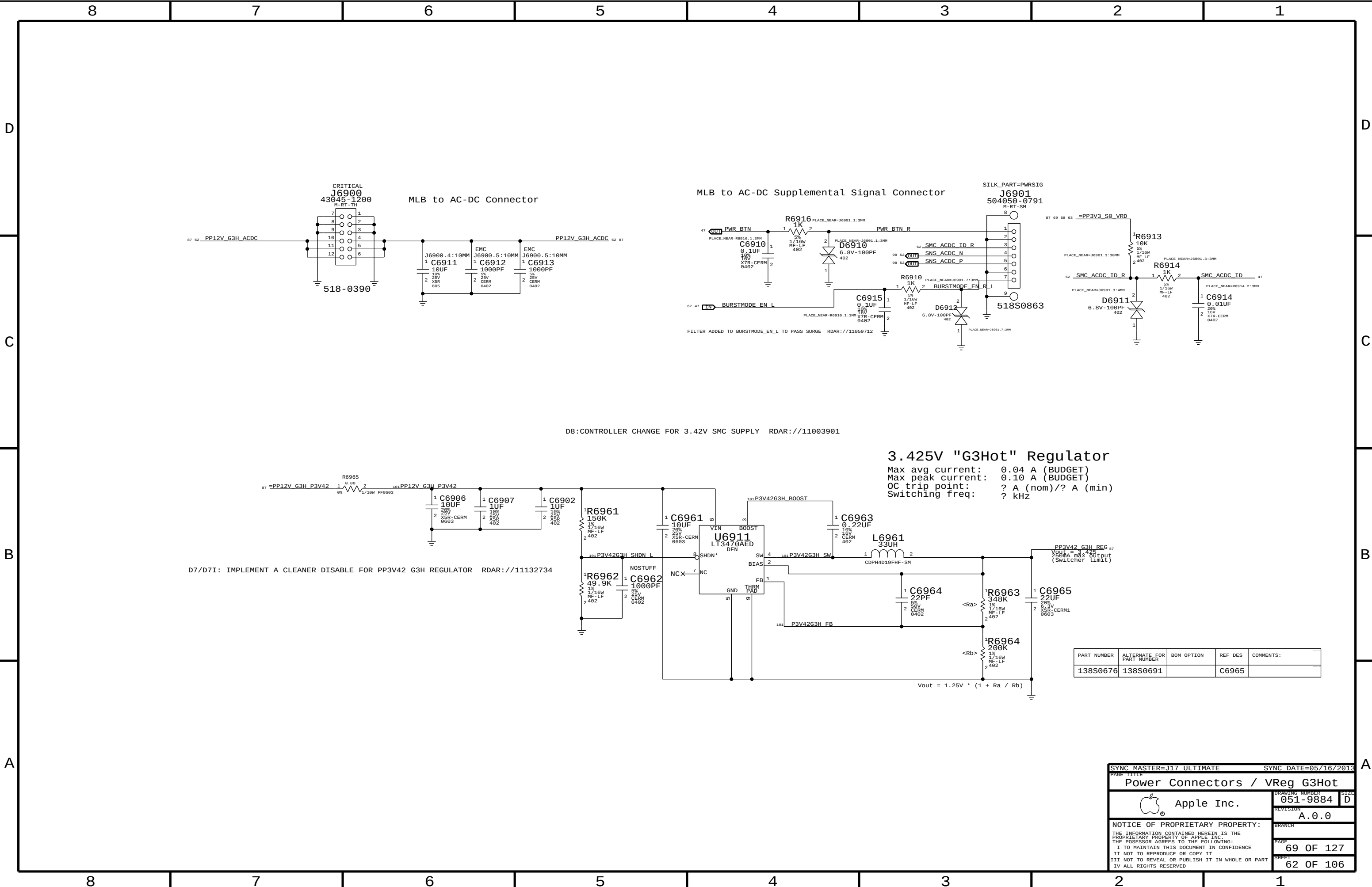


SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
AUDIO	*	0.1 MM	?
SPKROUT	*	0.2 MM	?

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
AUDIODIFF	*	Y	0.1 MM	0.1 MM	10 MM	0.1 MM	0.1 MM
SPKROUTDIFF	*	Y	0.6 MM	0.25 MM	10 MM	0.2 MM	0.2 MM

ELECTRICAL_CONSTRAINT_SET		NET_TYPE			
		PHYSICAL	SPACING		
R6810	AUDIO_RTEPAIR	AUDIOTIEE	AUDIO	AUD_L01_L_P	54 55
R6811	AUDIO_RTEPAIR	AUDIOTIEE	AUDIO	AUD_L01_L_N	54 55
R6812	AUDIO_RTEPAIR	AUDIOTIEE	AUDIO	AUD_L01_L_C_P	55
R6813	AUDIO_RTEPAIR	AUDIOTIEE	AUDIO	AUD_L01_L_C_N	55
R6814	AUDIO_RTEPAIR	AUDIOTIEE	AUDIO	AUD_L01_R_P	54 55
R6815	AUDIO_RTEPAIR	AUDIOTIEE	AUDIO	AUD_L01_R_N	54 55
R6816	AUDIO_RTEPAIR	AUDIOTIEE	AUDIO	AUD_L01_R_C_P	55
R6817	AUDIO_RTEPAIR	AUDIOTIEE	AUDIO	AUD_L01_R_C_N	55
R6818	AUDIO_RTEPAIR	AUDIOTIEE	AUDIO	AUD_L02_L_P	54 56
R6819	AUDIO_RTEPAIR	AUDIOTIEE	AUDIO	AUD_L02_L_N	54 56
R6820	AUDIO_RTEPAIR	AUDIOTIEE	AUDIO	AUD_L02_R_P	54 57
R6821	AUDIO_RTEPAIR	AUDIOTIEE	AUDIO	AUD_L02_R_N	54 57
R6822	AUDIO_RTEPAIR	AUDIOTIEE	AUDIO	AUD_RAMP_LINC_P	57
R6823	AUDIO_RTEPAIR	AUDIOTIEE	AUDIO	AUD_RAMP_LINC_N	57
R6824	AUDIO_RTEPAIR	AUDIOTIEE	AUDIO	AUD_RAMP_RINC_P	57
R6825	AUDIO_RTEPAIR	AUDIOTIEE	AUDIO	AUD_RAMP_RINC_N	57
R6826	AUDIO_RTEPAIR	AUDIOTIEE	AUDIO	AUD_RAMP_LIN_P	57
R6827	AUDIO_RTEPAIR	AUDIOTIEE	AUDIO	AUD_RAMP_LIN_N	57
R6828	AUDIO_RTEPAIR	AUDIOTIEE	AUDIO	AUD_RAMP_RIN_P	57
R6829	AUDIO_RTEPAIR	AUDIOTIEE	AUDIO	AUD_RAMP_RIN_N	57
R6830	AUDIO_RTEPAIR	AUDIOTIEE	AUDIO	AUD_LAMP_LINC_P	56
R6831	AUDIO_RTEPAIR	AUDIOTIEE	AUDIO	AUD_LAMP_LINC_N	56
R6832	AUDIO_RTEPAIR	AUDIOTIEE	AUDIO	AUD_LAMP_RINC_P	56
R6833	AUDIO_RTEPAIR	AUDIOTIEE	AUDIO	AUD_LAMP_RINC_N	56
R6834	AUDIO_RTEPAIR	AUDIOTIEE	AUDIO	AUD_LAMP_LIN_P	56
R6835	AUDIO_RTEPAIR	AUDIOTIEE	AUDIO	AUD_LAMP_LIN_N	56
R6836	AUDIO_RTEPAIR	AUDIOTIEE	AUDIO	AUD_LAMP_RIN_P	56
R6837	AUDIO_RTEPAIR	AUDIOTIEE	AUDIO	AUD_LAMP_RIN_N	56
R6838	AUDIO_RTEPAIR	AUDIOTIEE	AUDIO	MAX97220_INL_P	55
R6839	AUDIO_RTEPAIR	AUDIOTIEE	AUDIO	MAX97220_INL_N	55
R6840	AUDIO_RTEPAIR	AUDIOTIEE	AUDIO	MAX97220_INR_P	55
R6841	AUDIO_RTEPAIR	AUDIOTIEE	AUDIO	MAX97220_INR_N	55
R6842	SPKROUT_RTEPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_RMFR_OUT_P	57 59
R6843	SPKROUT_RTEPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_RMFR_OUT_N	57 59
R6844	SPKROUT_RTEPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_RTWT_OUT_P	57 59
R6845	SPKROUT_RTEPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_RTWT_OUT_N	57 59
R6846	SPKROUT_RTEPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_LWFR_OUT_P	56 59
R6847	SPKROUT_RTEPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_LWFR_OUT_N	56 59
R6848	SPKROUT_RTEPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_LTWT_OUT_P	56 59
R6849	SPKROUT_RTEPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_LTWT_OUT_N	56 59
R6850	AUDIO_RTEPAIR	AUDIOTIEE	AUDIO	AUD_MIC_INL_P	54 58
R6851	AUDIO_RTEPAIR	AUDIOTIEE	AUDIO	AUD_MIC_INL_N	54 58
R6852	AUDIO_RTEPAIR	AUDIOTIEE	AUDIO	AUD_HS_MIC_RC_P	58
R6853	AUDIO_RTEPAIR	AUDIOTIEE	AUDIO	AUD_HS_MIC_RC_N	58
R6854	AUDIO_RTEPAIR	AUDIOTIEE	AUDIO	AUD_HS_MIC_P	58
R6855	AUDIO_RTEPAIR	AUDIOTIEE	AUDIO	AUD_HS_MIC_N	58
R6856	AUDIO_RTEPAIR	AUDIOTIEE	AUDIO	AUD_31_MIC_P	58
R6857	AUDIO_RTEPAIR	AUDIOTIEE	AUDIO	AUD_31_MIC_N	58

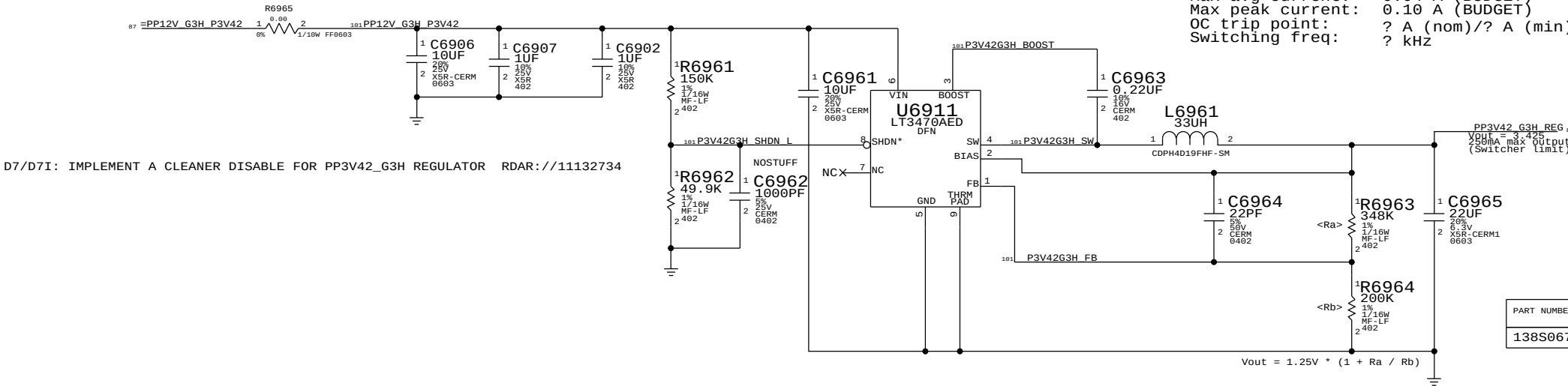
SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/16/2013	
PAGE TITLE			
AUDIO: Speaker ID			
 Apple Inc.	DRAWING NUMBER		SIZE
	051-9884		D
		REVISION	
		A.0.0	
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D8:CONTROLLER CHANGE FOR 3.42V SMC SUPPLY RDAR://11059712

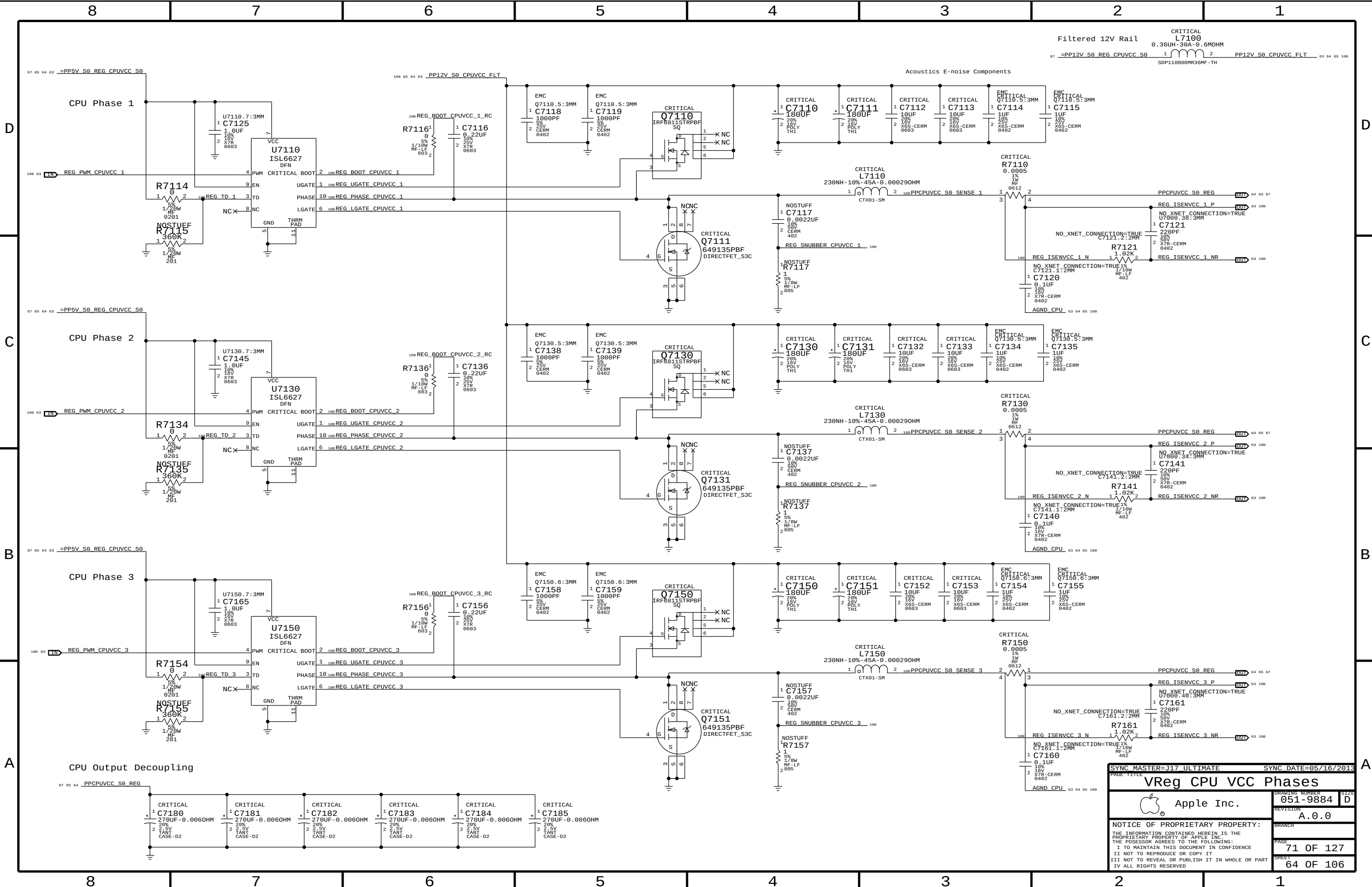
3.425V "G3Hot" Regulator

Max avg current: 0.04 A (BUDGET)
Max peak current: 0.10 A (BUDGET)
OC trip point: ? A (nom)/? A (min)
Switching freq: ? kHz



PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S0676	138S0691		C6965	

SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/16/2013	
PAGE TITLE		Power Connectors / VReg G3Hot	
Apple Inc.		DRAWING NUMBER	051-9884
		REVISION	A.0.0
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VReg CPU VCC Phases

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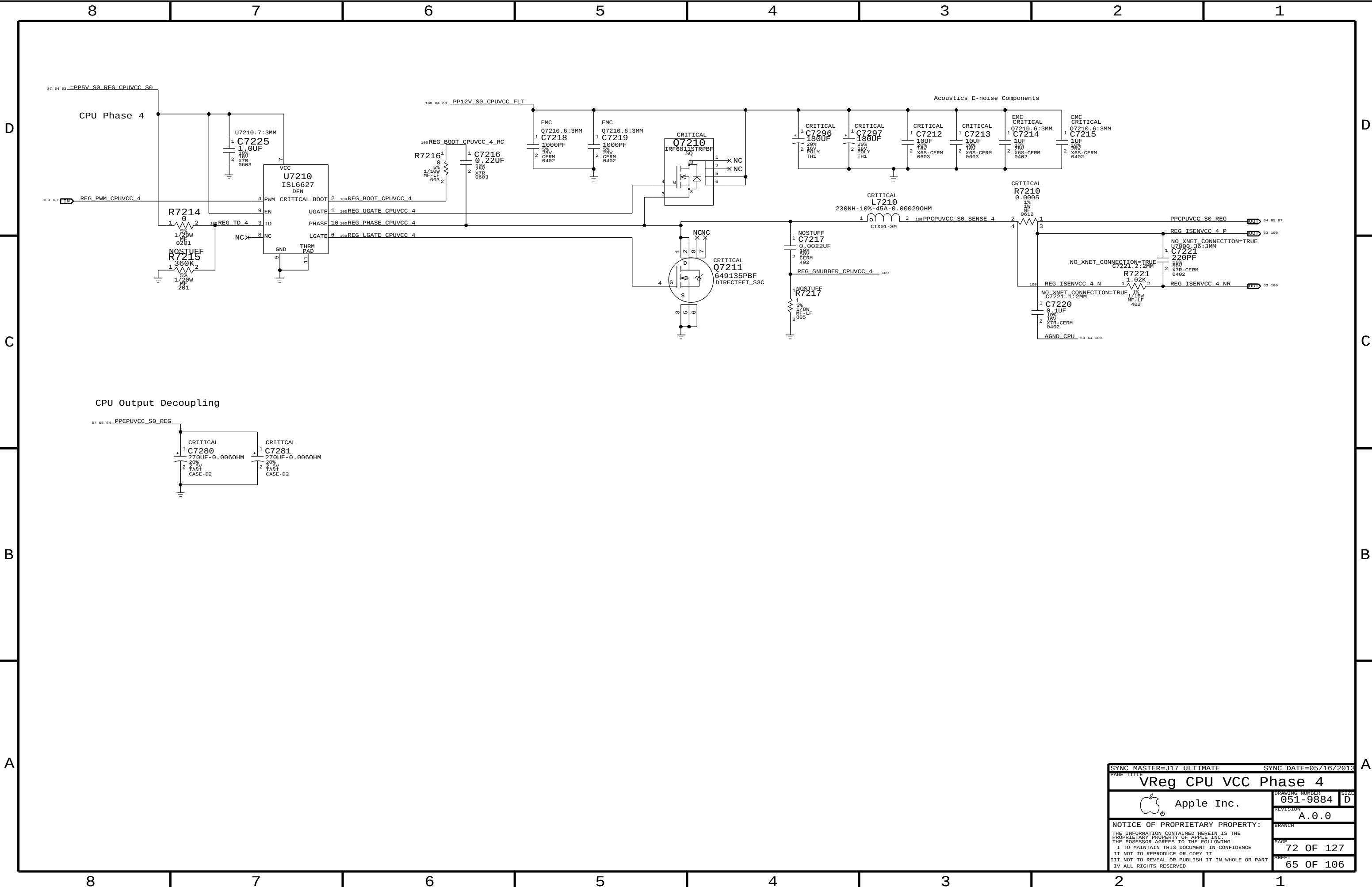
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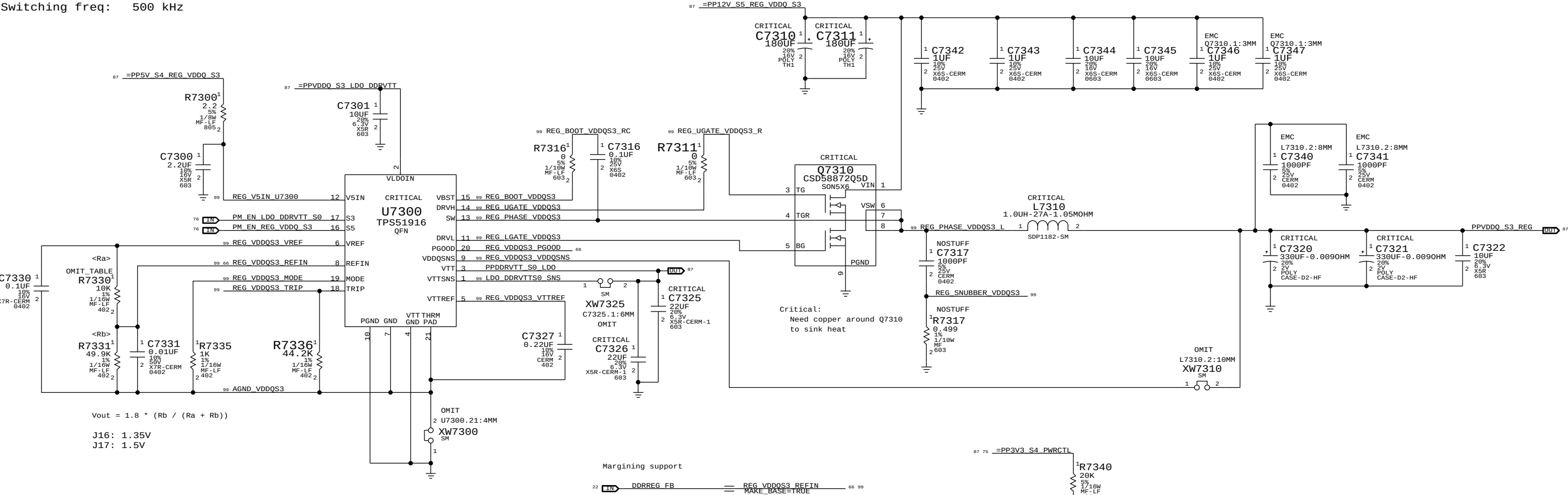


VDDQ (1.5V / 1.35V) S3 Regulator

OC trip point: $30.4 \text{ A VDDQ} = \frac{R7336}{8 \text{ E5} * \text{Rds}(Q7310)} + \frac{0.65625}{L7310 * f(\text{switch})}$

3 A VTT (FIXED)
10 mA VTTREF (FIXED)

Switching freq: 500 kHz



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
11450335	1	RES, 16.5K, 402	R7330	VDDQ:P1V35
11450315	1	RES, 10K, 402	R7330	VDDQ:P1V5

3.3V S5 Regulator

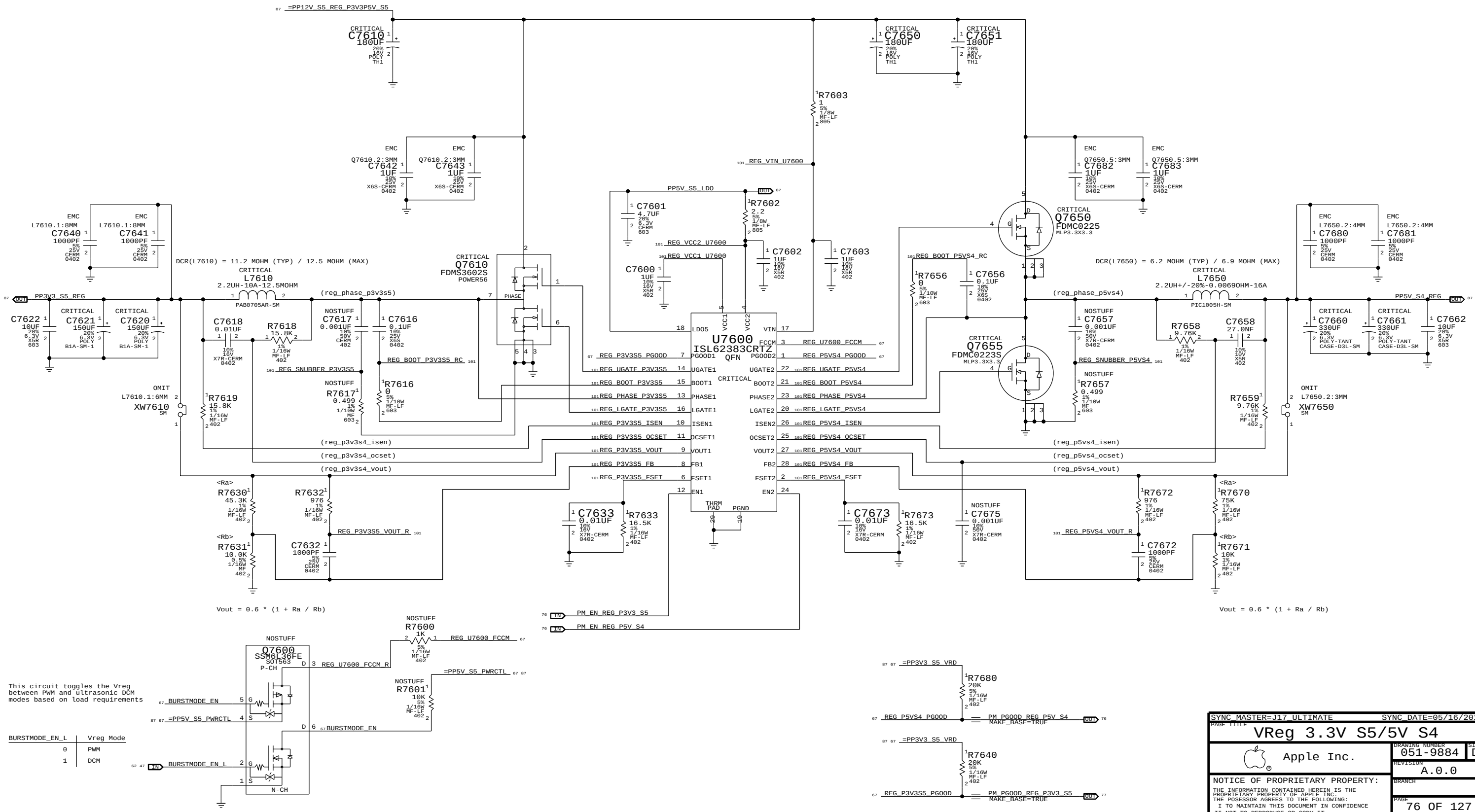
OC trip point: $12.5 \text{ A} = \frac{R7618 * 10 \text{ E-6}}{\text{DCR}(\text{L7610})}$

Switching freq: $356 \text{ kHz} = \frac{1}{170 \text{ E-12} * R7633}$

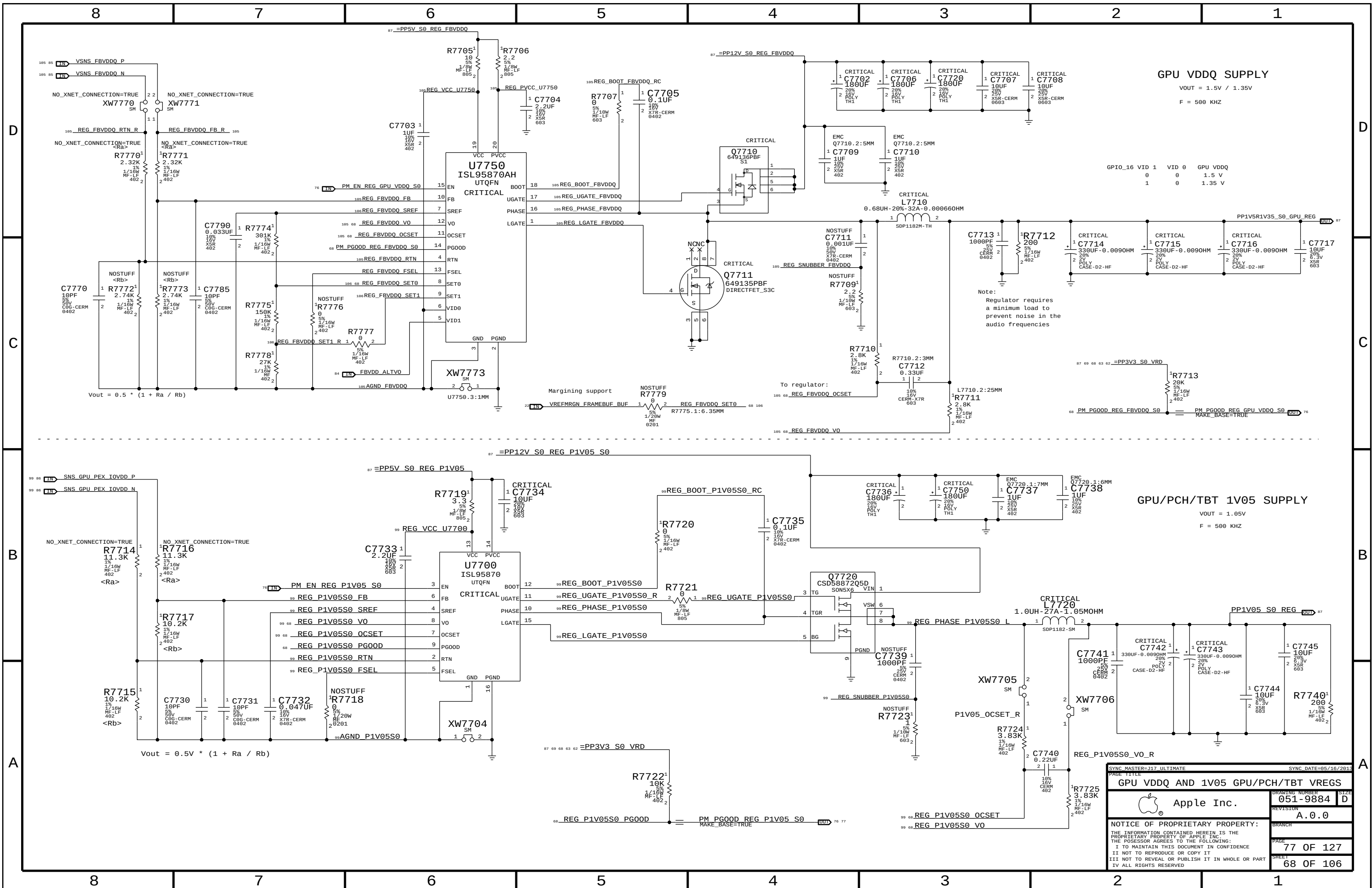
5V S4 Regulator

OC trip point: $14.1 \text{ A} = \frac{R7658 * 10 \text{ E-6}}{\text{DCR}(\text{L7650})}$

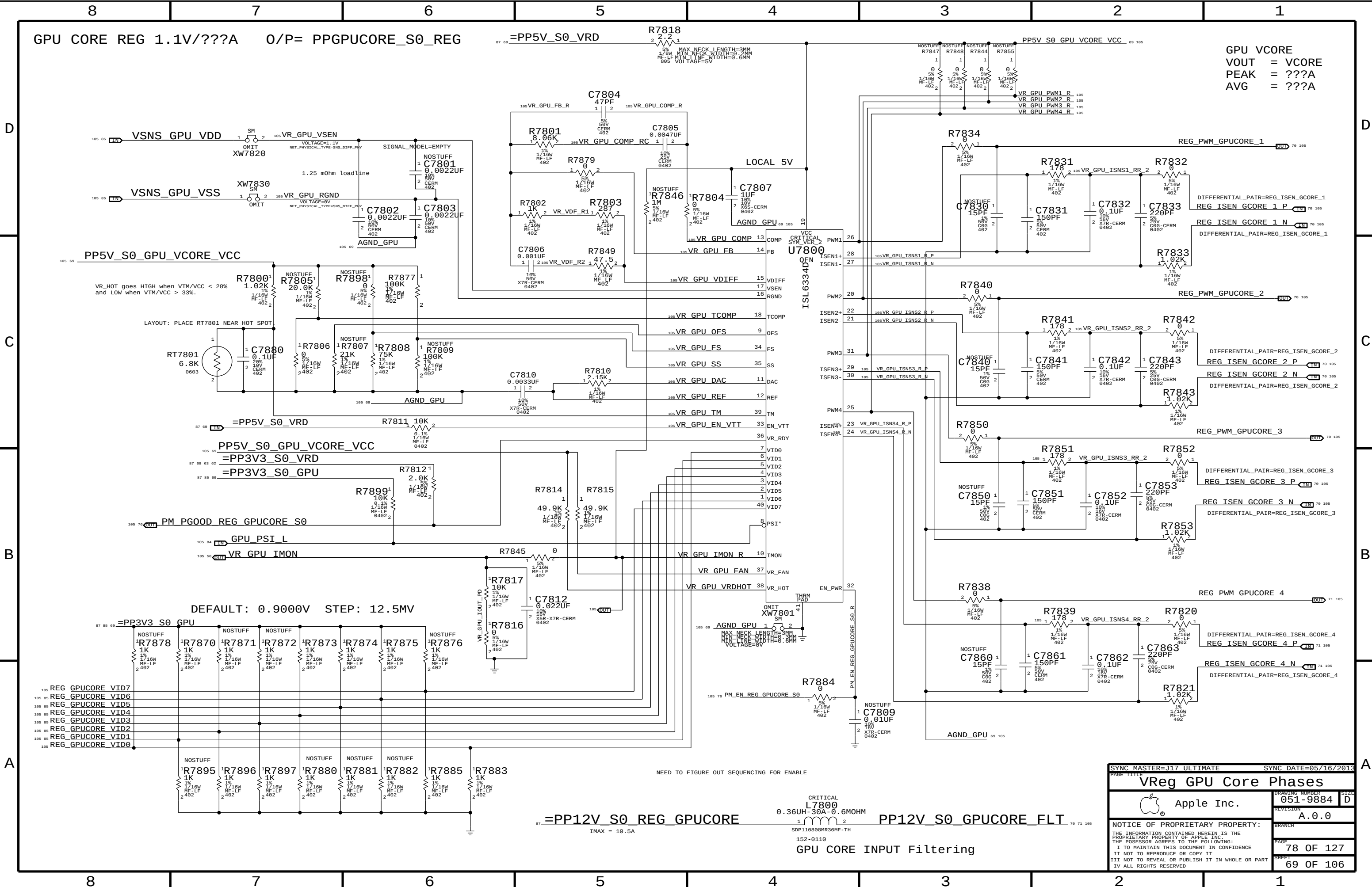
Switching freq: $356 \text{ kHz} = \frac{1}{170 \text{ E-12} * R7673}$



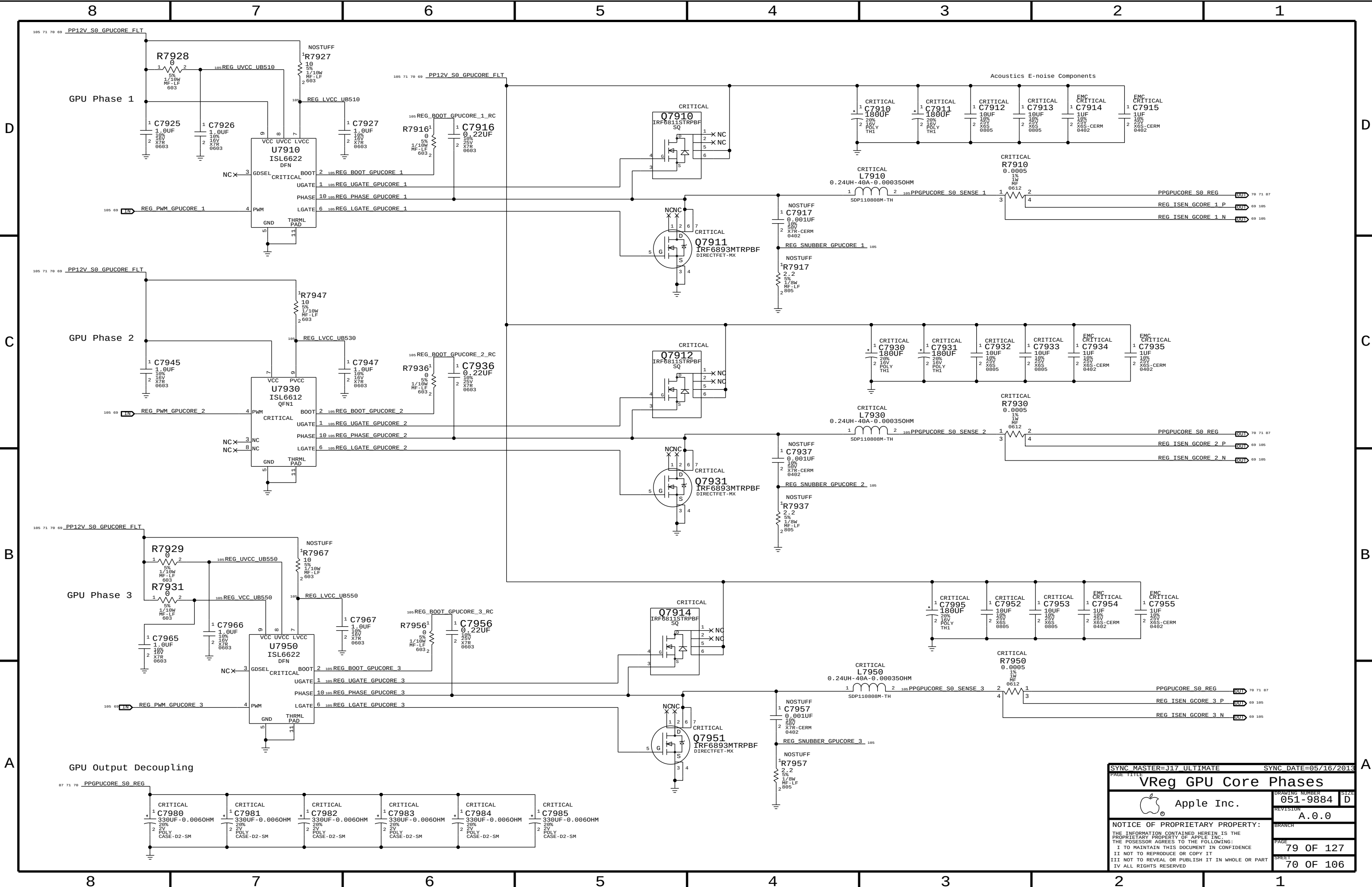
SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/16/2013	
PAGE TITLE		VReg 3.3V S5/5V S4	
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PAGE TITLE		GPU VDDQ AND 1V05 GPU/PCH/TBT VREGS	
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VReg GPU Core Phases		051-9884	
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VReg GPU Core Phases

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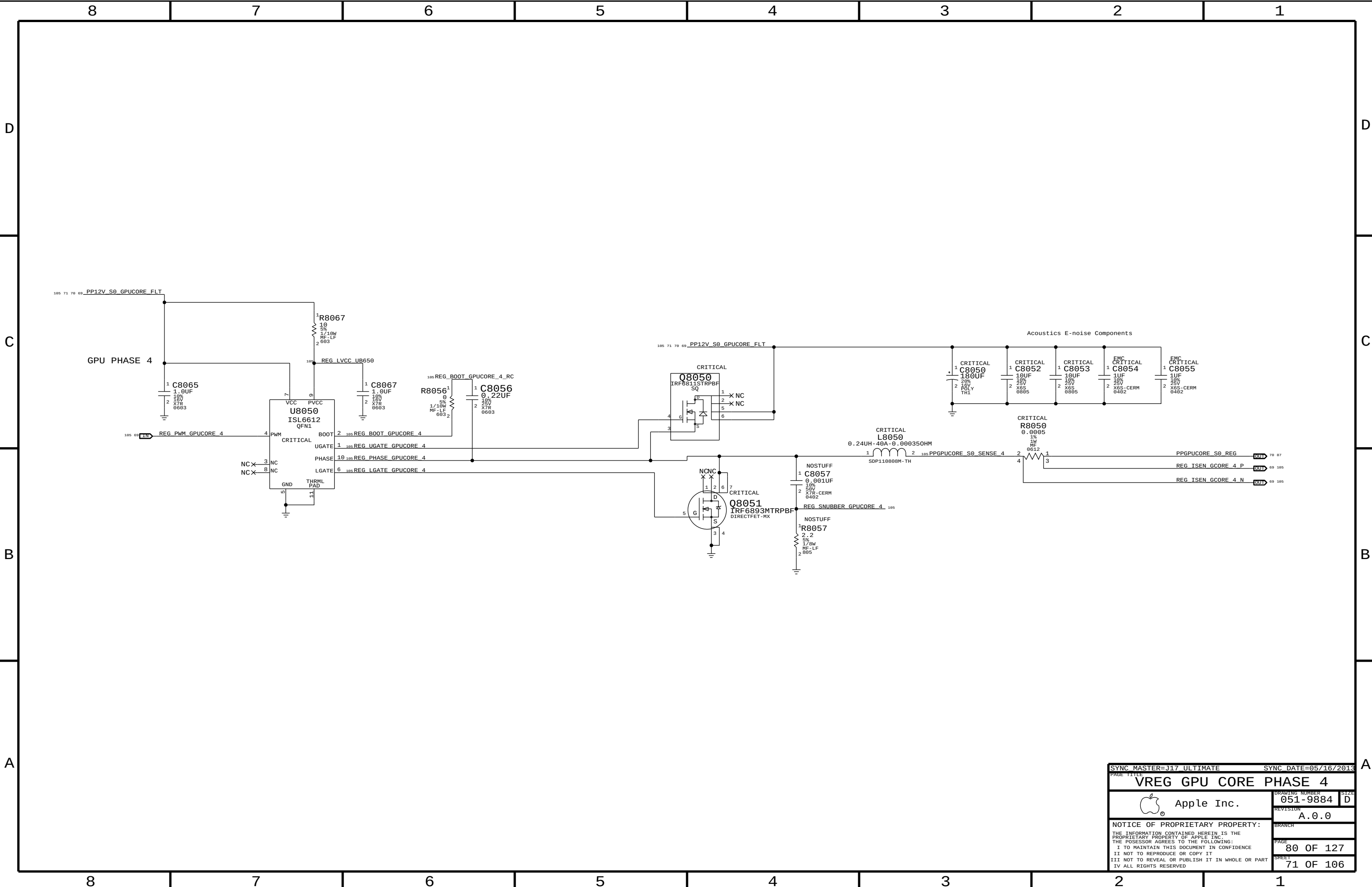
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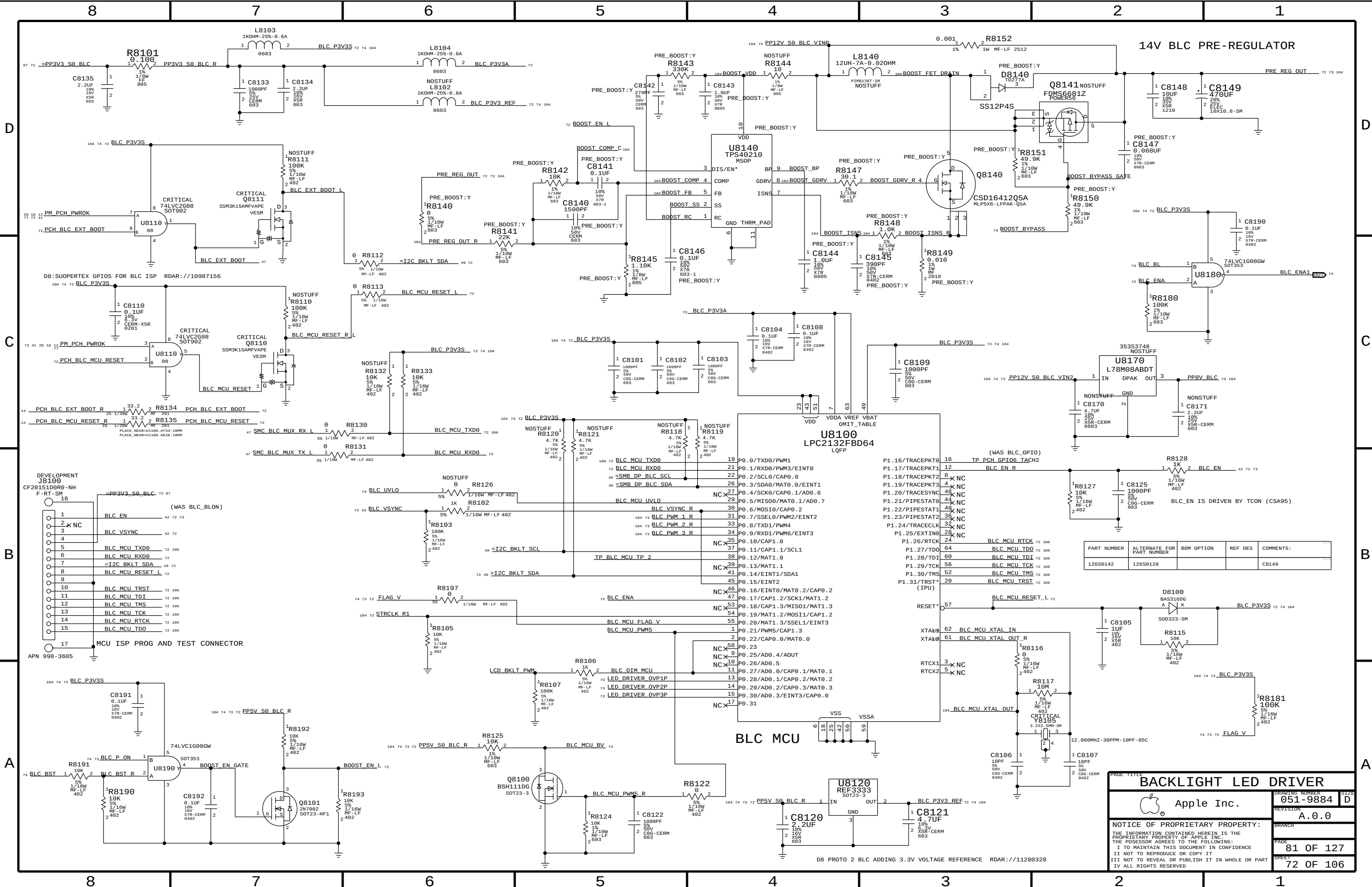
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PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
126S0142	126S0128			C8149

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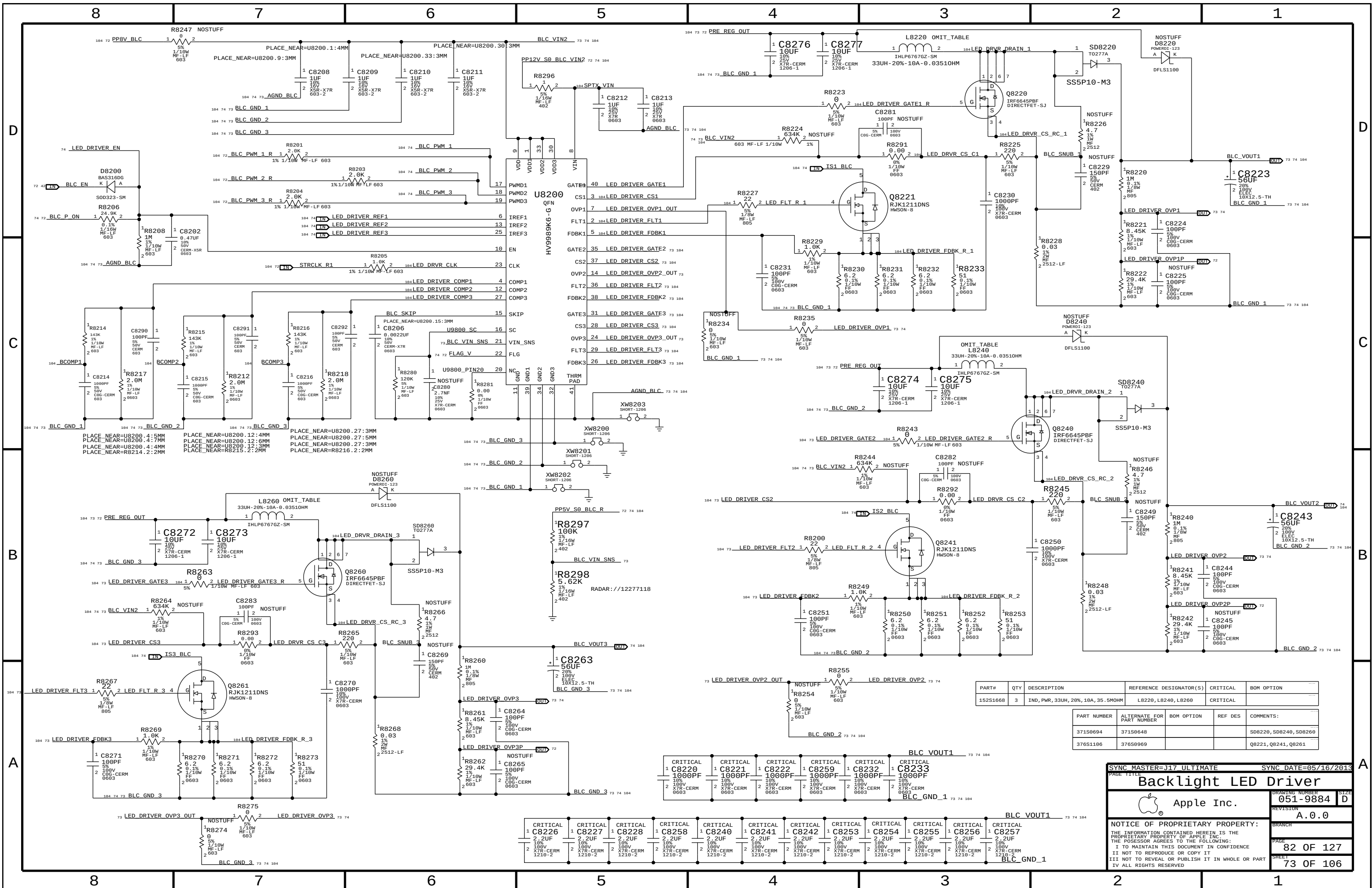
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BLC MCU

14V BLC PRE-REGULATOR



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
152S1668	3	IND, PWR, 33UH, 20%, 10A, 35.5MOHM	L8220, L8240, L8260	CRITICAL	

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
371S0694	371S0648			SD8220, SD8240, SD8260
376S1106	376S0969			Q8221, Q8241, Q8261

SYNC MASTER=J17 ULTIMATE

SYNC DATE=05/16/2013

Backlight LED Driver

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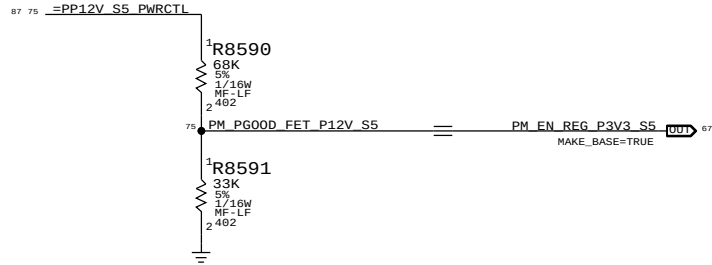
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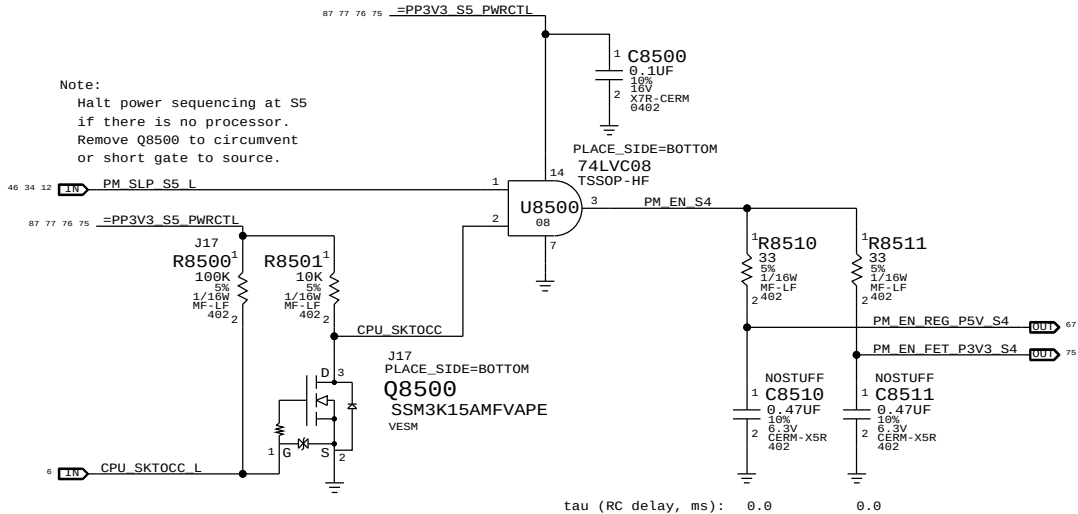
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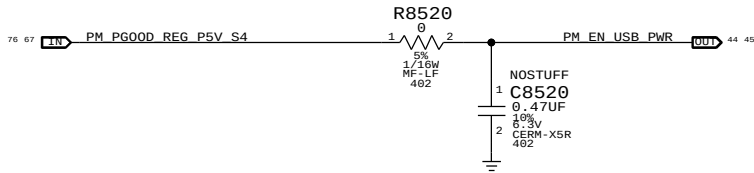
S5 Enable



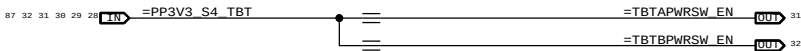
S4 Enables



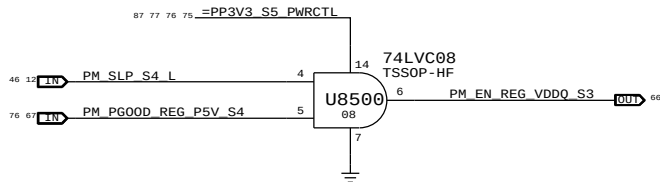
S4 USB Enable



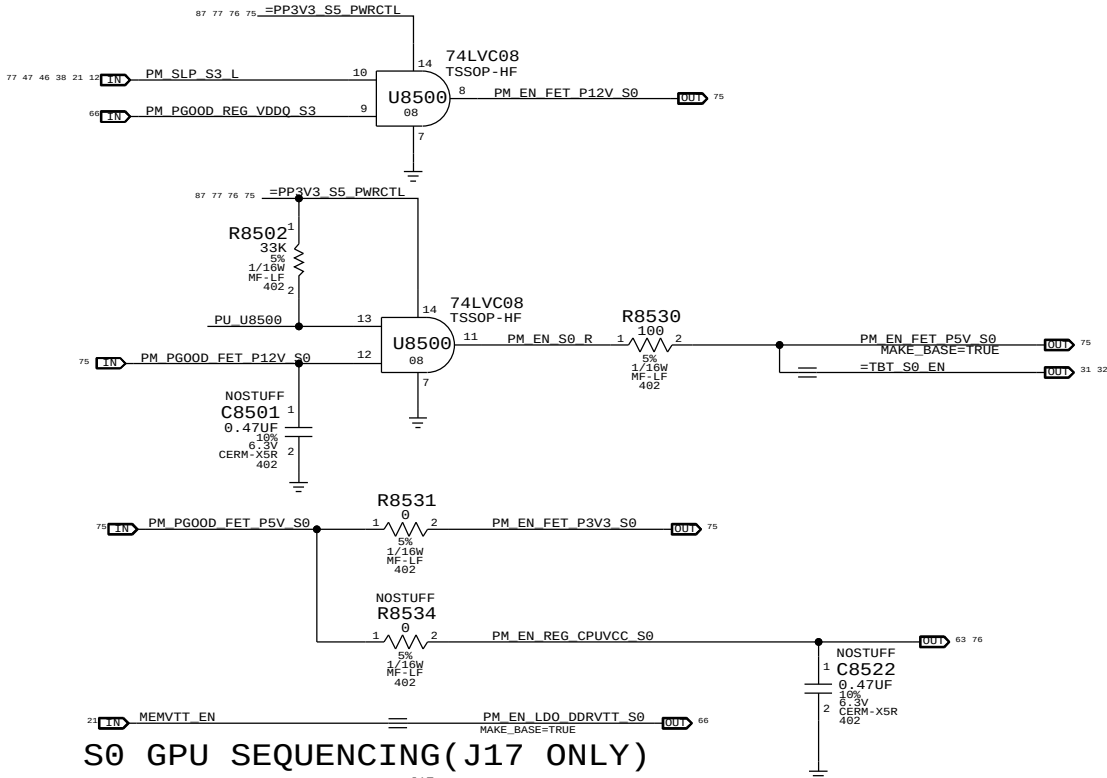
S4 TBT S4 Port Enable



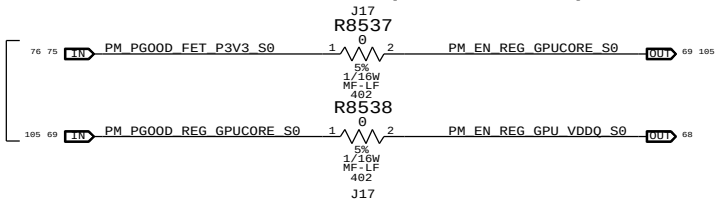
S3 VDDQ Enable



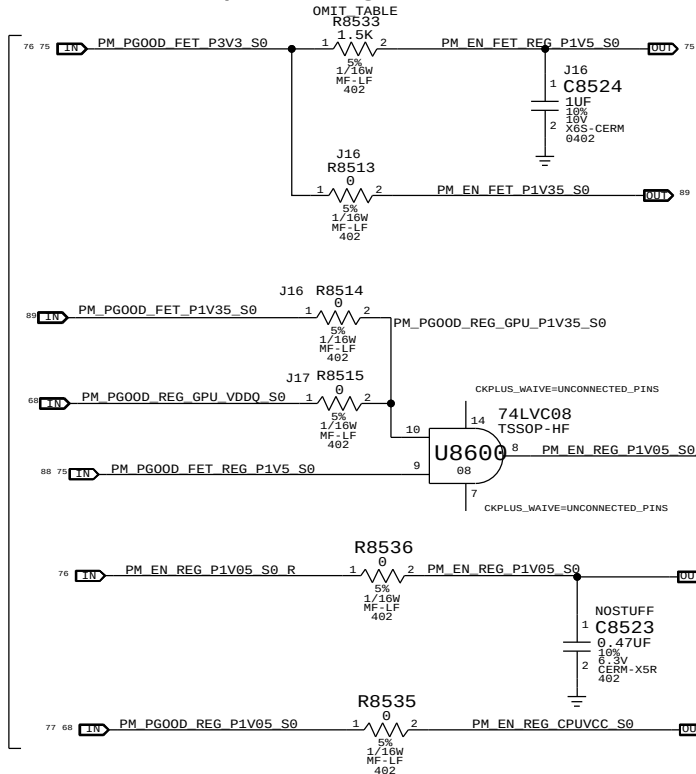
S0 Enables



S0 GPU SEQUENCING(J17 ONLY)



S0 PCH Sequencing



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
116S0070	1	RES,1.5K,0402,5%	R8533	J16
116S0094	1	RES,00HM,0402,5%	R8533	J17

Rail definitions

Platform: All processor non-Core and non-Graphics (5 V, 3.3 V, 1.5 V, 1.05V for PCH/TBT/GPU)
Uncore: VDDQ

Notes on sequencing requirements

Intel:

- No hard specification on platform rails
- SMC guarantees timing on PCH DPWROK and PWROK
- VCC3_3 may power up before VCC, VCC must ramp to 0.6V within 25ms of VCC3V3 ramping to 2.6V
- VCC1_5 may power up before VCC, VCC must ramp to 0.6V within 25ms of VCC1V5 ramping to 1.35V
- VCC may power down before VCC3_3, VCC3_3 must ramp down to 2.6V within 35ms
- VCC may power down before VCC1_5, VCC1_5 must ramp down to 1.35V within 35ms

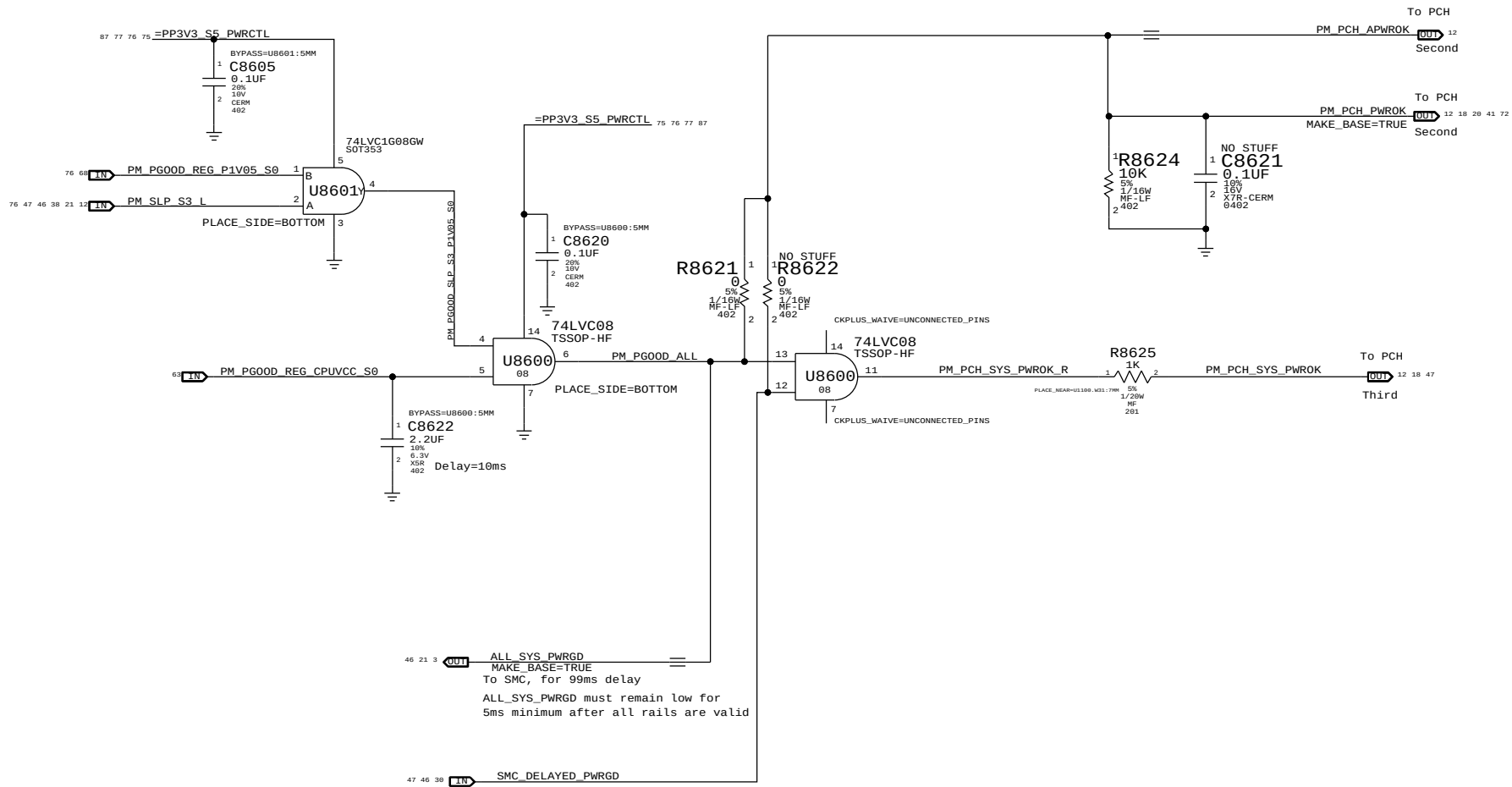
NVIDIA:

- 3V3_S0 must ramp first
- VDDQ MUST RAMP AFTER GPUOP-HF
- PEX_VDD with IFPC/D/E/F_IOVDD (1.05V) must ramp after VDDQ
- All rails must reach their target voltages in more than 40 uS

SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/16/2013	
PAGE TITLE		PM Regulator Enables	
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ALL_SYS_PWRGD, PCH_PWROK & SYS_PWROK Generation

PCH Power Goods

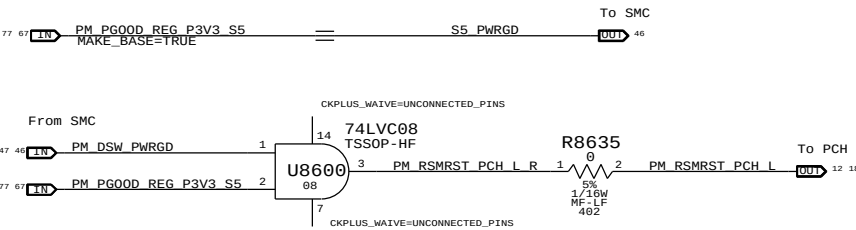


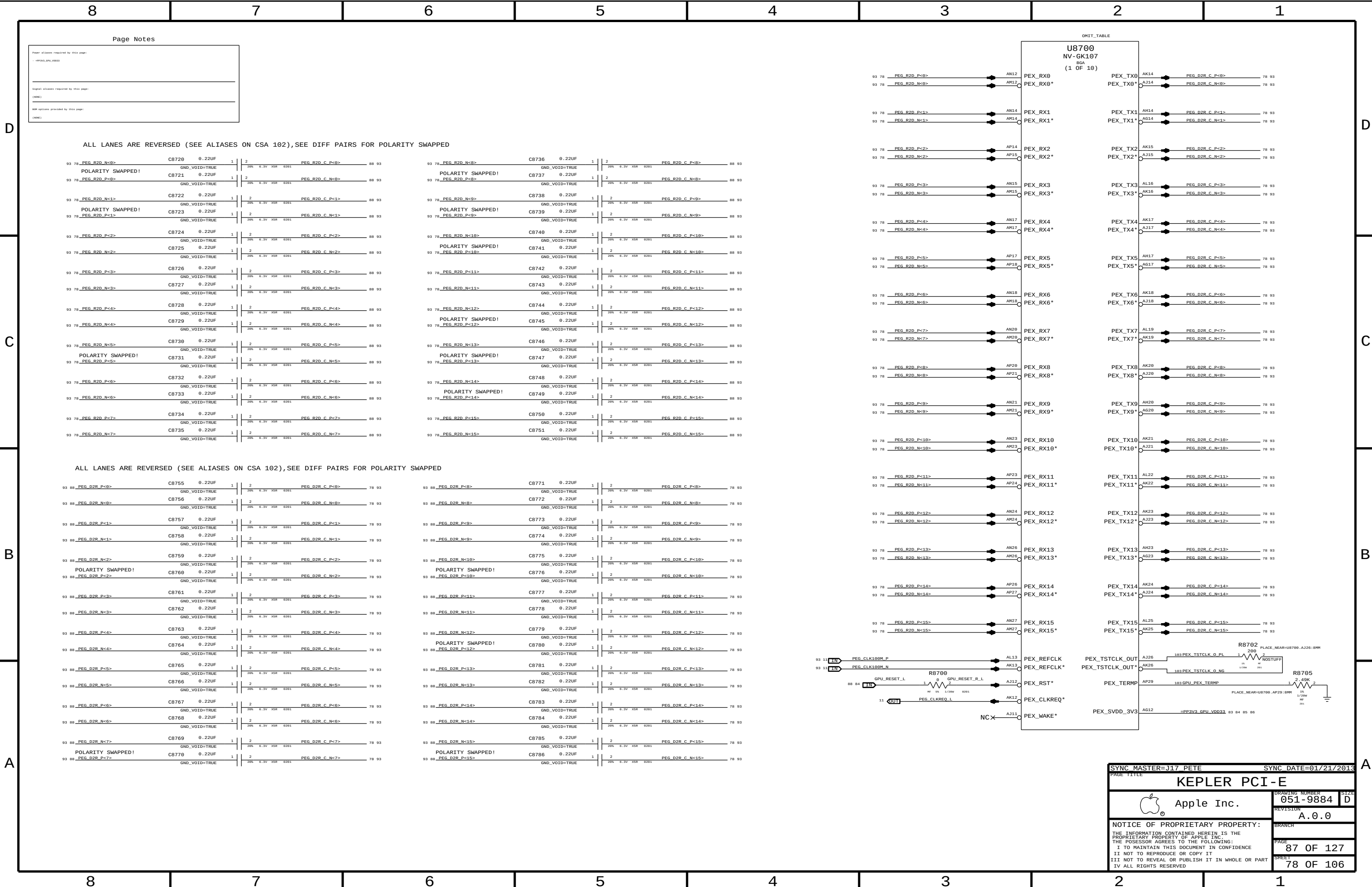
Resume Reset

Intel Doc# 29517 Maho Bay PDG, Section 22.13
Intel Doc# 29562 Panther Point EDS, Section 8.7 and 8.8
Note:
The iMac J16/J17 designs does not support Deep Sx modes so both DPWROK and RSMRST# signals are shorted together

Requirements:
Power on:
Asserted at least 10 ms after all suspend well power is valid
Power off or loss of AC:
Transition to 0.8V or less before VccSUS3_3 drops to 2.90 V
to allow PCH to switch suspend well to battery without excessive loading

Method:
The SMC guarantees proper assertion and de-assertion of RSMRST# for normal operation via PM_DSW_PWRGD.
RSMRST# is asserted when power good from regulator is de-asserted in the event AC is lost. Power good de-assertion should happen quickly enough to meet Intel spec.





Page Notes

Power aliases required by this page:

...PP3V3_GPU_VDD33

Signal aliases required by this page:

(NONE)

ROM options provided by this page:

(NONE)

OMIT_TABLE

U8700

NV-GK107

BGA

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PEX_RX0

PEX_RX0*

PEX_RX1

PEX_RX1*

PEX_RX2

PEX_RX2*

PEX_RX3

PEX_RX3*

PEX_RX4

PEX_RX4*

PEX_RX5

PEX_RX5*

PEX_RX6

PEX_RX6*

PEX_RX7

PEX_RX7*

PEX_RX8

PEX_RX8*

PEX_RX9

PEX_RX9*

PEX_RX10

PEX_RX10*

PEX_RX11

PEX_RX11*

PEX_RX12

PEX_RX12*

PEX_RX13

PEX_RX13*

PEX_RX14

PEX_RX14*

PEX_RX15

PEX_RX15*

PEX_REFCLK

PEX_REFCLK*

PEX_RST*

PEX_CLKREQ*

PEX_WAKE*

PEX_TX0

PEX_TX0*

PEX_TX1

PEX_TX1*

PEX_TX2

PEX_TX2*

PEX_TX3

PEX_TX3*

PEX_TX4

PEX_TX4*

PEX_TX5

PEX_TX5*

PEX_TX6

PEX_TX6*

PEX_TX7

PEX_TX7*

PEX_TX8

PEX_TX8*

PEX_TX9

PEX_TX9*

PEX_TX10

PEX_TX10*

PEX_TX11

PEX_TX11*

PEX_TX12

PEX_TX12*

PEX_TX13

PEX_TX13*

PEX_TX14

PEX_TX14*

PEX_TX15

PEX_TX15*

PEX_TSTCLK_OUT

PEX_TSTCLK_OUT*

PEX_TERM

PEX_SVDD_3V3

SYNC MASTER=J17 PETE

SYNC DATE=01/21/2013

PAGE TITLE

KEPLER PCI-E

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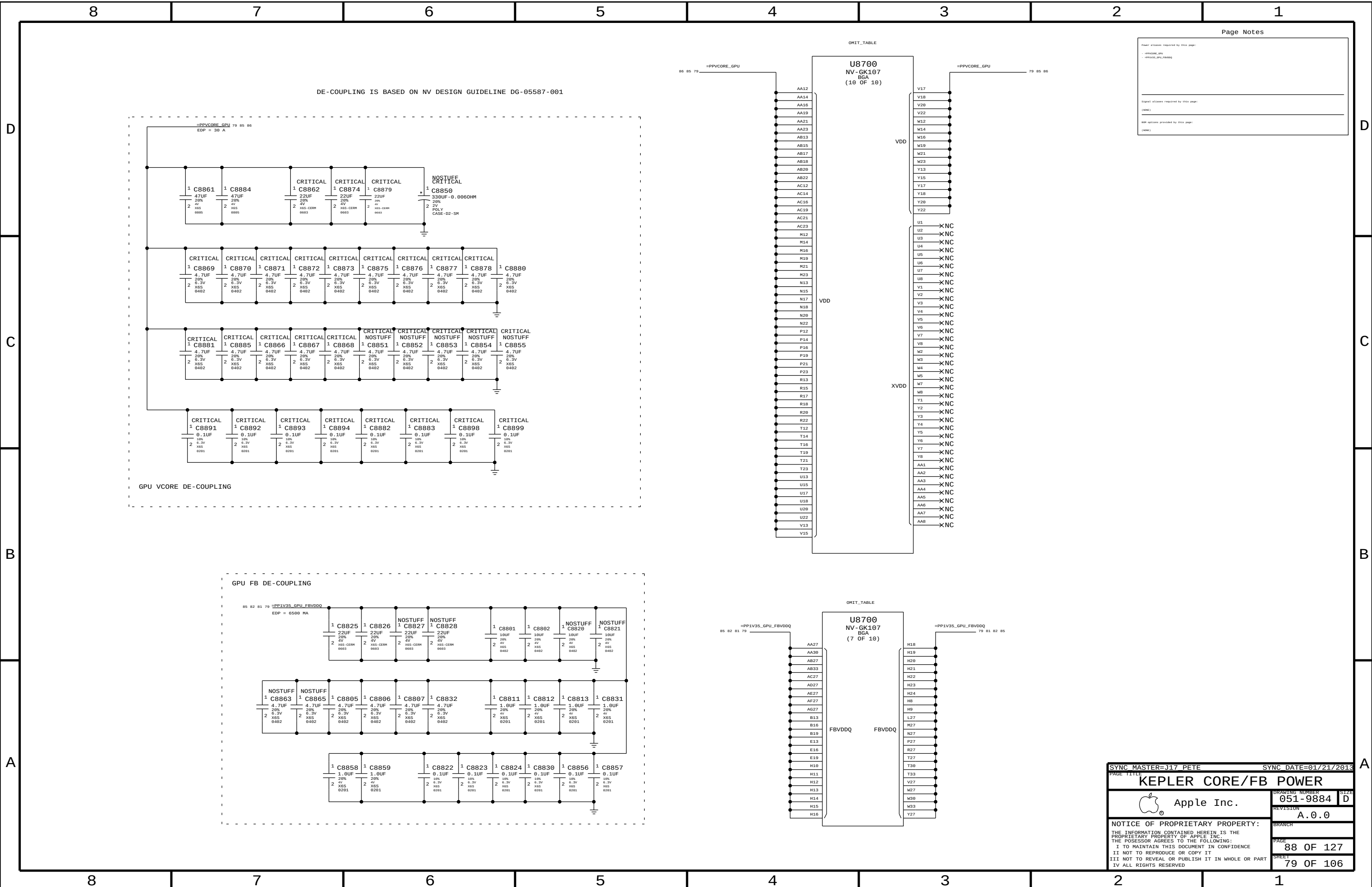
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D

C

B

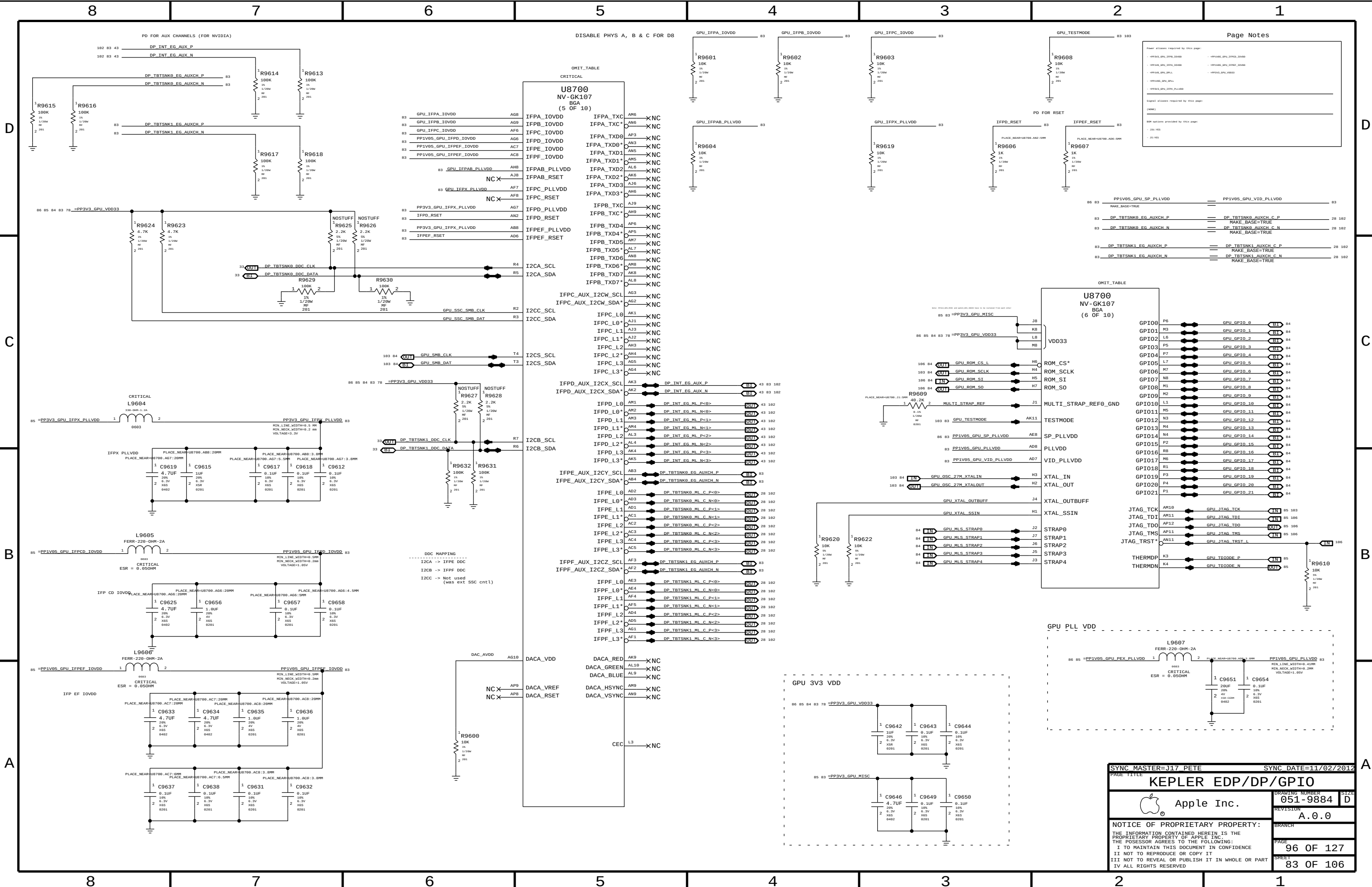
A

D

C

B

A



Page Notes

Power allows required by this page:

Signal allows required by this page:

SW options provided by this page:

U8700 NV-GK107 BGA (6 OF 10)

GPIO0	P6	GPU_GPIO_0	B1	84
GPIO1	M3	GPU_GPIO_1	B1	84
GPIO2	L6	GPU_GPIO_2	B1	84
GPIO3	P5	GPU_GPIO_3	B1	84
GPIO4	P7	GPU_GPIO_4	B1	84
GPIO5	L7	GPU_GPIO_5	B1	84
GPIO6	M7	GPU_GPIO_6	B1	84
GPIO7	N8	GPU_GPIO_7	B1	84
GPIO8	M1	GPU_GPIO_8	B1	84
GPIO9	M2	GPU_GPIO_9	B1	84
GPIO10	L1	GPU_GPIO_10	B1	84
GPIO11	M5	GPU_GPIO_11	B1	84
GPIO12	N3	GPU_GPIO_12	B1	84
GPIO13	M4	GPU_GPIO_13	B1	84
GPIO14	N4	GPU_GPIO_14	B1	84
GPIO15	P2	GPU_GPIO_15	B1	84
GPIO16	R8	GPU_GPIO_16	B1	84
GPIO17	M6	GPU_GPIO_17	B1	84
GPIO18	R1	GPU_GPIO_18	B1	84
GPIO19	P3	GPU_GPIO_19	B1	84
GPIO20	P4	GPU_GPIO_20	B1	84
GPIO21	P1	GPU_GPIO_21	B1	84
JTAG_TCK	AM10	GPU_JTAG_TCK	AN	85 103
JTAG_TDI	AP12	GPU_JTAG_TDI	AN	85 106
JTAG_TDO	AP11	GPU_JTAG_TDO	AN	85 106
JTAG_TMS	AN11	GPU_JTAG_TMS	AN	85 106
JTAG_TRST*	AN11	GPU_JTAG_TRST_L	AN	106
THERMDP	K3	GPU_TDIODE_P	AN	85
THERMDN	K4	GPU_TDIODE_N	AN	85

GPU PLL VDD

L9607 FERR-220-OHM-2A

CRITICAL ESR = 0.050OHM

PP1V05_GPU_PEX_PLLVDD

PP1V05_GPU_PLLVDD

C9651

C9654

GPU 3V3 VDD

PP3V3_GPU_VDD33

C9642

C9643

C9644

C9646

C9649

C9650

SYNC MASTER=J17 PETE SYNC DATE=11/02/2012

KEPLER EDP/DP/GPIO

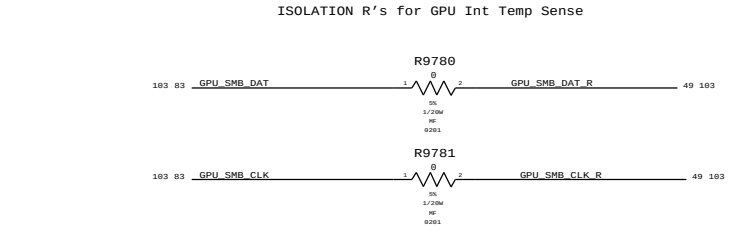
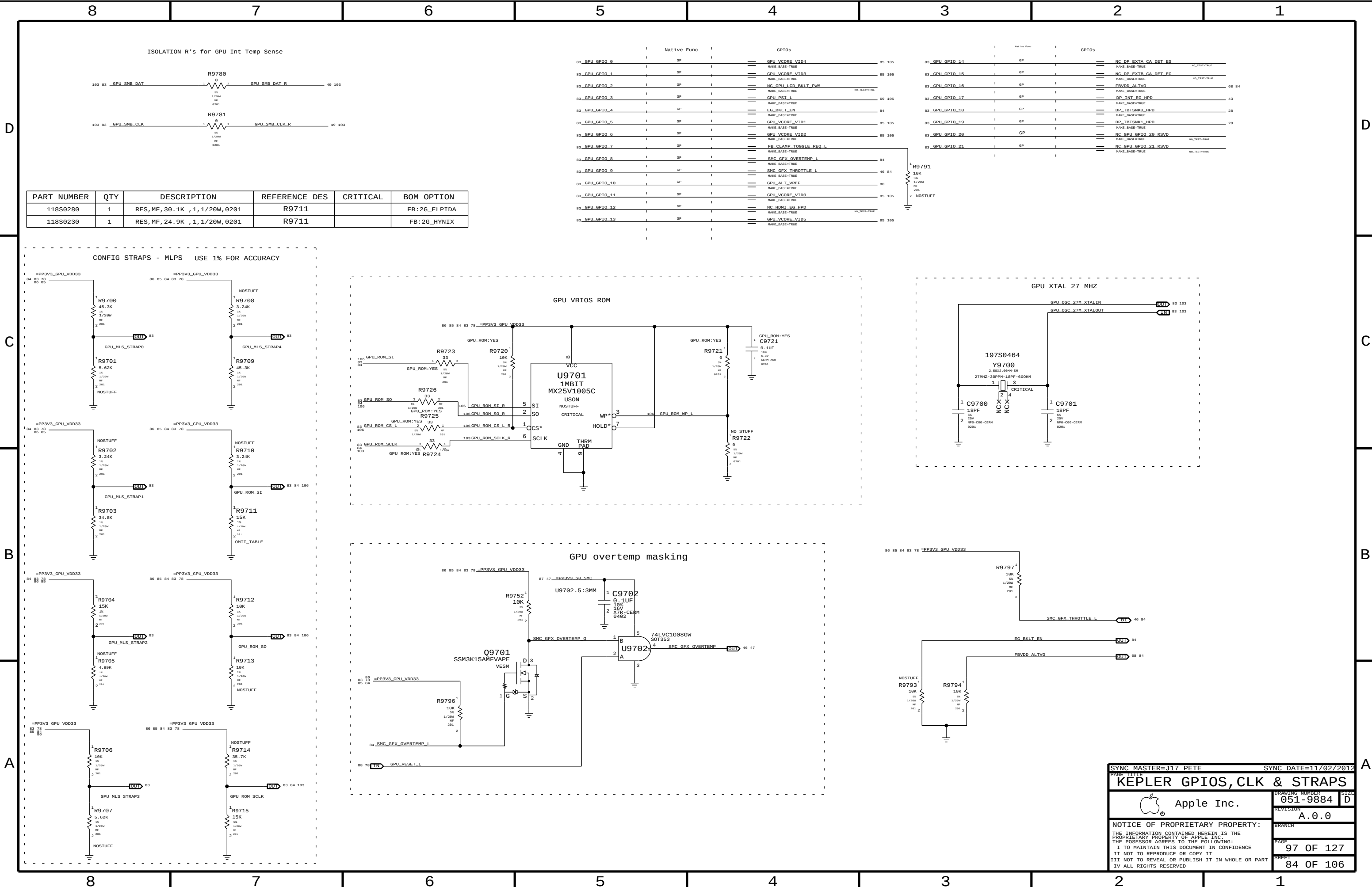
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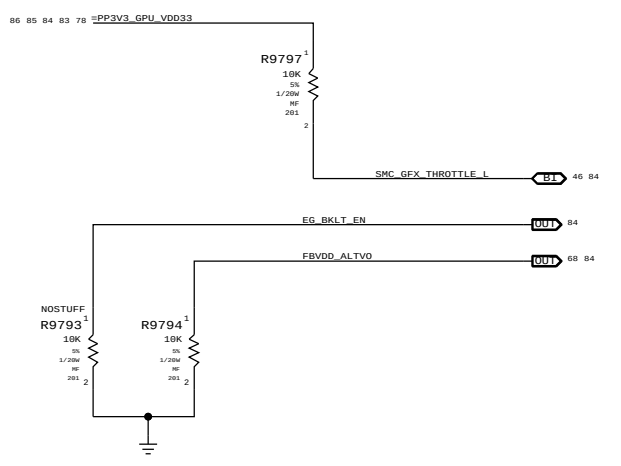
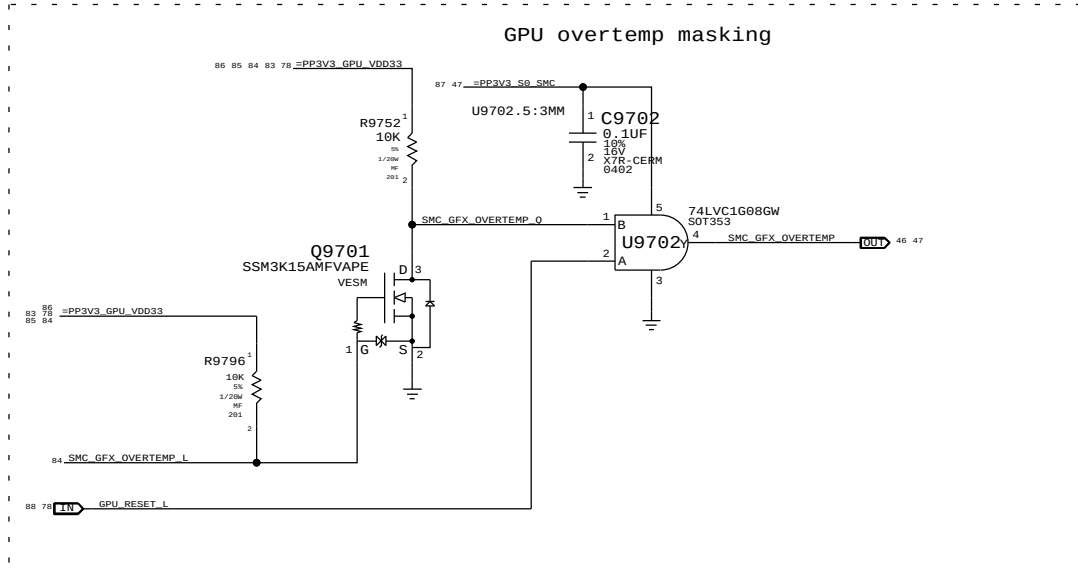
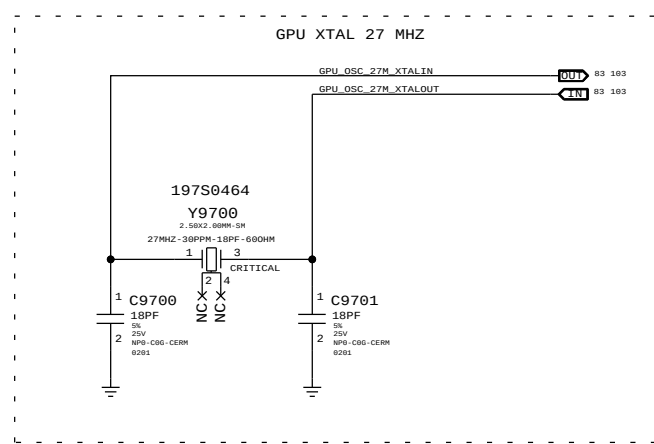
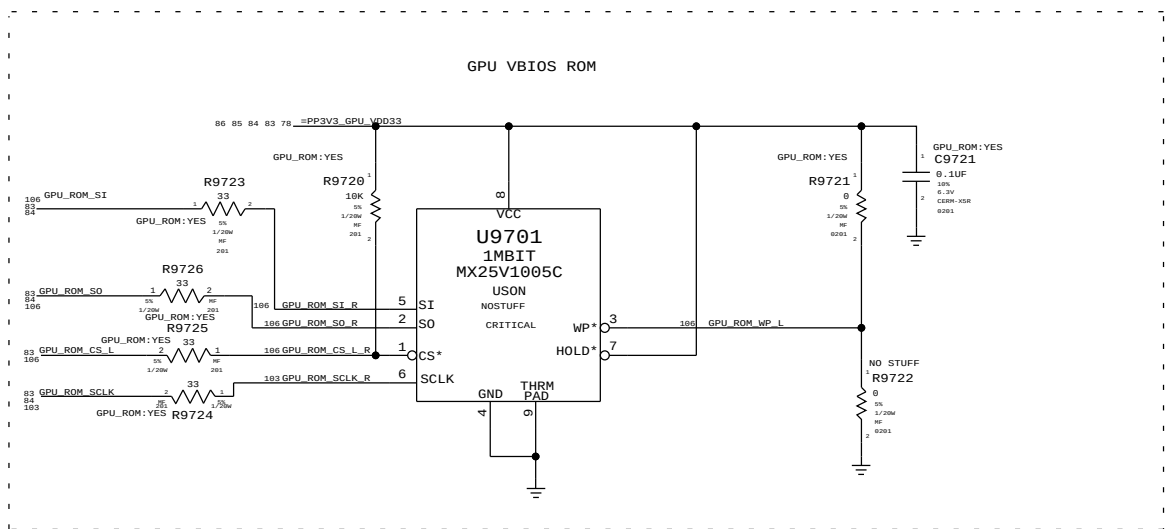
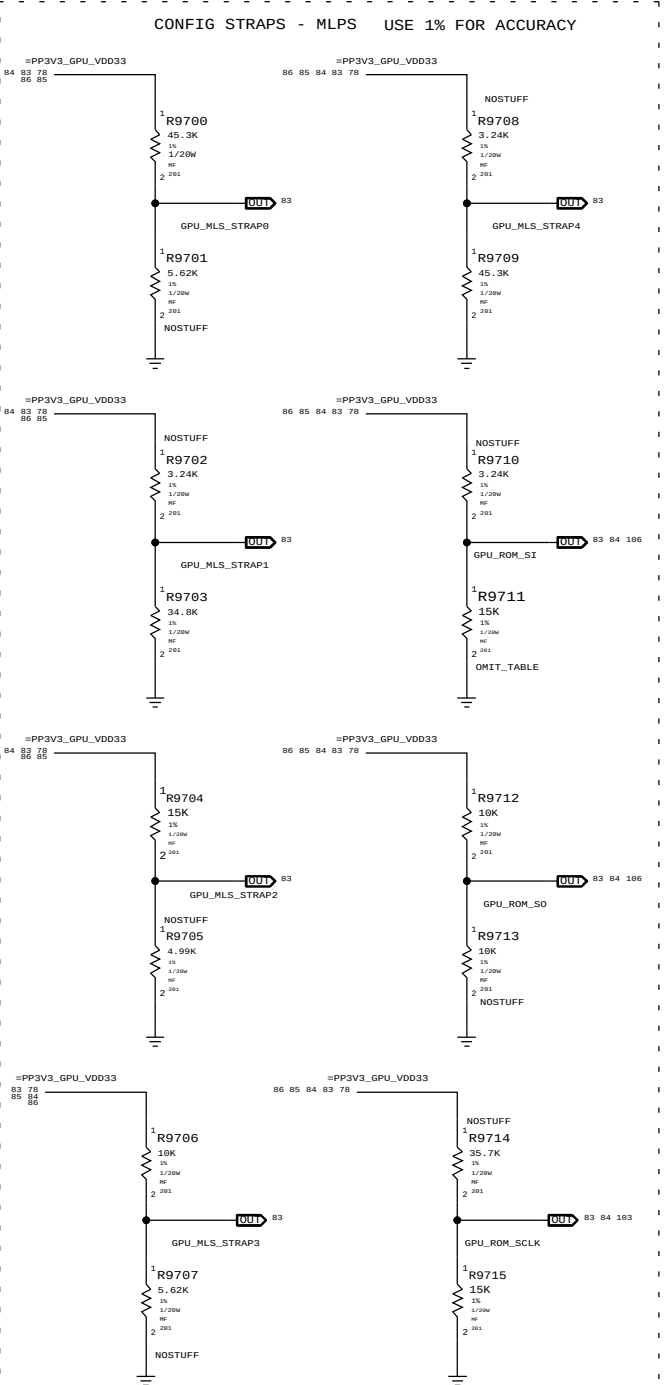
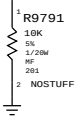
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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
118S0280	1	RES,MF,30.1K ,1,1/20W,0201	R9711		FB:2G_ELPIDA
118S0230	1	RES,MF,24.9K ,1,1/20W,0201	R9711		FB:2G_HYNIX

	Native Func		GPIOs	
83_GPU_GPIO_0	GP		GPU_VCORE_VID4	85 105
83_GPU_GPIO_1	GP		MAKE_BASE=TRUE	
83_GPU_GPIO_2	GP		GPU_VCORE_VID3	85 105
83_GPU_GPIO_3	GP		MAKE_BASE=TRUE	
83_GPU_GPIO_4	GP		NC_GPU_LCD_BKLT_PWM	NO_TEST+TRUE
83_GPU_GPIO_5	GP		GPU_PSI_L	69 105
83_GPU_GPIO_6	GP		EG_BKLT_EN	84
83_GPU_GPIO_7	GP		MAKE_BASE=TRUE	
83_GPU_GPIO_8	GP		GPU_VCORE_VID1	85 105
83_GPU_GPIO_9	GP		GPU_VCORE_VID2	85 105
83_GPU_GPIO_10	GP		FB_CLAMP_TOGGLE_REQ_L	
83_GPU_GPIO_11	GP		SMC_GFX_OVERTEMP_L	84
83_GPU_GPIO_12	GP		SMC_GFX_THROTTLE_L	46 84
83_GPU_GPIO_13	GP		GPU_ALT_VREF	80
			MAKE_BASE=TRUE	
			GPU_VCORE_VID0	85 105
			MAKE_BASE=TRUE	
			NC_HDMI_EG_HPD	NO_TEST+TRUE
			MAKE_BASE=TRUE	
			GPU_VCORE_VID5	85 105
			MAKE_BASE=TRUE	

	Native Func		GPIOs	
83_GPU_GPIO_14	GP		NC_DP_EXTN_CA_DET_EG	
83_GPU_GPIO_15	GP		MAKE_BASE=TRUE	
83_GPU_GPIO_16	GP		NC_DP_EXTN_CA_DET_EG	
83_GPU_GPIO_17	GP		MAKE_BASE=TRUE	
83_GPU_GPIO_18	GP		FBVDD_ALTVDD	68 84
83_GPU_GPIO_19	GP		MAKE_BASE=TRUE	
83_GPU_GPIO_20	GP		DP_INT_EG_HPD	43
83_GPU_GPIO_21	GP		MAKE_BASE=TRUE	
			DP_TBTSNK0_HPD	28
			MAKE_BASE=TRUE	
			DP_TBTSNK1_HPD	28
			MAKE_BASE=TRUE	
			NC_GPU_GPIO_28_RSVD	
			MAKE_BASE=TRUE	
			NC_GPU_GPIO_21_RSVD	
			MAKE_BASE=TRUE	



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KEPLER GPIOs,CLK & STRAPS

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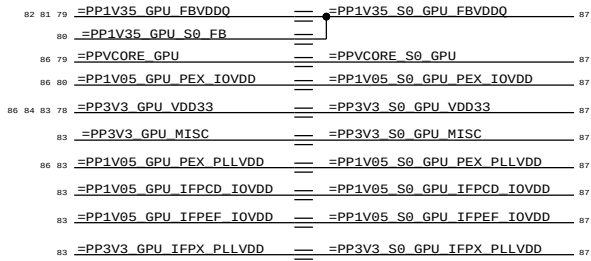
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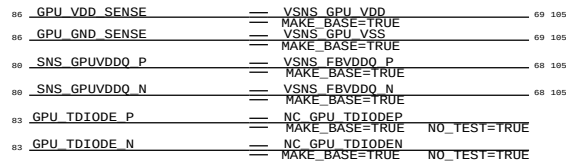
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GPU POWER ALIAS

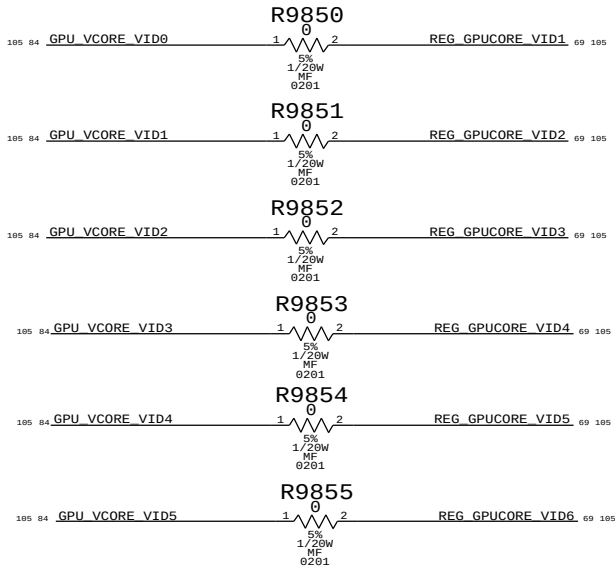


GPU SIGNAL & SENSE ALIAS



C

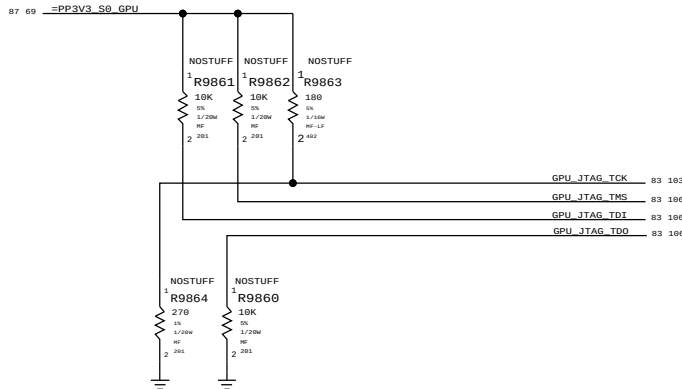
GPU VIDS ALIAS (VR VID0 IS TIED LOW)



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B

PU/PD IS BASED ON RECOMMENDATION FROM NVIDIA FOR THEIR GPU JTAG DEBUGGER

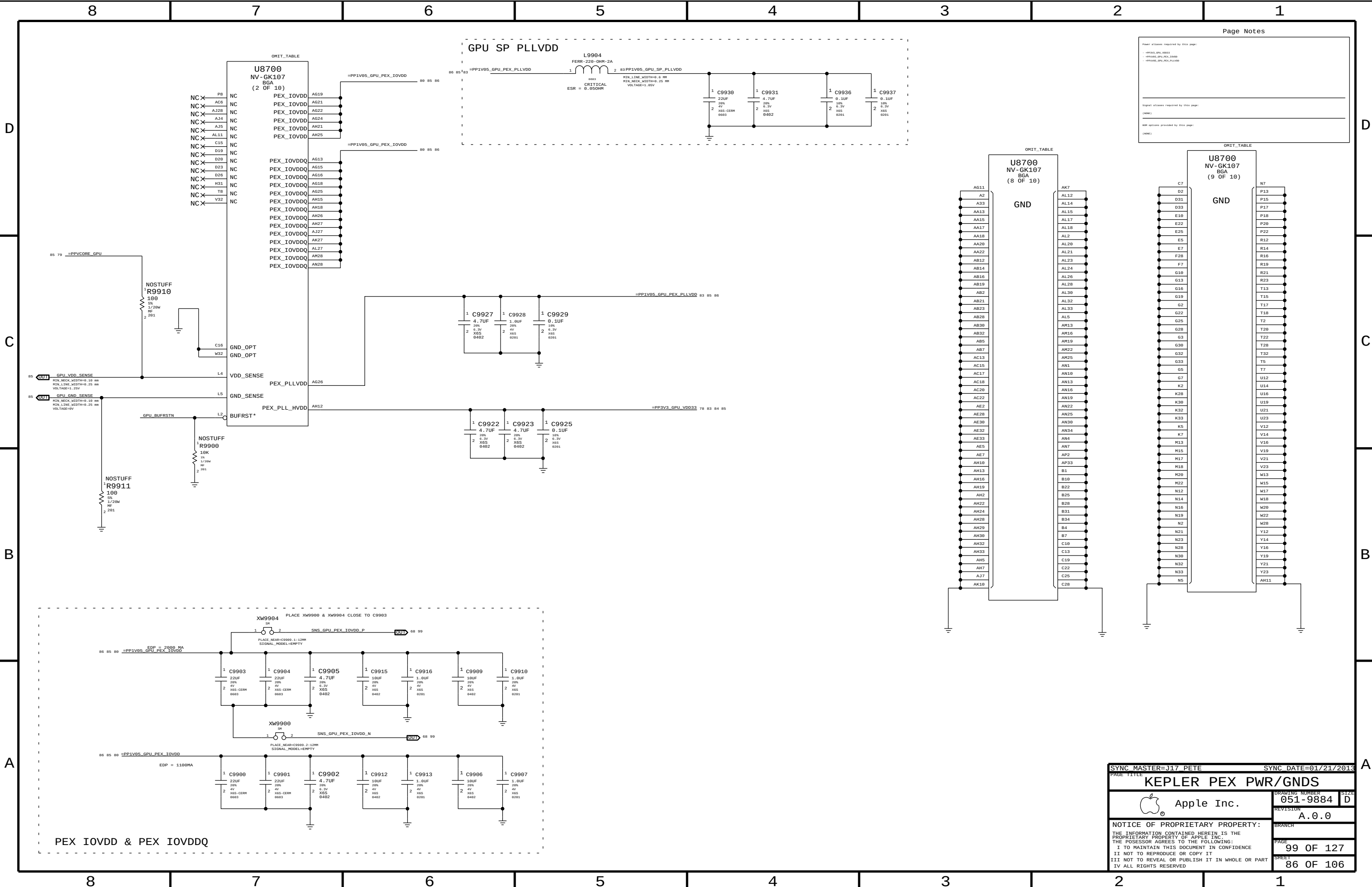


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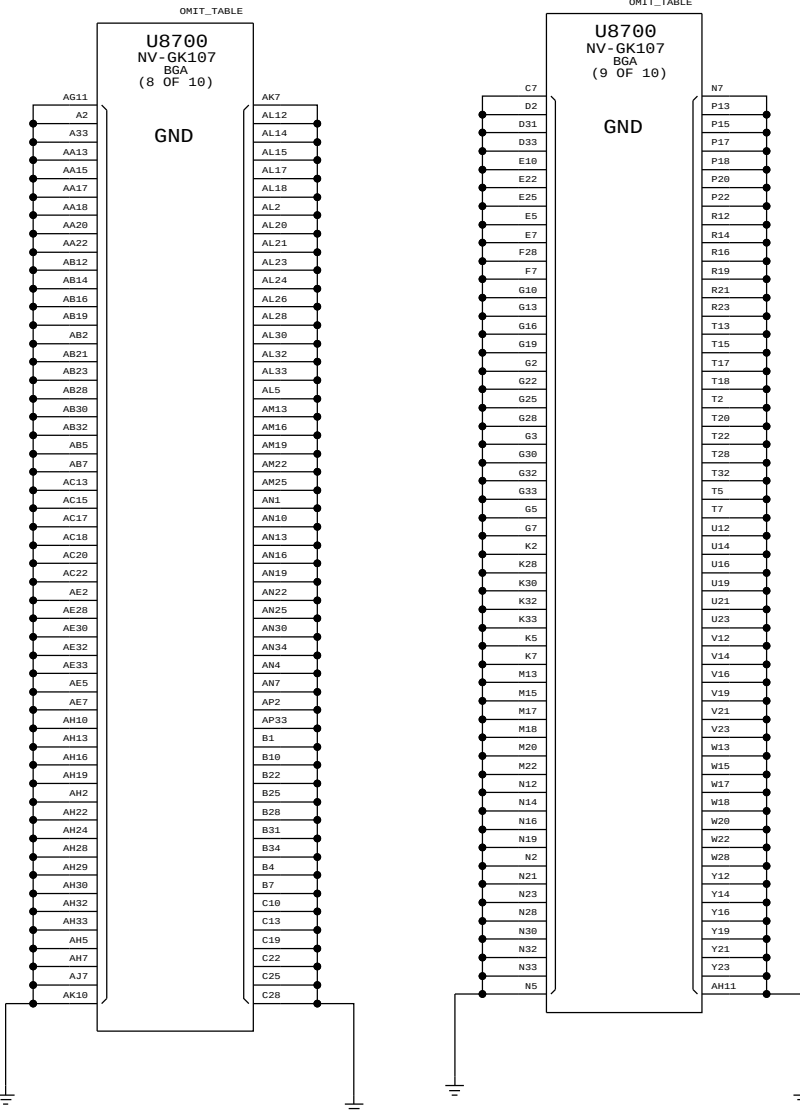
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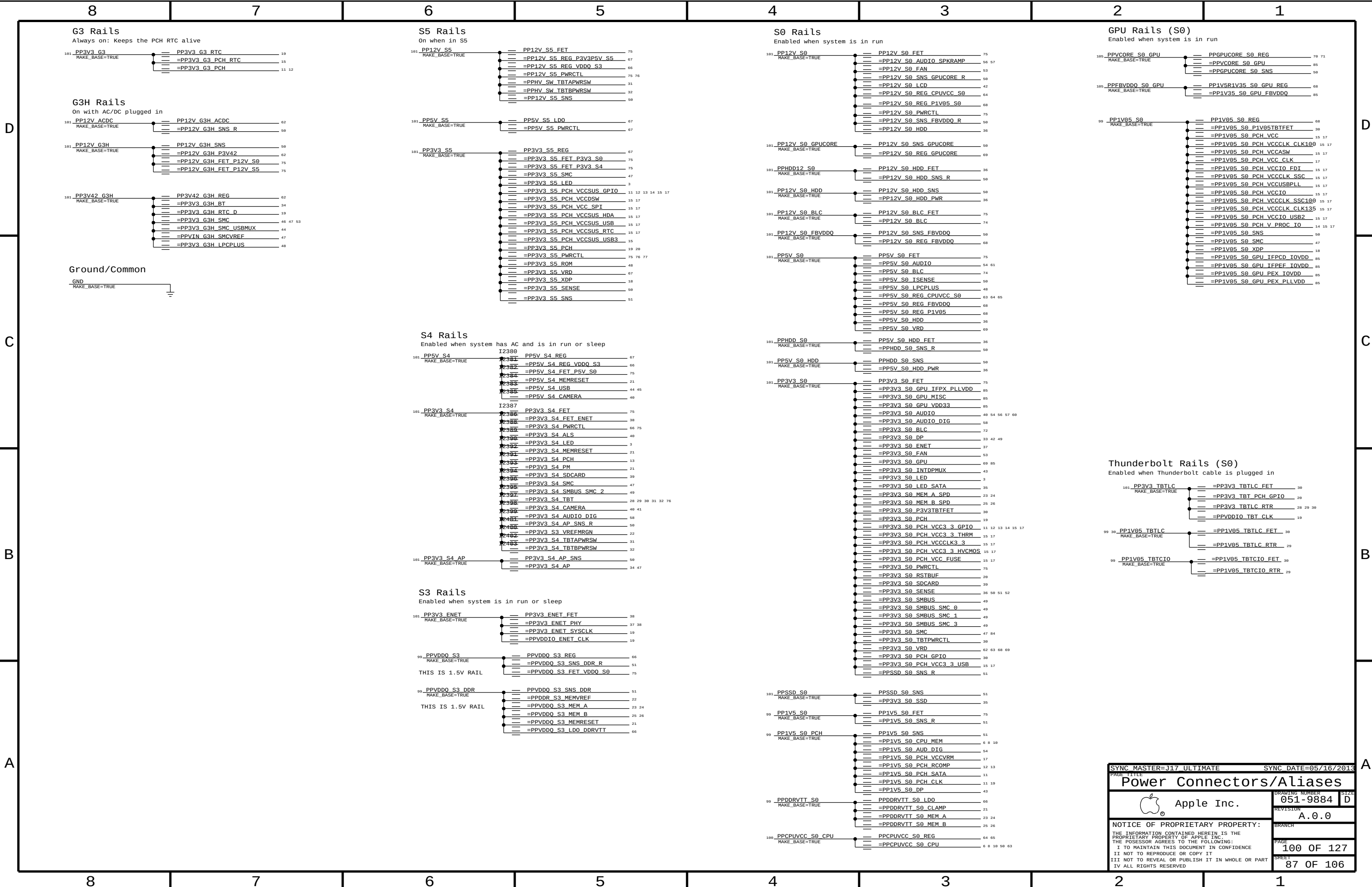
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Page Notes	
Power aliases required by this page:	
- PP1V05_GPU_VDDQ1	
- PP1V05_GPU_PEX_PLLVDD	
- PP1V05_GPU_PEX_PLLVDD	
Signal aliases required by this page:	
(NONE)	
SOP options provided by this page:	
(NONE)	



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Power Connectors/Aliases

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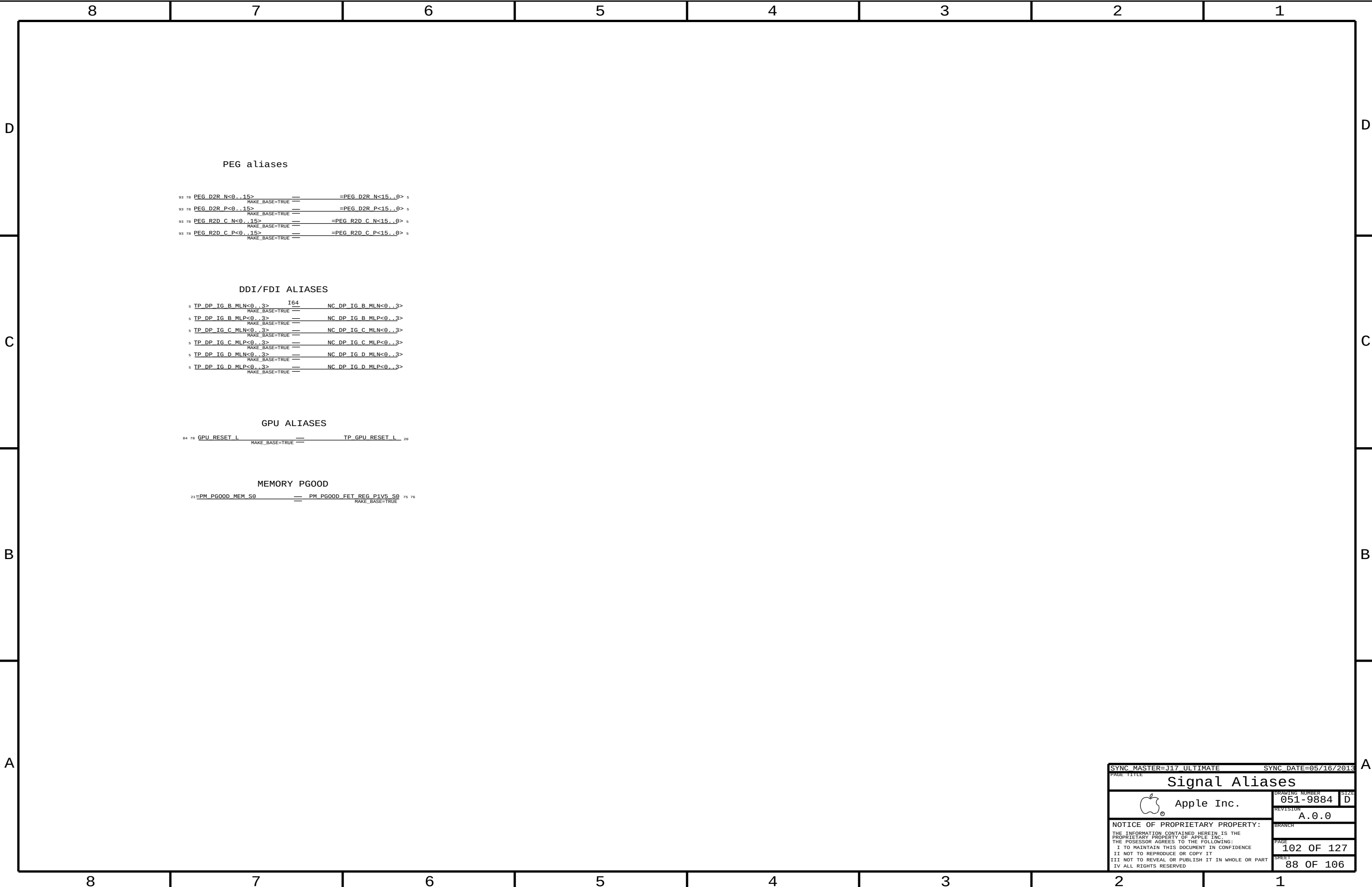
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PEG aliases

93 78 PEG_D2R_N<0..15> == =PEG_D2R_N<15..0> 5
MAKE_BASE=TRUE ==
93 78 PEG_D2R_P<0..15> == =PEG_D2R_P<15..0> 5
MAKE_BASE=TRUE ==
93 78 PEG_R2D_C_N<0..15> == =PEG_R2D_C_N<15..0> 5
MAKE_BASE=TRUE ==
93 78 PEG_R2D_C_P<0..15> == =PEG_R2D_C_P<15..0> 5
MAKE_BASE=TRUE ==

DDI/FDI ALIASES

5 TP_DP_IG_B_MLN<0..3> I64 == NC_DP_IG_B_MLN<0..3>
MAKE_BASE=TRUE ==
5 TP_DP_IG_B_MLP<0..3> == NC_DP_IG_B_MLP<0..3>
MAKE_BASE=TRUE ==
5 TP_DP_IG_C_MLN<0..3> == NC_DP_IG_C_MLN<0..3>
MAKE_BASE=TRUE ==
5 TP_DP_IG_C_MLP<0..3> == NC_DP_IG_C_MLP<0..3>
MAKE_BASE=TRUE ==
5 TP_DP_IG_D_MLN<0..3> == NC_DP_IG_D_MLN<0..3>
MAKE_BASE=TRUE ==
5 TP_DP_IG_D_MLP<0..3> == NC_DP_IG_D_MLP<0..3>
MAKE_BASE=TRUE ==

GPU ALIASES

84 78 GPU_RESET_L == TP_GPU_RESET_L 20
MAKE_BASE=TRUE ==

MEMORY PG00D

21 ==PM_PG00D_MEM_S0 == PM_PG00D_FET_REG_P1V5_S0 75 76
MAKE_BASE=TRUE ==

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Signal Aliases

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C							C
B							B
A							A
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Functional / ICT Test

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J17 BOARD SPECIFIC PHYSICAL AND SPACING CONSTRAINTS

BOARD LAYERS	BOARD AREAS	BOARD UNITS (MIL or MM)	ALLEGRO VERSION
TOP, ISL2, ISL3, ISL4, ISL5, ISL6, ISL7, ISL8, ISL9, ISL10, ISL11, BOTTOM	NO_TYPE, BGA, BGA_TBT	MM	16.2

General Physical Rule Definitions

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	0.1 MM	=50_OHM_SE	10 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	10 MM	=DEFAULT	=DEFAULT

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
34_OHM_SE	*	Y	0.185 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
34_OHM_SE	ISL5, ISL8	Y	0.205 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
34_OHM_SE	TOP, BOTTOM	Y	0.220 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
39_OHM_SE	*	Y	0.150 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
39_OHM_SE	ISL5, ISL8	Y	0.165 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
39_OHM_SE	TOP, BOTTOM	Y	0.175 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
42_OHM_SE	*	Y	0.130 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
42_OHM_SE	ISL5, ISL8	Y	0.145 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
42_OHM_SE	TOP, BOTTOM	Y	0.155 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
45_OHM_SE	*	Y	0.115 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
45_OHM_SE	ISL5, ISL8	Y	0.126 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
45_OHM_SE	TOP, BOTTOM	Y	0.135 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	*	Y	0.090 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
50_OHM_SE	ISL5, ISL8	Y	0.100 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
50_OHM_SE	TOP, BOTTOM	Y	0.105 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
55_0HM_SE	*	Y	0.075 MM	0.075 MM	=STANDARD	=STANDARD	=STANDARD
55_0HM_SE	ISL5, ISL8	Y	0.080 MM	0.080 MM	=STANDARD	=STANDARD	=STANDARD
55_0HM_SE	TOP, BOTTOM	Y	0.085 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
68_OHM_DIFF	*	Y	0.171 MM	0.085 MM	=STANDARD	0.130 MM	0.1 MM
68_OHM_DIFF	TOP,BOTTOM	Y	0.185 MM	0.085 MM	=STANDARD	0.150 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
80_OHM_DIFF	*	Y	0.136 MM	0.085 MM	=STANDARD	0.190 MM	0.1 MM
80_OHM_DIFF	TOP,BOTTOM	Y	0.141 MM	0.085 MM	=STANDARD	0.185 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
85_OHM_DIFF	*	Y	0.121 MM	0.085 MM	=STANDARD	0.190 MM	0.1 MM
85_OHM_DIFF	TOP,BOTTOM	Y	0.125 MM	0.085 MM	=STANDARD	0.190 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	*	Y	0.109 MM	0.085 MM	=STANDARD	0.200 MM	0.1 MM
90_OHM_DIFF	TOP,BOTTOM	Y	0.111 MM	0.085 MM	=STANDARD	0.200 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_OHM_DIFF	*	Y	0.086 MM	0.085 MM	=STANDARD	0.200 MM	0.1 MM
100_OHM_DIFF	TOP,BOTTOM	Y	0.090 MM	0.085 MM	=STANDARD	0.230 MM	0.1 MM

General Spacing Definitions

Default

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?

Fixed and Dielectric

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1:1_SPACING	*	0.1 MM	?
1X_DIELECTRIC	TOP,BOTTOM	0.071 MM	?
1X_DIELECTRIC	ISL3,ISL10	0.101 MM	?
1X_DIELECTRIC	*	0.076 MM	?

Board Stack-up

FINISHED BOARD THICKNESS: 1.94 MM

Layer	Material	Thickness
Top	Signal	0.5 oz (Cu plated)
2	Prepreg	0.071 MM
	Plane	1 oz
3	Core	0.101 MM
	Signal	0.5 oz
4	Prepreg	0.115 MM
	Plane	1 oz
5	Core	0.076 MM
	Signal	0.5 oz
6	Prepreg	0.380 MM
	Plane	1 oz
7	Core	0.076 MM
	Plane	1 oz
8	Prepreg	0.380 MM
	Signal	0.5 oz
9	Core	0.076 MM
	Plane	1 oz
10	Prepreg	0.115 MM
	Signal	0.5 oz
11	Core	0.101 MM
	Plane	1 oz
Btm	Prepreg	0.071 MM
	Signal	0.5 oz (Cu plated)

BGA

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
BGA_P1MM	*	=STANDARD	?

Power and Common

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND_ISO	*	=STANDARD	8000
GND_P2MM	*	=2:1_SPACING	1000
PWR_P2MM	*	=2:1_SPACING	1100

GENERIC

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GENERIC_ISO	*	=1:1_SPACING	?

BGA Area Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA	BGA_P1MM

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PCI Express/DMI

PCIe-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALL ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCIE_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
PCIE_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF
PCIE_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
PCIE_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF
PCIE_COMP	*	Y	0.305 MM	0.105 MM	=STANDARD	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
PCIE3_PHY	*	PCIE_80D
CLK_PCIE_PHY	*	PCIE_90D
COMP_PCIE_PHY	*	PCIE_COMP

PCIe and DMI Compensation Rules (mils)

Table	Imp	Design	Iso	Design	Comments
4-5	50	50	15	15.75	PCIe. Impedance inferred from Table 4-7.
4-7	50	50	8	15.75	DMI. Numbers based on Intel stack-up.

PCIe-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCIE_ISO	*	=5:1_SPACING	?
COMP_PCIE_ISO	*	=4:1_SPACING	?

Spacing Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCIE	*	*	CLK_PCIE_ISO
COMP_PCIE	*	*	COMP_PCIE_ISO
PEG_R2D	PEG_R2D	*	PEG_SAME_DIR
PEG_D2R	PEG_D2R	*	PEG_SAME_DIR
PEG_D2R	PEG_D2R	*	PEG_ALT_DIR
PEG_D2R	*	*	PEG_ISO
PEG_R2D	*	*	PEG_ISO

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PEG_SAME_DIR	TOP,BOTTOM	=5X_DIELECTRIC	?
PEG_SAME_DIR	*	=3.5X_DIELECTRIC	?
PEG_ALT_DIR	*	=7X_DIELECTRIC	?
PEG_ISO	*	=4:1_SPACING	?

PEG Min Spacing Rules (mils) (Maho Bay PDG, Intel Doc# 473718)

Section	Imp	Design	Iso	Design	Comments
4.2.1	80	80	16	15.75	PCIe Gen3. Allow looser spacing for same direction on stripline per Anil

PCIe (CPU)

Electrical Constraint Set	Physical	Spacing
x16 Graphics		
E0E01 PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D
E0E02 PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D P<15>
E0E03 PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D N<15>
E0E04 PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D C P<15>
E0E05 PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_R2D C N<15>
E0E06 PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R P<15>
E0E07 PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R N<15>
E0E08 PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R C P<15>
E0E09 PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R C N<15>
E0E10 PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D P<14>
E0E11 PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D N<14>
E0E12 PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D C P<14>
E0E13 PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D C N<14>
E0E14 PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R P<14>
E0E15 PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R N<14>
E0E16 PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R C P<14>
E0E17 PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R C N<14>
E0E18 PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D P<13>
E0E19 PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D N<13>
E0E20 PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D C P<13>
E0E21 PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D C N<13>
E0E22 PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R P<13>
E0E23 PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R N<13>
E0E24 PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R C P<13>
E0E25 PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R C N<13>
E0E26 PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D P<12>
E0E27 PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D N<12>
E0E28 PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D C P<12>
E0E29 PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D C N<12>
E0E30 PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R P<12>
E0E31 PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R N<12>
E0E32 PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R C P<12>
E0E33 PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R C N<12>
E0E34 PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D P<11>
E0E35 PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D N<11>
E0E36 PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D C P<11>
E0E37 PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D C N<11>
E0E38 PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R P<11>
E0E39 PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R N<11>
E0E40 PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R C P<11>
E0E41 PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R C N<11>
E0E42 PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D P<10>
E0E43 PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D N<10>
E0E44 PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D C P<10>
E0E45 PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D C N<10>
E0E46 PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R P<10>
E0E47 PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R N<10>
E0E48 PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R C P<10>
E0E49 PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R C N<10>
E0E50 PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D P<9>
E0E51 PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D N<9>
E0E52 PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D C P<9>
E0E53 PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D C N<9>
E0E54 PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R P<9>
E0E55 PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R N<9>
E0E56 PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R C P<9>
E0E57 PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R C N<9>
E0E58 PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D P<8>
E0E59 PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D N<8>
E0E60 PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D C P<8>
E0E61 PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D C N<8>
E0E62 PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R P<8>
E0E63 PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R N<8>
E0E64 PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R C P<8>
E0E65 PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R C N<8>

PCIe (CPU)

Electrical Constraint Set		Physical	Spacing	
x16 Graphics				
1E000	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D P<7> 78
1E000	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D N<7> 78
1E000		PCIE3_PHY	PEG_R2D	PEG_R2D C P<7> 78 88
1E000		PCIE3_PHY	PEG_R2D	PEG_R2D C N<7> 78 88
1E000	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R P<7> 78 88
1E000	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R N<7> 78 88
1E000		PCIE3_PHY	PEG_D2R	PEG_D2R C P<7> 78
1E000		PCIE3_PHY	PEG_D2R	PEG_D2R C N<7> 78
1E000	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D P<6> 78
1E000	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D N<6> 78
1E000		PCIE3_PHY	PEG_R2D	PEG_R2D C P<6> 78 88
1E000		PCIE3_PHY	PEG_R2D	PEG_R2D C N<6> 78 88
1E000	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R P<6> 78 88
1E000	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R N<6> 78 88
1E000		PCIE3_PHY	PEG_D2R	PEG_D2R C P<6> 78 88
1E000		PCIE3_PHY	PEG_D2R	PEG_D2R C N<6> 78
1E000	PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D	PEG_R2D P<5> 78
1E000	PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D	PEG_R2D N<5> 78
1E000		PCIE3_PHY	PEG_R2D	PEG_R2D C P<5> 78 88
1E000		PCIE3_PHY	PEG_R2D	PEG_R2D C N<5> 78 88
1E000	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R P<5> 78 88
1E000	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R N<5> 78 88
1E000		PCIE3_PHY	PEG_D2R	PEG_D2R C P<5> 78 88
1E000		PCIE3_PHY	PEG_D2R	PEG_D2R C N<5> 78
1E000	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D P<4> 78
1E000	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D N<4> 78
1E000		PCIE3_PHY	PEG_R2D	PEG_R2D C P<4> 78 88
1E000		PCIE3_PHY	PEG_R2D	PEG_R2D C N<4> 78 88
1E000	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R P<4> 78 88
1E000	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R N<4> 78 88
1E000		PCIE3_PHY	PEG_D2R	PEG_D2R C P<4> 78 88
1E000		PCIE3_PHY	PEG_D2R	PEG_D2R C N<4> 78
1E000	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D P<3> 78
1E000	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D N<3> 78
1E000		PCIE3_PHY	PEG_R2D	PEG_R2D C P<3> 78 88
1E000		PCIE3_PHY	PEG_R2D	PEG_R2D C N<3> 78 88
1E000	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R P<3> 78 88
1E000	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R N<3> 78 88
1E000		PCIE3_PHY	PEG_D2R	PEG_D2R C P<3> 78 88
1E000		PCIE3_PHY	PEG_D2R	PEG_D2R C N<3> 78
1E000	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D P<2> 78
1E000	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D N<2> 78
1E000		PCIE3_PHY	PEG_R2D	PEG_R2D C P<2> 78 88
1E000		PCIE3_PHY	PEG_R2D	PEG_R2D C N<2> 78 88
1E000	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R P<2> 78 88
1E000	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R N<2> 78 88
1E000		PCIE3_PHY	PEG_D2R	PEG_D2R C P<2> 78 88
1E000		PCIE3_PHY	PEG_D2R	PEG_D2R C N<2> 78
1E000	PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D	PEG_R2D P<1> 78
1E000	PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D	PEG_R2D N<1> 78
1E000		PCIE3_PHY	PEG_R2D	PEG_R2D C P<1> 78 88
1E000		PCIE3_PHY	PEG_R2D	PEG_R2D C N<1> 78 88
1E000	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R P<1> 78 88
1E000	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R N<1> 78 88
1E000		PCIE3_PHY	PEG_D2R	PEG_D2R C P<1> 78 88
1E000		PCIE3_PHY	PEG_D2R	PEG_D2R C N<1> 78
1E000	PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D	PEG_R2D P<0> 78
1E000	PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D	PEG_R2D N<0> 78
1E000		PCIE3_PHY	PEG_R2D	PEG_R2D C P<0> 78 88
1E000		PCIE3_PHY	PEG_R2D	PEG_R2D C N<0> 78 88
1E000	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R P<0> 78 88
1E000	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R N<0> 78 88
1E000		PCIE3_PHY	PEG_D2R	PEG_D2R C P<0> 78 88
1E000		PCIE3_PHY	PEG_D2R	PEG_D2R C N<0> 78
CPU PCIe Clocks				
1E000	PEG_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PEG CLK100M P 11 78
1E000	PEG_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PEG CLK100M N 11 78
CPU PCIe Compensation				
1E000		COMP_PCIE_PHY	COMP_PCIE	CPU_PEG_RCOMP 5
CPU eDP Compensation				
1E000		COMP_PCIE_PHY	COMP_PCIE	CPU_EDP_RCOMP 5

SYNC MASTER=J17 NICK		SYNC DATE=12/14/2012	
PAGE TITLE			
CPU PCIe Constraints			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-9884	D
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Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
PCIE_PHY	*	PCIE_85D
COMP_DMI_PHY	*	DMI_COMP

PCIE-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE_SAME_DIR	TOP,BOTTOM	=5X_DIELECTRIC	?
PCIE_SAME_DIR	*	=3.5X_DIELECTRIC	?
PCIE_ALT_DIR	*	=7X_DIELECTRIC	?
PCIE_ISO	*	=4:1_SPACING	?

TBT x4 PCIE Spacing Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCIE_TBT_R2D	PCIE_TBT_R2D	*	PCIE_SAME_DIR
PCIE_TBT_D2R	PCIE_TBT_D2R	*	PCIE_SAME_DIR
PCIE_TBT_D2R	PCIE_TBT_R2D	*	PCIE_ALT_DIR
PCIE_TBT_D2R	*	*	PCIE_ISO
PCIE_TBT_R2D	*	*	PCIE_ISO

PCH x1 PCIE Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCIE	*	*	PCIE_ISO

DMI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DMI_SAME_DIR	TOP,BOTTOM	=5X_DIELECTRIC	?
DMI_SAME_DIR	*	=4X_DIELECTRIC	?
DMI_ALT_DIR	*	=5X_DIELECTRIC	?
DMI_ISO	*	=4X_DIELECTRIC	?

DMI x4 PCIE Spacing Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DMI_N2S	DMI_N2S	*	DMI_SAME_DIR
DMI_S2N	DMI_S2N	*	DMI_SAME_DIR
DMI_N2S	DMI_S2N	*	DMI_ALT_DIR
DMI_N2S	*	*	DMI_ISO
DMI_S2N	*	*	DMI_ISO

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DMI_COMP	*	Y	0.2032 MM	0.2032 MM	3 MM	=STANDARD	=STANDARD

PCie (PCH)

Electrical Constraint Set	Physical	Spacing
x4 Thunderbolt		
PCIE_GEN2_R2D	PCIE_PHY	PCIE_TBT_R2D
PCIE_GEN2_R2D	PCIE_PHY	PCIE_TBT_R2D
PCIE_GEN2_R2D	PCIE_PHY	PCIE_TBT_R2D
PCIE_GEN2_R2D	PCIE_PHY	PCIE_TBT_R2D
PCIE_GEN2_D2R	PCIE_PHY	PCIE_TBT_D2R
PCIE_GEN2_D2R	PCIE_PHY	PCIE_TBT_D2R
PCIE_GEN2_D2R	PCIE_PHY	PCIE_TBT_D2R
PCIE_GEN2_D2R	PCIE_PHY	PCIE_TBT_D2R
PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE
PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE
x1 AirPort		
PCIE_GEN2_R2D_CONN_AP	PCIE_PHY	PCIE
PCIE_GEN2_R2D_CONN_AP	PCIE_PHY	PCIE
PCIE_GEN2_R2D_CONN_AP	PCIE_PHY	PCIE
PCIE_GEN2_D2R_CONN_AP	PCIE_PHY	PCIE
PCIE_GEN2_D2R_CONN_AP	PCIE_PHY	PCIE
PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE
PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE
x1 Caesar IV		
PCIE_GEN2_R2D	PCIE_PHY	PCIE
PCIE_GEN2_R2D	PCIE_PHY	PCIE
PCIE_GEN2_R2D	PCIE_PHY	PCIE
PCIE_GEN2_D2R	PCIE_PHY	PCIE
PCIE_GEN2_D2R	PCIE_PHY	PCIE
PCIE_GEN2_D2R	PCIE_PHY	PCIE
PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE
PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE
x2 SSD		
PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE
PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE
PCH PCIE Compensation		
COMP_DMI_PHY	COMP_PCIE	PCH_PCIE_RCOMP

CPU DP REF CLK

Electrical Constraint Set	Physical	Spacing
CPU DP REF CLK		
CPU_CLK135_P11	PCIE_PHY	CLK_PCIE
CPU_CLK135_P11	PCIE_PHY	CLK_PCIE
CPU_CLK135_P11	PCIE_PHY	CLK_PCIE
CPU_CLK135_P11	PCIE_PHY	CLK_PCIE

DMI

Electrical Constraint Set	Physical	Spacing
DMI		
DMI_N2S	PCIE_PHY	DMI_N2S
DMI_N2S	PCIE_PHY	DMI_N2S
DMI_S2N	PCIE_PHY	DMI_S2N
DMI_S2N	PCIE_PHY	DMI_S2N
PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE
PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE
DMI Compensation		
COMP_DMI_PHY	COMP_PCIE	PCH_DMI_RCOMP

SATA

SATA-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SATA_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
SATA_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
SATA_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SATA_PHY	*	SATA_85D
COMP_SATA_PHY	*	SATA_50S
SATA_PHY_90	*	SATA_90D

SATA-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA_ISO	*	=6:1_SPACING	?
COMP_SATA_ISO	*	=4:1_SPACING	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SATA	*	*	SATA_ISO
COMP_SATA	*	*	COMP_SATA_ISO

FDI

FDI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
FDI_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
COMP_FDI	*	Y	0.25 MM	0.25 MM	3 MM	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
FDI_SE_PHY	*	FDI_50S
COMP_FDI_PHY	*	COMP_FDI

FDI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
FDI_ISO	*	=3:1_SPACING	?
COMP_FDI_ISO	*	=4:1_SPACING	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
FDI	*	*	FDI_ISO
COMP_FDI	*	*	COMP_FDI_ISO

XDP

XDP-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
XDP_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
XDP_PHY	*	XDP_55S

XDP-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
XDP_ISO	*	=2:1_SPACING	?
CLK_JTAG_ISO	*	=4:1_SPACING	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
XDP	*	*	XDP_ISO
CLK_JTAG	*	*	CLK_JTAG_ISO

SATA Min Spacing Rules (mils) (Maho Bay PDG, Intel Doc# 473718)

Section	Imp	Design	Iso	Design	Comments
15.2.1	90	95	20	23.62	SATA Gen2, SATA Gen3

SATA Compensation Rules (mils)

Table	Imp	Design	Iso	Design	Comments
15-3	50	50	15	15.75	SATA Gen2, SATA Gen3

FDI Min Spacing Rules (mils) (Maho Bay PDG, Intel Doc# 473718)

Table	Imp	Design	Iso	Design	Comments
6-1/6-2	85	85	12	11.81	FDI main length

FDI Compensation Rules (mils)

Table	Trace	Design	Iso	Design	Comments
6-4	10	11.81	-	15.75	Using PCIe guidelines

Desktop Debug Design Guide (Intel Doc# 430883)

Section	Imp	Design	Iso	Design	Comments
1.5	45-65	55	-	15.75	Isolation is for JTAG clocks. All signals default are 50 Ohm SE.

SATA

Electrical Constraint Set	Physical	Spacing		
PCH SATA Port 0 (HDD)				
H000 SATA_R2D	SATA_PHY_90	SATA	SATA HDD R2D P	35
H009 SATA_R2D	SATA_PHY_90	SATA	SATA HDD R2D N	35
H00B	SATA_PHY_90	SATA	SATA HDD R2D C P	11 35
H009	SATA_PHY_90	SATA	SATA HDD R2D C N	11 35
H04E SATA_D2R	SATA_PHY_90	SATA	SATA HDD D2R P	11 35
H050 SATA_D2R	SATA_PHY_90	SATA	SATA HDD D2R N	11 35
H051	SATA_PHY_90	SATA	SATA HDD D2R C P	35
H059	SATA_PHY_90	SATA	SATA HDD D2R C N	35
PCH SATA Port 1 (SSD)				
H080 SATA_SSD_R2D	SATA_PHY	SATA	SSD R2D P<0...1>	11 35
H080 SATA_SSD_R2D	SATA_PHY	SATA	SSD R2D N<0...1>	11 35
H081 SATA_SSD_R2D	SATA_PHY	SATA	SSD R2D C P<0...1>	35
H080 SATA_SSD_R2D	SATA_PHY	SATA	SSD R2D C N<0...1>	35
H089 SATA_SSD_D2R	SATA_PHY	SATA	SSD D2R P<0...1>	11 35
H089 SATA_SSD_D2R	SATA_PHY	SATA	SSD D2R N<0...1>	11 35
PCH SATA Compensation				
H100	COMP_SATA_PHY	COMP_SATA	PCH SATA RCOMP	11

FDI

Electrical Constraint Set	Physical	Spacing	
FDI			
LESS	FDI_SF_PHY	FDI	FDI_CSXNC 5 12
LESS	FDI_SF_PHY	FDI	FDI_INT 5 12
FDI Compensation			
LESS	COMP_FDI_PHY	COMP_FDI	PCH_FDI_RCOMP 12

XDP

Electrical Constraint Set	Physical	Spacing	
CPU XDP			
E877	XDP_PHY	XDP	XDP BPM L<7..0> 6 18
E88	XDP_PHY	XDP	CPU CFG<17..0> 6 18 89
E89 ITP_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE	ITPCPU CLK100M P
E90 ITP_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE	ITPCPU CLK100M N
E91	CLK_PCIE_PHY	CLK_PCIE	ITPXPDP CLK100M P 11 89
E93	CLK_PCIE_PHY	CLK_PCIE	ITPXPDP CLK100M N 11 89
E92	CLK_PCIE_PHY	CLK_PCIE	XDP CPU CLK100M P
E94	CLK_PCIE_PHY	CLK_PCIE	XDP CPU CLK100M N
E95	XDP_PHY	CLK_JTAG	XDP CPU TCK 6 18
E96	XDP_PHY	XDP	XDP CPU TMS 6 18
E97	XDP_PHY	XDP	XDP CPU TDI 6 18
E98	XDP_PHY	XDP	XDP CPU TDO 6 18
PCH XDP			
E99	XDP_PHY	CLK_JTAG	XDP PCH TCK 11 18
E90	XDP_PHY	XDP	XDP PCH TMS 11 18
E91	XDP_PHY	XDP	XDP PCH TDI 11 18
E92	XDP_PHY	XDP	XDP PCH TDO 11 18

PCH

PCH-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_PCH_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

PCH-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCH	*	=4:1_SPACING	?
COMP_PCH	*	=2:1_SPACING	?

PCI

PCI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_PCI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

PCI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCI	*	=2:1_SPACING	?

LPC

LPC-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

LPC-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC	*	=1.5:1_SPACING	?
CLK_LPC	*	=2:1_SPACING	?

HDA

HDA-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

HDA-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA	*	=2x_DIELECTRIC	?

Crystal

Crystal-specific Physical Rules

[illegible]

Crystal-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
XTAL	*	=4X_DIELECTRIC	?

SPI

SPI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI	*	=2:1_SPACING	?

PCI

Electrical Constraint Set	Physical	Spacing
PCI Clock		
100M	CLK_PCT_55S	CLK_PCT
100M	CLK_PCT_55S	CLK_PCT

LPC

Electrical Constraint Set	Physical	Spacing	
LPC			
	LPC_55S	LPC	LPC_AD<3..0> 13 46 48
	LPC_55S	LPC	LPC_AD_R<3..0> 13
	LPC_55S	LPC	LPC_FRAME_L 13 46 48
	LPC_55S	LPC	LPC_FRAME_R_L 13
LPC Clocks			
	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_LPCPLUS 19 48
	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_LPCPLUS_R 11 19
	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_SMC 19 46
	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_SMC_R 11 19

PCH Clocks

Electrical Constraint Set	Physical	Spacing	
PCH Reference Clock			
	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_SB 11 19
	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_SB_R 11
PCH RTC 32K			
	CLK_XTAL	XTAL	PCH_CLK32K_RTCX1 11 19
	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2 11 19
	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2_R 19
SMC 32K			
	CLK_PCH_55S	CLK_PCH	PM_CLK32K_SUSCLK_R 12 47
	CLK_PCH_55S	CLK_PCH	SMC_CLK32K 46 47

25 MHz Reference Clocks

Electrical Constraint Set	Physical	Spacing		
25M Reference Crystal				
	CLK_XTAL	XTAL	SYSCLK_CLK25M_X1	19
	CLK_XTAL	XTAL	SYSCLK_CLK25M_X2	19
	CLK_XTAL	XTAL	SYSCLK_CLK25M_X2_R	19
25M Reference Clocks				
	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_ENET	19 37
	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_ENET_R	19
	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_TBT	19 28
	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_TBT_R	28

HDA

Electrical Constraint Set	Physical	Spacing	
HDA			
H35P	HDA_55S	HDA	HDA_BIT_CLK 11 54
H35P	HDA_55S	HDA	HDA_BIT_CLK_R 11
H35B	HDA_55S	HDA	HDA_RST_L 11 54
H35P	HDA_55S	HDA	HDA_RST_R_L 11
H35B	HDA_55S	HDA	HDA_SDOUT 11 54
H35P	HDA_55S	HDA	HDA_SDOUT_R 11 19
H35P	HDA_55S	HDA	HDA_SYNC 11 54
H35B	HDA_55S	HDA	HDA_SYNC_R 11
H35P	HDA_55S	HDA	HDA_SDIN0 11 54
H35P	HDA_55S	HDA	AUD_SDI_R 54
SPDIF			
H37P		HDA	AUD_SPDIF_CHIP 54
H37P		HDA	AUD_SPDIF_OUT 54 58

SPI Bootrom

Electrical Constraint Set	Physical	Spacing	
SPI ROM			
H09D	SPI_56S	SPT	SPI_CLK_R 13-48
H09E	SPI_56S	SPT	SPI_CLK 48
H09F	SPI_56S	SPT	SPI_ALT_CLK 48
H09G	SPI_56S	SPT	SPI_SMC_CLK 46-48
H09H	SPI_56S	SPT	SPI_MLB_CLK 48
H09I	SPI_56S	SPT	SPI_CS0_R_L 13-48
H09J	SPI_56S	SPT	SPI_CS0_L 48
H09K	SPI_56S	SPT	SPI_ALT_CS_L 48
H09L	SPI_56S	SPT	SPI_SMC_CS_L 46-48
H09M	SPI_56S	SPT	SPI_MLB_CS_L 48
H09N	SPI_56S	SPT	SPI_MOSI_R 13-48
H09O	SPI_56S	SPT	SPI_MOSI 48
H09P	SPI_56S	SPT	SPI_ALT_MOSI 48
H09Q	SPI_56S	SPT	SPI_SMC_MOSI 46-48
H09R	SPI_56S	SPT	SPI_MLB_MOSI 48
H09S	SPI_56S	SPT	SPI_MISO 13-48
H09T	SPI_56S	SPT	SPI_ALT_MISO 48
H09U	SPI_56S	SPT	SPI_SMC_MISO 46-48
H09V	SPI_56S	SPT	SPI_MLB_MISO 48
H09W	SPI_56S	SPT	SPIROM_USE_MLB 14-48

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3.3V S5/5V S4

Physical	Spacing	Voltage	DIDT	NO_TEST	
Input Bus					
U624 POWER	POWER	12V			REG VIN U7600 67
U626 POWER	POWER	5V			REG VCC1 U7600 67
U625 POWER	POWER	5V			REG VCC2 U7600 67
3.3V S5					
U646 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG PHASE P3V3S5 67
U647 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG B00T P3V3S5 67
U653 VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG B00T P3V3S5_RC 67
U648 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG UGATE P3V3S5 67
U650 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG LGATE P3V3S5 67
U652 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG SNUBBER P3V3S5 67
U609 VR_CTL_PHY	VR_CTL				REG P3V3S5_ISEN 67
U607 VR_CTL_PHY	VR_CTL				REG P3V3S5_OCSET 67
U610 VR_CTL_PHY	VR_CTL				REG P3V3S5_FSET 67
U608 VR_CTL_PHY	VR_CTL				REG P3V3S5_VOUT 67
U607 VR_CTL_PHY	VR_CTL				REG P3V3S5_VOUT_R 67
U607 VR_CTL_PHY	VR_CTL				REG P3V3S5_FB 67
5V S3					
U613 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG PHASE P5VS4 67
U615 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG B00T P5VS4 67
U616 VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG B00T P5VS4_RC 67
U615 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG UGATE P5VS4 67
U616 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG LGATE P5VS4 67
U614 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG SNUBBER P5VS4 67
U610 VR_CTL_PHY	VR_CTL				REG P5VS4_ISEN 67
U610 VR_CTL_PHY	VR_CTL				REG P5VS4_OCSET 67
U610 VR_CTL_PHY	VR_CTL				REG P5VS4_FSET 67
U610 VR_CTL_PHY	VR_CTL				REG P5VS4_VOUT 67
U610 VR_CTL_PHY	VR_CTL				REG P5VS4_VOUT_R 67
U610 VR_CTL_PHY	VR_CTL				REG P5VS4_FB 67
Output Bus					
U615 POWER	POWER	5V			PP5V_S5 67
U616 POWER	POWER	5V			PP5V_S4 67
U616 POWER	POWER	3.3V			PP3V3_S5 67
FET Switched					
U620 POWER	POWER	5V			PP5V_S0 67
U624 POWER	POWER	3.3V			PP3V3_S4 67
U620 POWER	POWER	3.3V			PP3V3_S0 67
U625 POWER	POWER	3.3V			PP3V3_S0_SSD 67
U620 POWER	POWER	3.3V			PP3V3_ENET 67
U620 POWER	POWER	3.3V			PP3V3_TBTLIC 67
Sensed					
U625 POWER	POWER	3.3V			PPSSD_S0 67
U627 POWER	POWER	3.3V			PP3V3_S4_AP 67

3.42V G3H

Physical	Spacing	Voltage	DIDT	NO_TEST	
3.42V G3H					
U630 POWER	VR_SWITCH	12V	TRUE		P3V42G3H_BOOST 62
U630 POWER	VR_SWITCH	12V	TRUE		P3V42G3H_SW 62
U630 VR_CTL_PHY	VR_CTL				P3V42G3H_FB 62
U630 VR_CTL_PHY	VR_CTL				P3V42G3H_SHDN_L 62
Output Bus					
U630 POWER	POWER	3.425V			PP3V42_G3H 67

3.3V G3

Physical	Spacing	Voltage	DIDT	NO_TEST	
U627 POWER	POWER	3.3V			PP3V3_G3 67

HDD S0

Physical	Spacing	Voltage	DIDT	NO_TEST	
FET Switched					
U620 POWER	POWER	5V			PPHDD_S0 67
Sensed					
U630 POWER	POWER	5V			PP5V_S0_HDD 67

12V

Physical	Spacing	Voltage	DIDT	NO_TEST	
Input Bus					
U620 POWER	POWER	12V			PP12V_ACDC 67
FET Switched					
U630 POWER	POWER	12V			PP12V_S5 67
U630 POWER	POWER	12V			PP12V_S0 67
U627 POWER	POWER	12V			PP12V_S0_BLC 67
U630 POWER	POWER	12V			PPHDD12_S0 67
Sensed					
U630 POWER	POWER	12V			PP12V_G3H 67
U627 POWER	POWER	12V			PP12V_G3H_P3V42 62
U630 POWER	POWER	12V			PP12V_S0_GPUCORE 67
U630 POWER	POWER	12V			PP12V_S0_FBVDDQ 67
U627 POWER	POWER	12V			PP12V_S0_HDD 67

Ground/Common

Physical	Spacing	Voltage	DIDT	NO_TEST	
Common					
U630 GND	GND	0V			GND

SYNC_MASTER=J17_ULTIMATE

SYNC_DATE=05/16/2013

Platform VReg Constraints

 Apple Inc.

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GDDR5

GDDR5-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
GDDR_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
GDDR_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	12 MM	=STANDARD	=STANDARD
GDDR_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
GDDR_MA_PHY	*	GDDR_45S
GDDR_ADBI_PHY	*	GDDR_45S
GDDR_CTRL_PHY	*	GDDR_45S
GDDR_CLK_PHY	*	GDDR_80D
GDDR_DQ_PHY	*	GDDR_45S
GDDR_EDC_PHY	*	GDDR_45S
GDDR_DBI_PHY	*	GDDR_45S
GDDR_WCK_PHY	*	GDDR_80D

Main Segment Min Spacing Rules for 4.5 Gbps or Less (AMD Doc# 49919)

Trace-to-Trace				Isolation					
Table	Micro	Design	Strip	Design	Micro	Design	Strip	Design	Comments
5-6/5-7	2:1	2:1	2:1	2:1	5:1	5:1	5:1	5:1	Memory address (MA). Implemented 4.5 Gbps or less rules for K70.
5-6/5-7	2:1	2:1	2:1	2:1	5:1	5:1	5:1	5:1	Address dynamic bus inversion (ADBI)
5-6/5-7	2:1	2:1	2:1	2:1	5:1	5:1	5:1	5:1	Control (CTRL)
5-6/5-7	5:1	5:1	5:1	5:1	5:1	5:1	5:1	5:1	Clock (CLK)
5-6/5-7	3:1	3:1	3:1	3:1	5:1	5:1	5:1	5:1	Data (DQ)
5-6/5-7	7:1	7:1	7:1	7:1	7:1	7:1	7:1	7:1	Error detection pins (EDC). Using larger isolation rules,
5-6/5-7	3:1	3:1	3:1	3:1	5:1	5:1	5:1	5:1	Data dynamic bus inversion (DBI)
5-6/5-7	5:1	5:1	5:1	5:1	5:1	5:1	5:1	5:1	Forwarded clock (WCK)

GDDR5-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR_ISO	*	=3x_DIELECTRIC	?
GDDR_ISO	TOP, BOTTOM	=5x_DIELECTRIC	?
GDDR_MA2MA	*	=2x_DIELECTRIC	?
GDDR_MA2MA	TOP, BOTTOM	=3x_DIELECTRIC	?
GDDR_ADBI2ADBI	*	=2x_DIELECTRIC	?
GDDR_ADBI2ADBI	TOP, BOTTOM	=2x_DIELECTRIC	?
GDDR_CTRL2CTRL	*	=2x_DIELECTRIC	?
GDDR_CTRL2CTRL	TOP, BOTTOM	=2x_DIELECTRIC	?
GDDR_CLK2CLK	*	=3x_DIELECTRIC	?
GDDR_CLK2CLK	TOP, BOTTOM	=5x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR_DQ2DQ	*	=3x_DIELECTRIC	?
GDDR_DQ2DQ	TOP, BOTTOM	=3x_DIELECTRIC	?
GDDR_EDC_ISO	*	=3X_DIELECTRIC	?
GDDR_EDC_ISO	TOP, BOTTOM	=5X_DIELECTRIC	?
GDDR_EDC2EDC	*	=3X_DIELECTRIC	?
GDDR_EDC2EDC	TOP, BOTTOM	=5X_DIELECTRIC	?
GDDR_DBI2DBI	*	=3x_DIELECTRIC	?
GDDR_DBI2DBI	TOP, BOTTOM	=3x_DIELECTRIC	?
GDDR_WCK2WCK	*	=3X_DIELECTRIC	?
GDDR_WCK2WCK	TOP, BOTTOM	=5x_DIELECTRIC	?

Constraints ($x \in \{A, B\}$, $y \in \{0, 1\}$)
Memory Address: $MA_{xy}[8:0]$

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_MA	*	*	GDDR_ISO
GDDR_*_*_MA	=SAME	*	GDDR_MA2MA

Address Dynamic Bus Inversion: ADBIx

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_ADBI	*	*	GDDR_ISO
GDDR_*_*_ADBI	=SAME	*	GDDR_ADBI2ADBI

Control: Reset, CKExy, CSxy, WExy, RASxy, CASxy

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_CTRL	*	*	GDDR_ISO
GDDR_*_*_CTRL	*	*	GDDR_ISO
GDDR_*_*_CTRL	=SAME	*	GDDR_CTRL2CTRL

Clock: CKxy

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_CLK	*	*	GDDR_ISO
GDDR_*_*_CLK	=SAME	*	GDDR_CLK2CLK

GPU

GPU-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_GPU_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

GPU-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_GPU_ISO	*	=4:1_SPACING	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_GPU	*	*	CLK_GPU_ISO

GDDR5 Frame Buffer A

Electrical Constraint Set		Physical	Spacing		
Memory Address					
MEM0	GDDR A0 MA	GDDR MA PHY	GDDR A 0 MA	FB A0 A<8..0>	NO TEST:TRUE
MEM1	GDDR A1 MA	GDDR MA PHY	GDDR A 1 MA	FB A1 A<8..0>	NO TEST:TRUE
Address Dynamic Bus Inv					
FE00	GDDR A0 ADBT	GDDR ADBT PHY	GDDR A 0 ADBT	FB A0 ABT L	NO TEST:TRUE
FE01	GDDR A1 ADBT	GDDR ADBT PHY	GDDR A 1 ADBT	FB A1 ABT L	NO TEST:TRUE
Control					
FE04	GDDR A0 CKE	GDDR CTRL PHY	GDDR A 0 CTRL	FB A0 CKE L	NO TEST:TRUE
FE05	GDDR A0 CTRL	GDDR CTRL PHY	GDDR A 0 CTRL	FB A0 CS L	NO TEST:TRUE
FE06	GDDR A0 CTRL	GDDR CTRL PHY	GDDR A 0 CTRL	FB A0 WE L	NO TEST:TRUE
FE07	GDDR A0 CTRL	GDDR CTRL PHY	GDDR A 0 CTRL	FB A0 CAS L	NO TEST:TRUE
FE08	GDDR A0 CTRL	GDDR CTRL PHY	GDDR A 0 CTRL	FB A0 RAS L	NO TEST:TRUE
FE09	GDDR A1 CKE	GDDR CTRL PHY	GDDR A 1 CTRL	FB A1 CKE L	NO TEST:TRUE
FE0A	GDDR A1 CTRL	GDDR CTRL PHY	GDDR A 1 CTRL	FB A1 CS L	NO TEST:TRUE
FE0B	GDDR A1 CTRL	GDDR CTRL PHY	GDDR A 1 CTRL	FB A1 WE L	NO TEST:TRUE
FE0C	GDDR A1 CTRL	GDDR CTRL PHY	GDDR A 1 CTRL	FB A1 CAS L	NO TEST:TRUE
FE0D	GDDR A1 CTRL	GDDR CTRL PHY	GDDR A 1 CTRL	FB A1 RAS L	NO TEST:TRUE
Clock					
FE0E	GDDR A0 CLK	GDDR CLK PHY	GDDR A 0 CLK	FB A0 CLK P	NO TEST:TRUE
FE0F	GDDR A0 CLK	GDDR CLK PHY	GDDR A 0 CLK	FB A0 CLK N	NO TEST:TRUE
FE10	GDDR A1 CLK	GDDR CLK PHY	GDDR A 1 CLK	FB A1 CLK P	NO TEST:TRUE
FE11	GDDR A1 CLK	GDDR CLK PHY	GDDR A 1 CLK	FB A1 CLK N	NO TEST:TRUE
Data					
FE12	GDDR A0 DQ BYTE0	GDDR DQ PHY	GDDR A 0 DQ	FB A0 DQ<7..0>	NO TEST:TRUE
FE13	GDDR A0 DQ BYTE1	GDDR DQ PHY	GDDR A 0 DQ	FB A0 DQ<15..8>	NO TEST:TRUE
FE14	GDDR A0 DQ BYTE2	GDDR DQ PHY	GDDR A 0 DQ	FB A0 DQ<23..16>	NO TEST:TRUE
FE15	GDDR A0 DQ BYTE3	GDDR DQ PHY	GDDR A 0 DQ	FB A0 DQ<31..24>	NO TEST:TRUE
FE16	GDDR A1 DQ BYTE0	GDDR DQ PHY	GDDR A 1 DQ	FB A1 DQ<7..0>	NO TEST:TRUE
FE17	GDDR A1 DQ BYTE1	GDDR DQ PHY	GDDR A 1 DQ	FB A1 DQ<15..8>	NO TEST:TRUE
FE18	GDDR A1 DQ BYTE2	GDDR DQ PHY	GDDR A 1 DQ	FB A1 DQ<23..16>	NO TEST:TRUE
FE19	GDDR A1 DQ BYTE3	GDDR DQ PHY	GDDR A 1 DQ	FB A1 DQ<31..24>	NO TEST:TRUE
Error Detection					
FE20	GDDR A0 EDC0	GDDR EDC PHY	GDDR A 0 EDC	FB A0 EDC<0>	NO TEST:TRUE
FE21	GDDR A0 EDC1	GDDR EDC PHY	GDDR A 0 EDC	FB A0 EDC<1>	NO TEST:TRUE
FE22	GDDR A0 EDC2	GDDR EDC PHY	GDDR A 0 EDC	FB A0 EDC<2>	NO TEST:TRUE
FE23	GDDR A0 EDC3	GDDR EDC PHY	GDDR A 0 EDC	FB A0 EDC<3>	NO TEST:TRUE
FE24	GDDR A1 EDC0	GDDR EDC PHY	GDDR A 1 EDC	FB A1 EDC<0>	NO TEST:TRUE
FE25	GDDR A1 EDC1	GDDR EDC PHY	GDDR A 1 EDC	FB A1 EDC<1>	NO TEST:TRUE
FE26	GDDR A1 EDC2	GDDR EDC PHY	GDDR A 1 EDC	FB A1 EDC<2>	NO TEST:TRUE
FE27	GDDR A1 EDC3	GDDR EDC PHY	GDDR A 1 EDC	FB A1 EDC<3>	NO TEST:TRUE
Data Dynamic Bus Inv					
FE28	GDDR A0 DBT0	GDDR DBT PHY	GDDR A 0 DBT	FB A0 DBT L<0>	NO TEST:TRUE
FE29	GDDR A0 DBT1	GDDR DBT PHY	GDDR A 0 DBT	FB A0 DBI L<1>	NO TEST:TRUE
FE2A	GDDR A0 DBT2	GDDR DBT PHY	GDDR A 0 DBT	FB A0 DBI L<2>	NO TEST:TRUE
FE2B	GDDR A0 DBT3	GDDR DBT PHY	GDDR A 0 DBT	FB A0 DBI L<3>	NO TEST:TRUE
FE2C	GDDR A1 DBT0	GDDR DBT PHY	GDDR A 1 DBT	FB A1 DBI L<0>	NO TEST:TRUE
FE2D	GDDR A1 DBT1	GDDR DBT PHY	GDDR A 1 DBT	FB A1 DBI L<1>	NO TEST:TRUE
FE2E	GDDR A1 DBT2	GDDR DBT PHY	GDDR A 1 DBT	FB A1 DBI L<2>	NO TEST:TRUE
FE2F	GDDR A1 DBT3	GDDR DBT PHY	GDDR A 1 DBT	FB A1 DBI L<3>	NO TEST:TRUE
Forwarded Clock					
FE30	GDDR A0 WCK0	GDDR WCK PHY	GDDR A 0 WCK	FB A0 WCLK P<0>	NO TEST:TRUE
FE31	GDDR A0 WCK0	GDDR WCK PHY	GDDR A 0 WCK	FB A0 WCLK N<0>	NO TEST:TRUE
FE32	GDDR A0 WCK1	GDDR WCK PHY	GDDR A 0 WCK	FB A0 WCLK P<1>	NO TEST:TRUE
FE33	GDDR A0 WCK1	GDDR WCK PHY	GDDR A 0 WCK	FB A0 WCLK N<1>	NO TEST:TRUE
FE34	GDDR A1 WCK0	GDDR WCK PHY	GDDR A 1 WCK	FB A1 WCLK P<0>	NO TEST:TRUE
FE35	GDDR A1 WCK0	GDDR WCK PHY	GDDR A 1 WCK	FB A1 WCLK N<0>	NO TEST:TRUE
FE36	GDDR A1 WCK1	GDDR WCK PHY	GDDR A 1 WCK	FB A1 WCLK P<1>	NO TEST:TRUE
FE37	GDDR A1 WCK1	GDDR WCK PHY	GDDR A 1 WCK	FB A1 WCLK N<1>	NO TEST:TRUE

GPU

Electrical Constraint Set	Physical	Spacing	
Clocks			
PEX0	CLK_GPU_55S	CLK_GPU	PEX_TSTCLK_0_PL 78
PEX0	CLK_GPU_55S	CLK_GPU	PEX_TSTCLK_0_NG 78
PEX0	CLK_PCTE_PHY	CLK_PCTE	GPU_TESTMODE 83
PEX0	CLK_PCTE_PHY	CLK_PCTE	GPU_PEX_TERM 78
PEX0	CLK_GPU_55S	CLK_GPU	FB_A0_CK_MID NO_TEST=TRUE 81
PEX0	CLK_GPU_55S	CLK_GPU	FB_A1_CK_MID NO_TEST=TRUE 81
PEX0	CLK_GPU_55S	CLK_GPU	FB_B0_CK_MID NO_TEST=TRUE 82
PEX0	CLK_GPU_55S	CLK_GPU	FB_B1_CK_MID NO_TEST=TRUE 82
PEX0	CLK_GPU_55S	CLK_GPU	FB_C0_CK_MID NO_TEST=TRUE 82
PEX0	CLK_GPU_55S	CLK_GPU	FB_C1_CK_MID NO_TEST=TRUE 82
PEX0	CLK_GPU_55S	CLK_GPU	FB_D0_CK_MID NO_TEST=TRUE 82
PEX0	CLK_GPU_55S	CLK_GPU	FB_D1_CK_MID NO_TEST=TRUE 82
PEX0	CLK_GPU_55S	CLK_GPU	GPU_JTAG_TCK 83 85
PEX0	CLK_GPU_55S	CLK_GPU	GPU_ROM_SCLK 83 84
PEX0	CLK_GPU_55S	CLK_GPU	GPU_ROM_SCLK_R 84
PEX0	CLK_GPU_55S	CLK_GPU	GPU_XTALOUT 83 84
PEX0	CLK_GPU_55S	CLK_GPU	GPU_OSC_27M_XTALOUT 83 84
PEX0	CLK_GPU_55S	CLK_GPU	GPU_OSC_27M_XTALIN 83 84
SMB			
ES00	SMB_PHY	SMB	GPU_SMB_CLK 83 84
ES00	SMB_PHY	SMB	GPU_SMB_DAT 83 84
ES00	SMB_PHY	SMB	GPU_SMB_CLK_R 49 84
ES00	SMB_PHY	SMB	GPU_SMB_DAT_R 49 84
PCIe Compensation			
ES00	PCIE_56S	COMP_PCIE	FB_CAL_PD_VDDQ 80
ES00	PCIE_56S	COMP_PCIE	FB_CAL_PU_GND 80
ES00	PCIE_56S	COMP_PCIE	FB_CAL_TERM_GND 80

GDDR5 Frame Buffer B

Electrical Constraint Set		Physical	Spacing		
Memory Address					
H009	GDDR_B0_MA	GDDR_MA_PHY	GDDR_B_0_MA	FB_B0_A<8..0>	NO TEST=TRUE
H009	GDDR_B1_MA	GDDR_MA_PHY	GDDR_B_1_MA	FB_B1_A<8..0>	NO TEST=TRUE
Address Dynamic Bus Inv					
H009	GDDR_B0_ADBT	GDDR_ADBT_PHY	GDDR_B_0_ADBT	FB_B0_ABT_L	NO TEST=TRUE
H009	GDDR_B1_ADBT	GDDR_ADBT_PHY	GDDR_B_1_ADBT	FB_B1_ABT_L	NO TEST=TRUE
Control					
H009	GDDR_B0_CKE	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_CKE_L	NO TEST=TRUE
H009	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_CS_L	NO TEST=TRUE
H009	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_WE_L	NO TEST=TRUE
H009	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_CAS_L	NO TEST=TRUE
H009	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_RAS_L	NO TEST=TRUE
H009	GDDR_B1_CKE	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_CKE_L	NO TEST=TRUE
H009	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_CS_L	NO TEST=TRUE
H009	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_WE_L	NO TEST=TRUE
H009	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_CAS_L	NO TEST=TRUE
H009	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_RAS_L	NO TEST=TRUE
Clock					
H009	GDDR_B0_CLK	GDDR_CLK_PHY	GDDR_B_0_CLK	FB_B0_CLK_P	NO TEST=TRUE
H009	GDDR_B0_CLK	GDDR_CLK_PHY	GDDR_B_0_CLK	FB_B0_CLK_N	NO TEST=TRUE
H009	GDDR_B1_CLK	GDDR_CLK_PHY	GDDR_B_1_CLK	FB_B1_CLK_P	NO TEST=TRUE
H009	GDDR_B1_CLK	GDDR_CLK_PHY	GDDR_B_1_CLK	FB_B1_CLK_N	NO TEST=TRUE
Data					
H009	GDDR_B0_DQ_BYTE0	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<7..0>	NO TEST=TRUE
H009	GDDR_B0_DQ_BYTE1	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<15..8>	NO TEST=TRUE
H009	GDDR_B0_DQ_BYTE2	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<23..16>	NO TEST=TRUE
H009	GDDR_B0_DQ_BYTE3	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<31..24>	NO TEST=TRUE
H009	GDDR_B1_DQ_BYTE0	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<7..0>	NO TEST=TRUE
H009	GDDR_B1_DQ_BYTE1	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<15..8>	NO TEST=TRUE
H009	GDDR_B1_DQ_BYTE2	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<23..16>	NO TEST=TRUE
H009	GDDR_B1_DQ_BYTE3	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<31..24>	NO TEST=TRUE
Error Detection					
H009	GDDR_B0_EDC0	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<0>	NO TEST=TRUE
H009	GDDR_B0_EDC1	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<1>	NO TEST=TRUE
H009	GDDR_B0_EDC2	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<2>	NO TEST=TRUE
H009	GDDR_B0_EDC3	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<3>	NO TEST=TRUE
H009	GDDR_B1_EDC0	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<0>	NO TEST=TRUE
H009	GDDR_B1_EDC1	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<1>	NO TEST=TRUE
H009	GDDR_B1_EDC2	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<2>	NO TEST=TRUE
H009	GDDR_B1_EDC3	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<3>	NO TEST=TRUE
Data Dynamic Bus Inv					
H009	GDDR_B0_DBT0	GDDR_DBT_PHY	GDDR_B_0_DBT	FB_B0_DBT_L<0>	NO TEST=TRUE
H009	GDDR_B0_DBT1	GDDR_DBT_PHY	GDDR_B_0_DBT	FB_B0_DBT_L<1>	NO TEST=TRUE
H009	GDDR_B0_DBT2	GDDR_DBT_PHY	GDDR_B_0_DBT	FB_B0_DBT_L<2>	NO TEST=TRUE
H009	GDDR_B0_DBT3	GDDR_DBT_PHY	GDDR_B_0_DBT	FB_B0_DBT_L<3>	NO TEST=TRUE
H009	GDDR_B1_DBT0	GDDR_DBT_PHY	GDDR_B_1_DBT	FB_B1_DBT_L<0>	NO TEST=TRUE
H009	GDDR_B1_DBT1	GDDR_DBT_PHY	GDDR_B_1_DBT	FB_B1_DBT_L<1>	NO TEST=TRUE
H009	GDDR_B1_DBT2	GDDR_DBT_PHY	GDDR_B_1_DBT	FB_B1_DBT_L<2>	NO TEST=TRUE
H009	GDDR_B1_DBT3	GDDR_DBT_PHY	GDDR_B_1_DBT	FB_B1_DBT_L<3>	NO TEST=TRUE
Forwarded Clock					
H009	GDDR_B0_WCK0	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_P<0>	NO TEST=TRUE
H009	GDDR_B0_WCK0	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_N<0>	NO TEST=TRUE
H009	GDDR_B0_WCK1	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_P<1>	NO TEST=TRUE
H009	GDDR_B0_WCK1	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_N<1>	NO TEST=TRUE
H009	GDDR_B1_WCK0	GDDR_WCK_PHY	GDDR_B_1_WCK	FB_B1_WCLK_P<0>	NO TEST=TRUE
H009	GDDR_B1_WCK0	GDDR_WCK_PHY	GDDR_B_1_WCK	FB_B1_WCLK_N<0>	NO TEST=TRUE
H009	GDDR_B1_WCK1	GDDR_WCK_PHY	GDDR_B_1_WCK	FB_B1_WCLK_P<1>	NO TEST=TRUE
H009	GDDR_B1_WCK1	GDDR_WCK_PHY	GDDR_B_1_WCK	FB_B1_WCLK_N<1>	NO TEST=TRUE

Frame Buffer Reset

Electrical Constraint Set		Physical	Spacing	
Reset				
H442	GDPR_A0_RESET	GDPR_S6S	GDPR_CTR1	FB_A0_RESET_L
H445	GDPR_A1_RESET	GDPR_S6S	GDPR_CTR1	FB_A1_RESET_L
H446	GDPR_B0_RESET	GDPR_S6S	GDPR_CTR1	FB_B0_RESET_L
H447	GDPR_B1_RESET	GDPR_S6S	GDPR_CTR1	FB_B1_RESET_L
H448	GDPR_C0_RESET	GDPR_S6S	GDPR_CTR1	FB_C0_RESET_L
H449	GDPR_C1_RESET	GDPR_S6S	GDPR_CTR1	FB_C1_RESET_L
H450	GDPR_D0_RESET	GDPR_S6S	GDPR_CTR1	FB_D0_RESET_L
H451	GDPR_D1_RESET	GDPR_S6S	GDPR_CTR1	FB_D1_RESET_L

SYNCH MASTER=D8 AARON		SYNCH DATE=03/13/2012	
PAGE TITLE		DRAWING NUMBER	
GDDR5/GPU Constraints		051-9884	
 Apple Inc.		SIZE	
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